

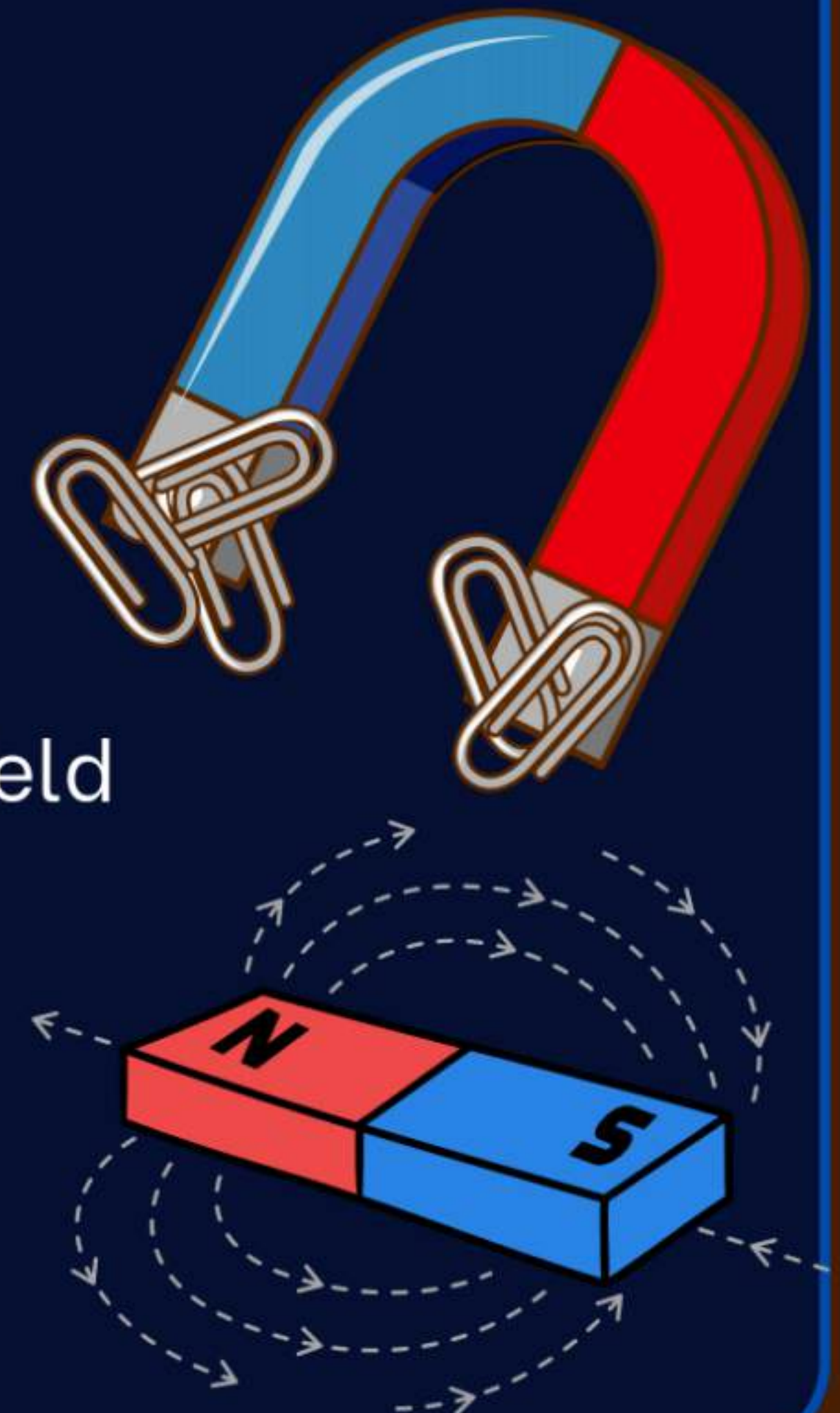


Class 10th - Physics

Magnetic Effect of Electric Current

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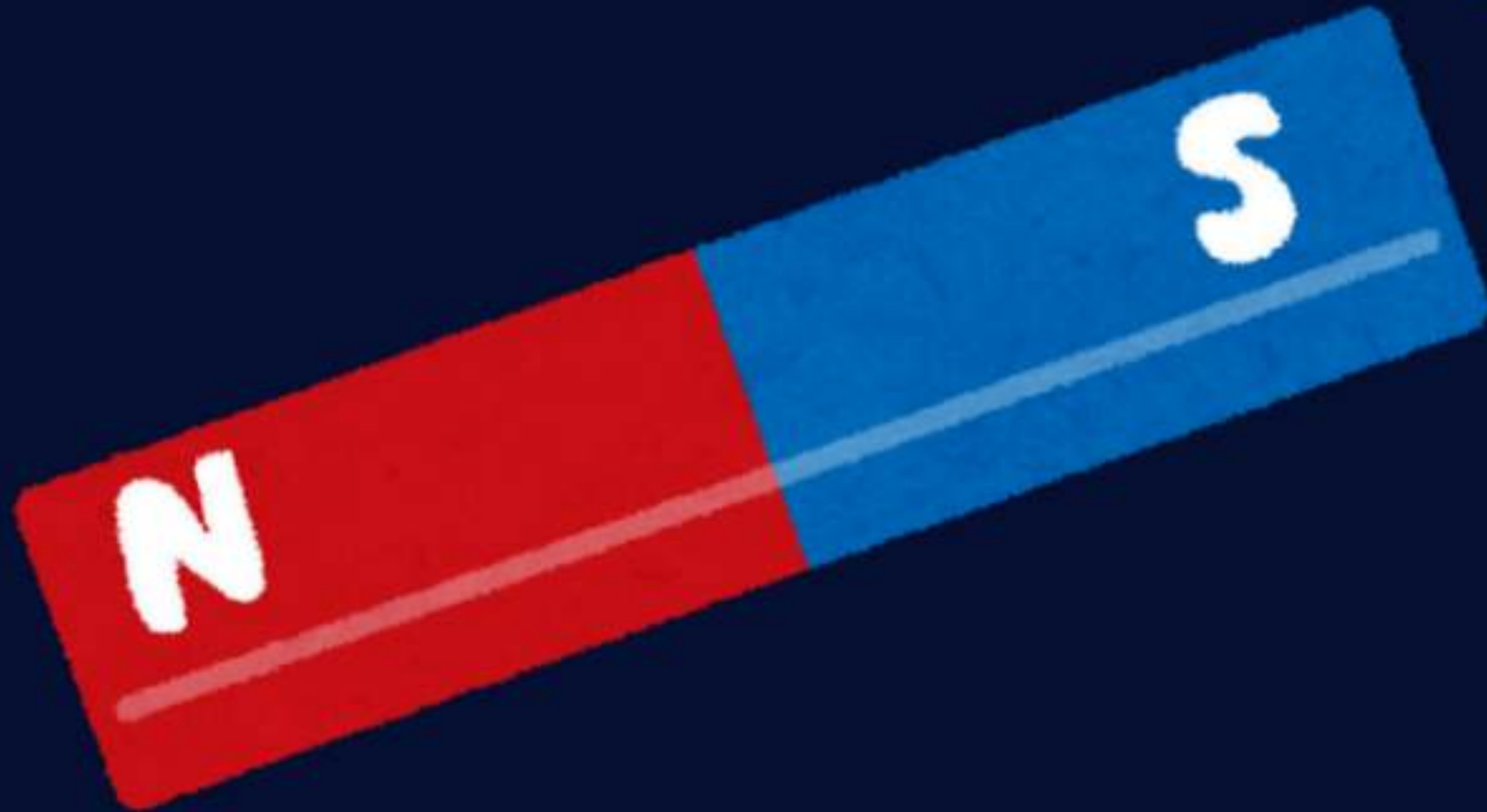


MAGNET

A magnet is a material that attracts objects made of iron, nickel, or cobalt and produces a region around it where magnetic force can be felt.

- Magnetism is the property of a material to attract or repel other materials.

A magnet has two poles – North Pole (N) and South Pole (S).

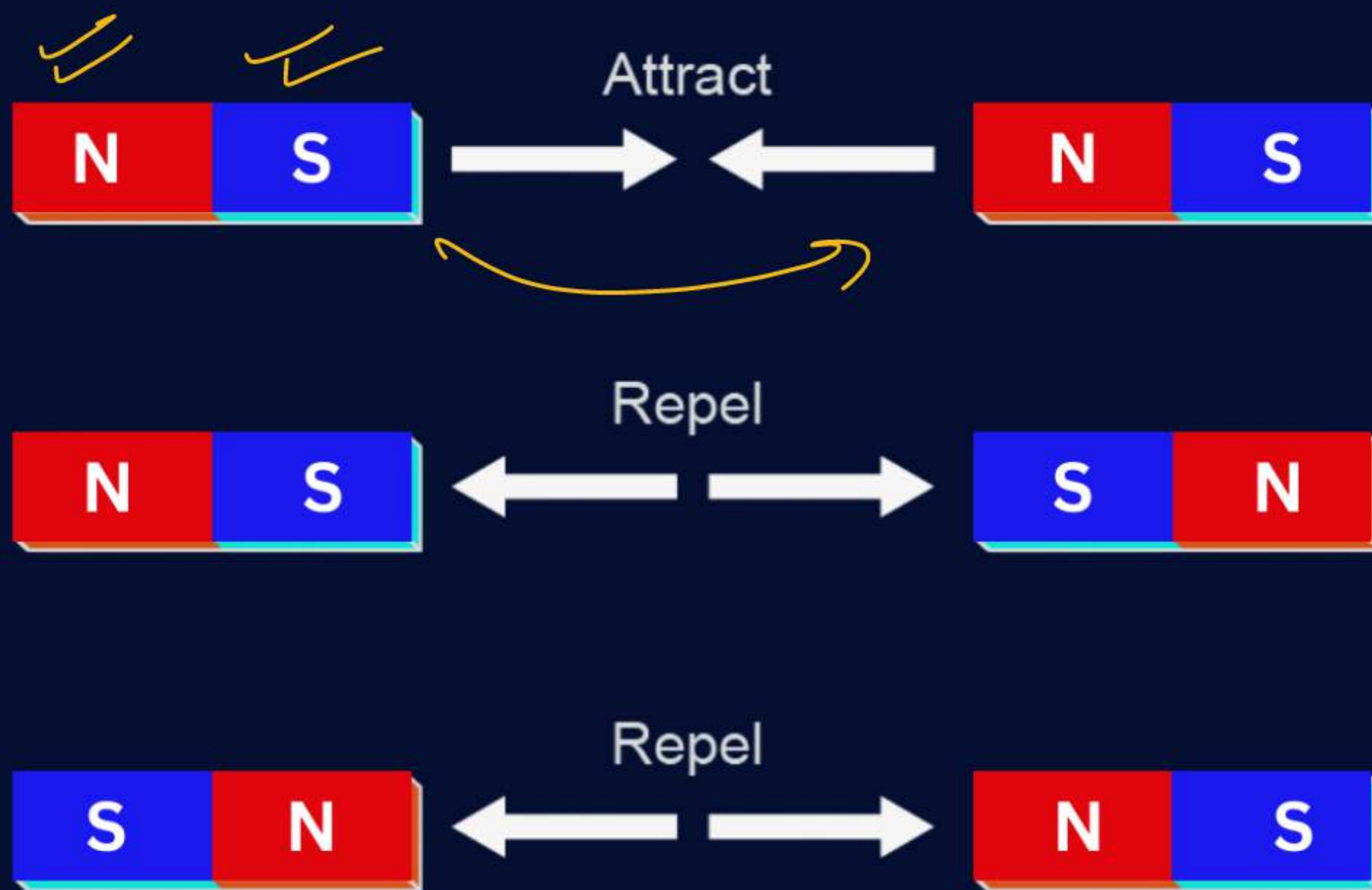


Bar magnet



Horseshoe magnet

BAR MAGNET: A bar magnet is a rectangular piece of a material, usually made of iron or steel, that has permanent magnetism. It has two poles – a north pole and a south pole – where the magnetic strength is the strongest.



A freely suspended bar magnet always rests in the north-south direction due to Earth's magnetic field.

Magnetic Compass

A magnetic compass is a small magnet in the form of a needle that can rotate freely in a horizontal plane.

- It is used to detect the presence and direction of a magnetic field.
- The needle aligns itself along the magnetic field lines, and the north end of the needle points towards the north direction of the magnetic field.



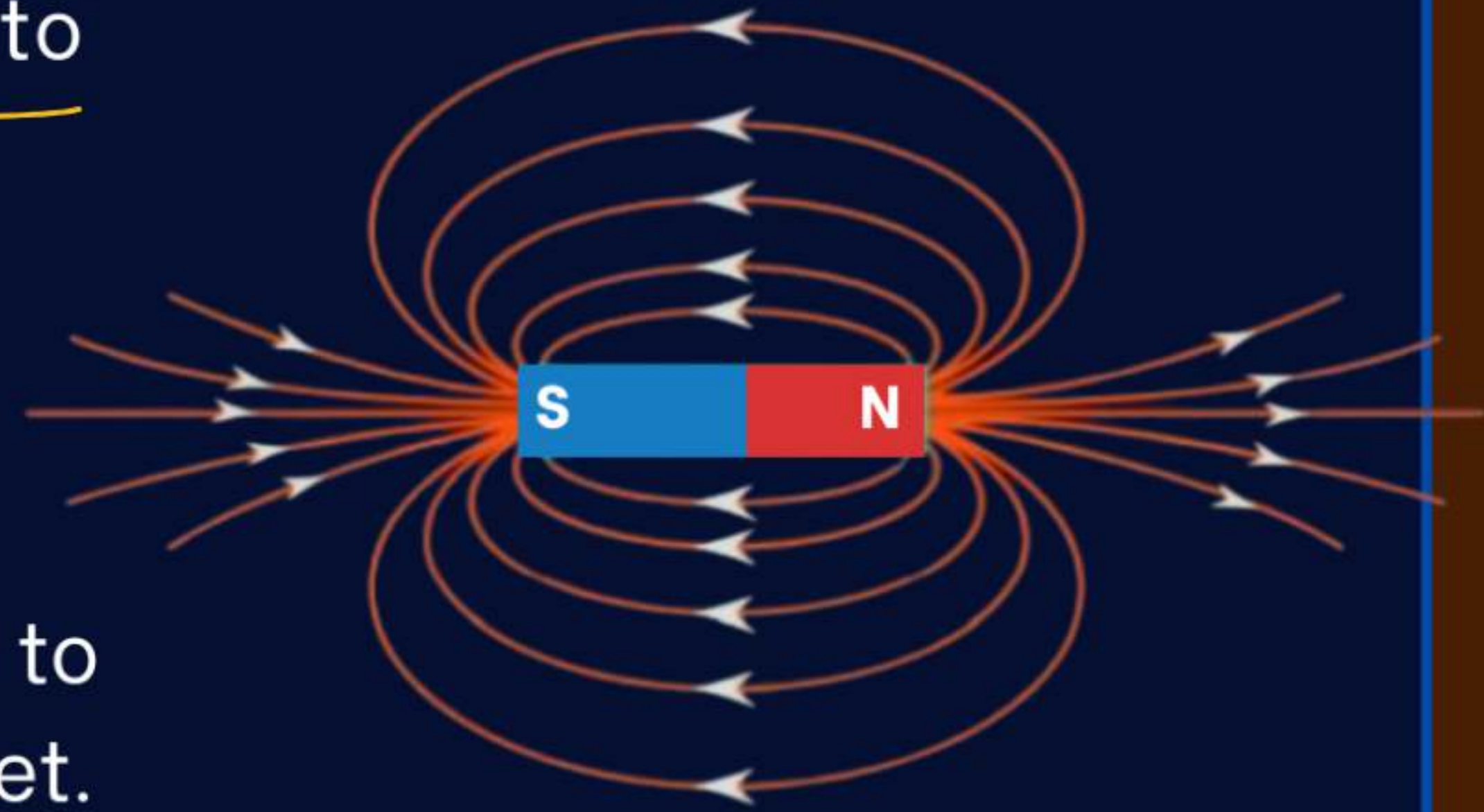
MAGNETIC FIELD :

The region surrounding a magnet, in which the force of the magnet can be detected, is said to have a magnetic field.

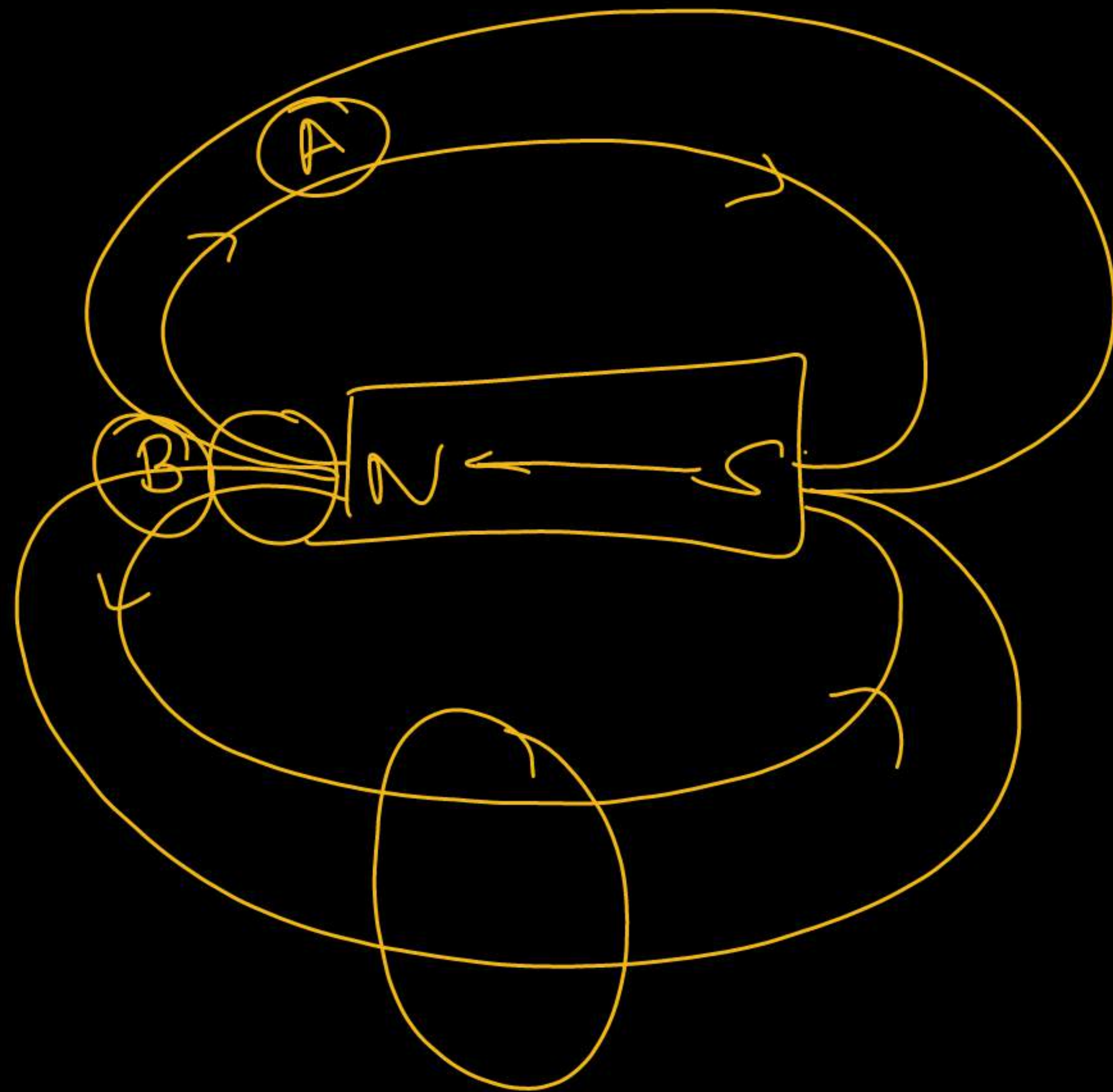
MAGNETIC FIELD LINES :

Magnetic field lines are imaginary lines used to represent the magnetic field around a magnet.

A magnetic field line is the path along which the north pole of a compass needle moves inside the magnetic field.



Field lines around a bar magnet



Properties of Magnetic Field Lines:



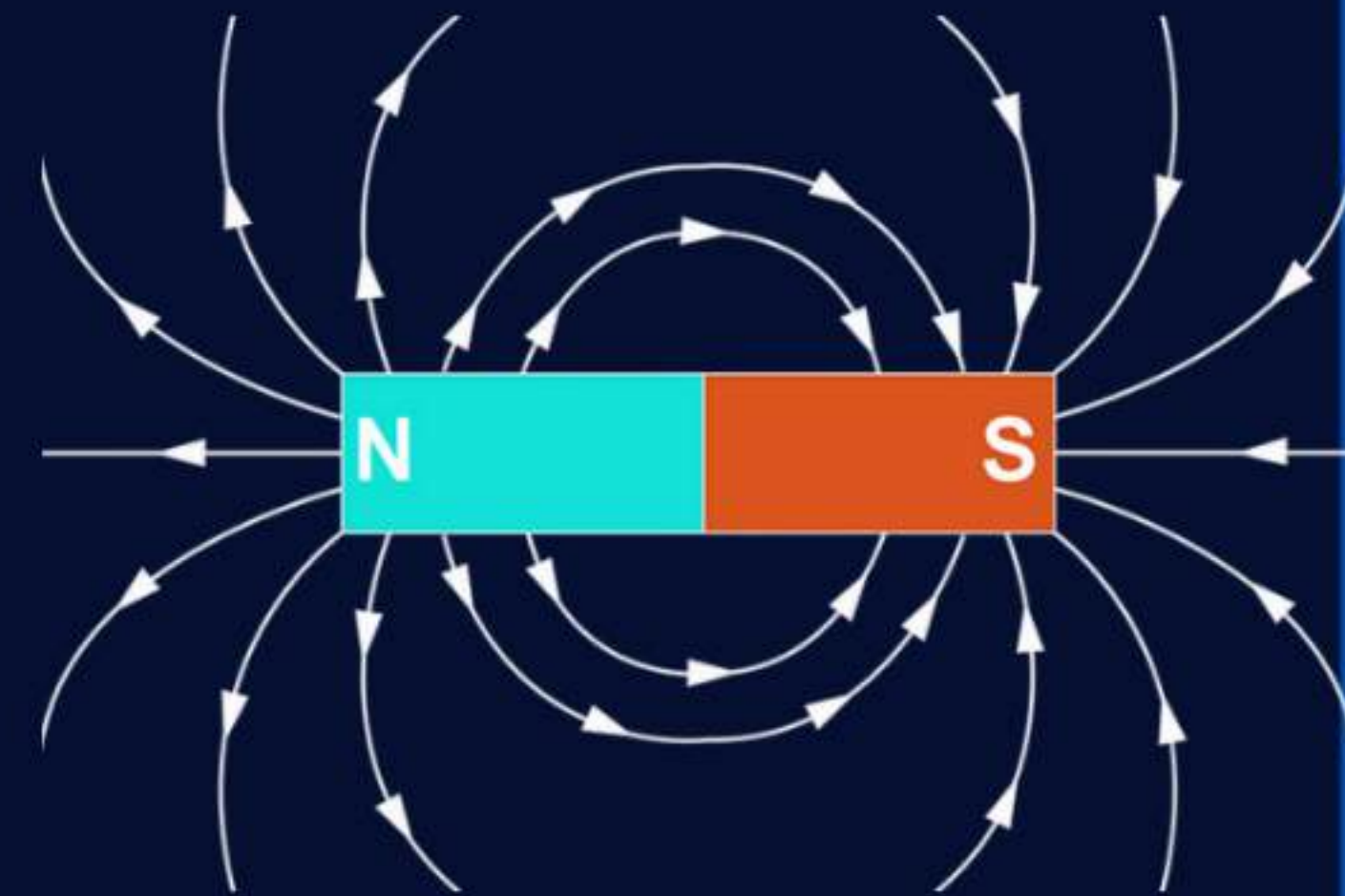
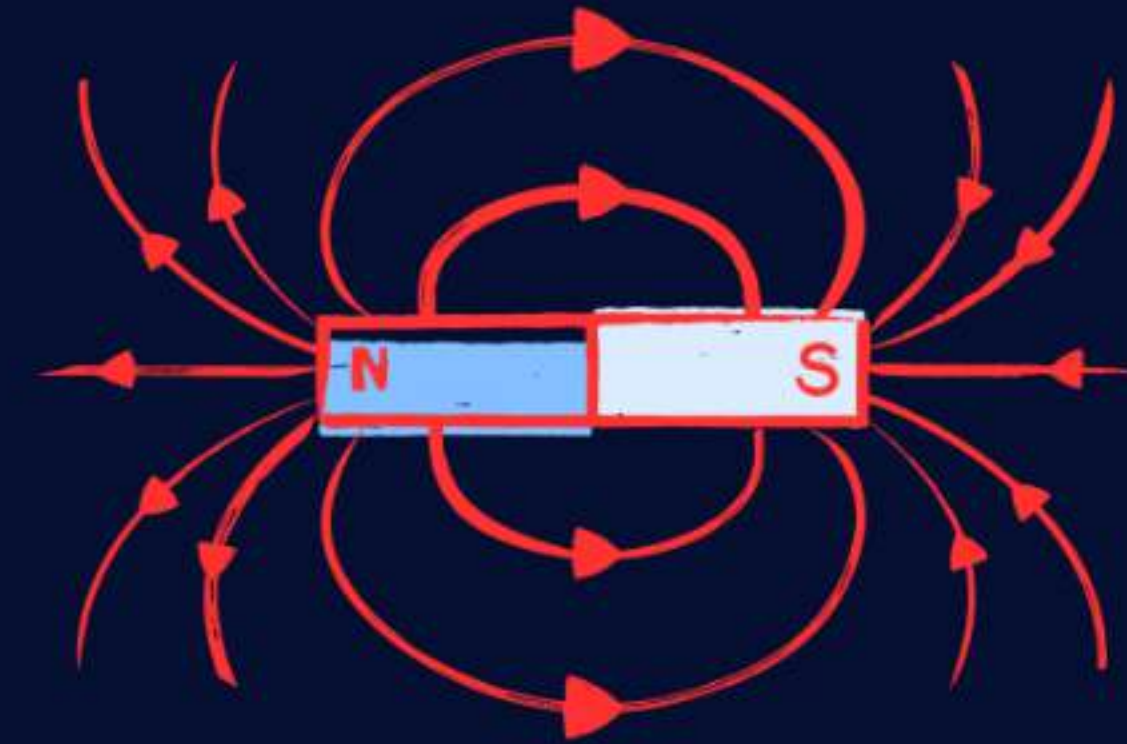
- **Closed continuous curves:** Magnetic field lines are closed continuous curves which start from the north pole and end at the south pole of a magnet, and inside the magnet they move from south to north.

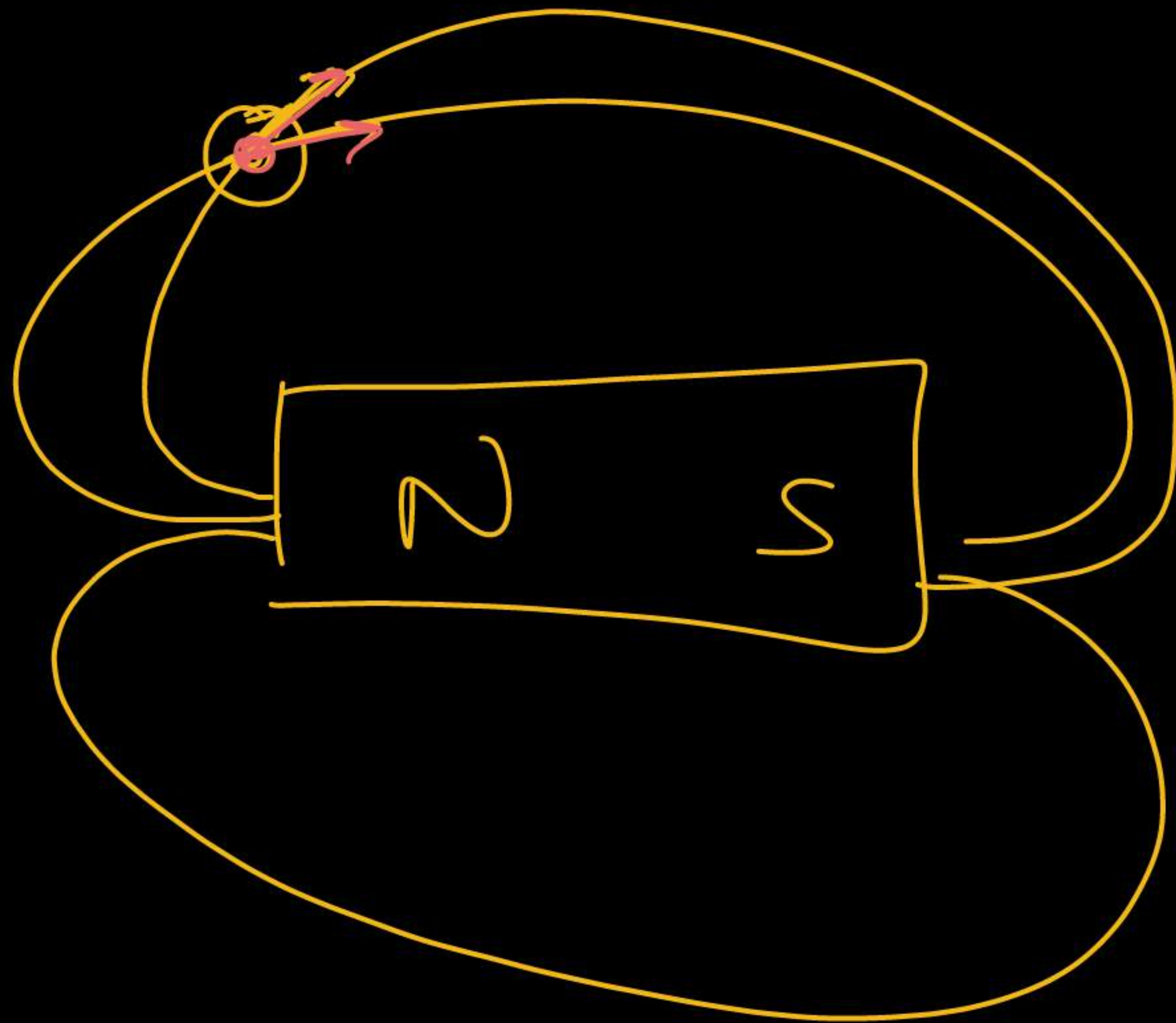
N-S
S-N

- **Direction of magnetic field lines:** The direction of magnetic field lines at any point gives the direction of the magnetic field at that point.

- **Closeness of the field lines:** The closeness of the field lines indicates the strength of the magnetic field – closer lines show a stronger field, and lines farther apart show a weaker field.

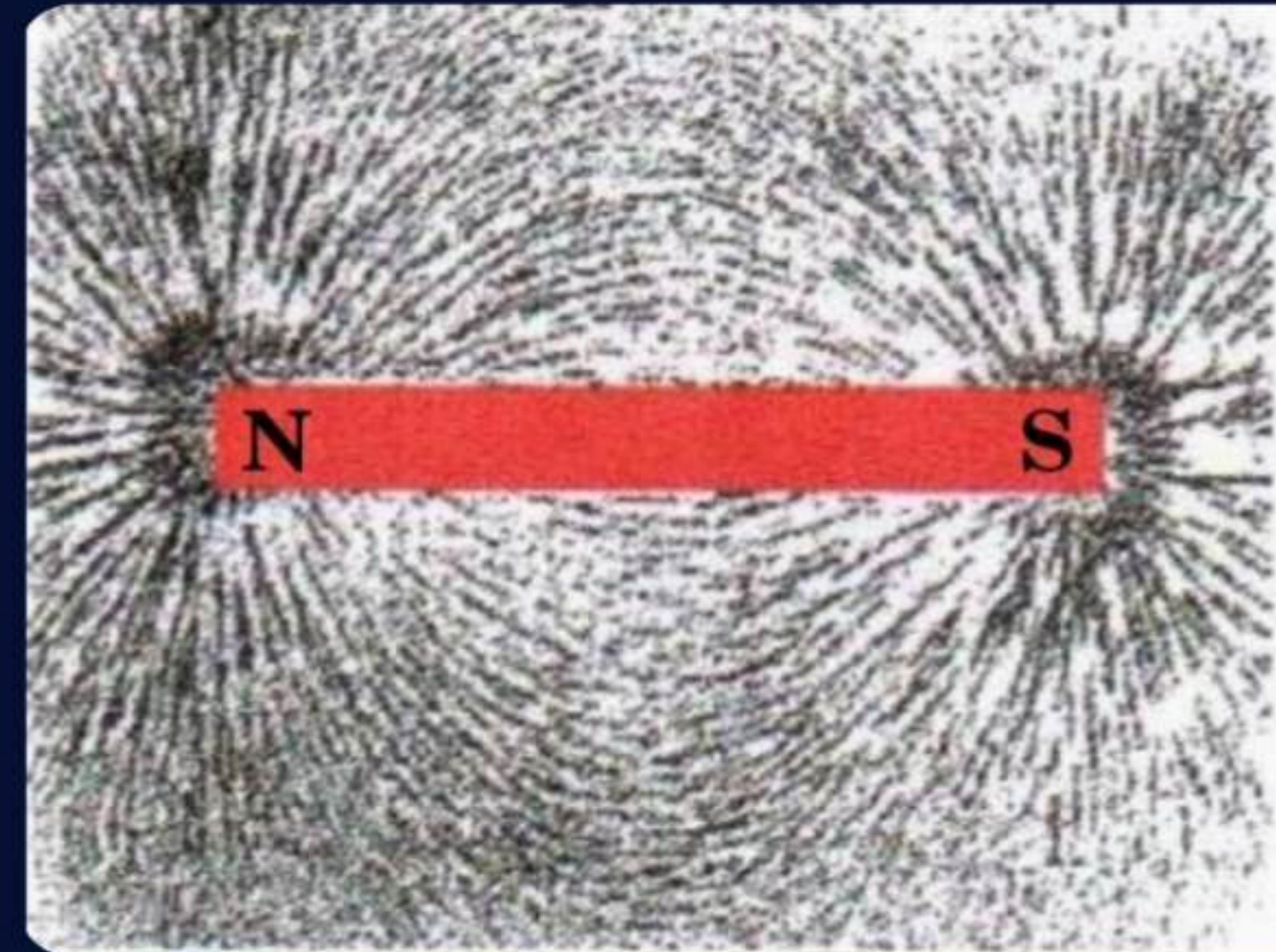
- **Never intersect each other:** Magnetic field lines **never intersect each other** because if they did, the compass needle would point in two directions at the same point, which is not possible.





ACTIVITY

- Fix a white sheet on a drawing board and place a bar magnet at the centre.
- Sprinkle iron filings uniformly around the magnet using a salt sprinkler and gently tap the board.
- **The iron filings arrange themselves in a definite pattern of curved lines around the magnet.**
These lines represent **magnetic field lines**.
- ✓ ***This shows that a magnetic field exists around a magnet, and the pattern gives the shape and direction of the magnetic field.***



2025

Q. Which one of the following statements is not true about a bar magnet?

- (a) It sets itself in north-south direction when suspended freely. ✗**
- (b) It has attractive power for iron filings. ✗**
- (c) It produces magnetic field lines. ✗**
- (d) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.**

2025

**Q. Assertion (A): No two magnetic field lines are found to cross each other.
Reason (R): The compass needle cannot point towards two directions at the point of intersection of two magnetic field lines.**

- (a) Both A and R are true, and R is the correct explanation of A.**
- (b) Both A and R are true, but R is not the correct explanation of A.**
- (c) A is true, but R is false.**
- (d) A is false, but R is true.**

Q1. Why do magnetic field lines never intersect each other? ✓

Because if they did, the compass needle at that point would point in two different directions, which is not possible.

Q2. Why are magnetic field lines denser near the poles of a magnet? ✓

Because the magnetic field is strongest near the poles, so the lines are closer together there.

Q3. Why do magnetic field lines emerge from the north pole and enter the south pole outside a bar magnet?

Because by convention, the direction of the magnetic field is taken as the direction in which the north pole of a compass needle moves, that is, from north to south outside the magnet.

Q4. Why is the magnetic field strongest near the poles of a bar magnet? ✓

Because magnetic field lines are crowded near the poles, showing that the field strength is maximum there.

Q5. Why do magnetic field lines not have free ends?

Because magnetic field lines form closed continuous loops – they start from the north pole and end at the south pole, then continue inside the magnet from south to north.

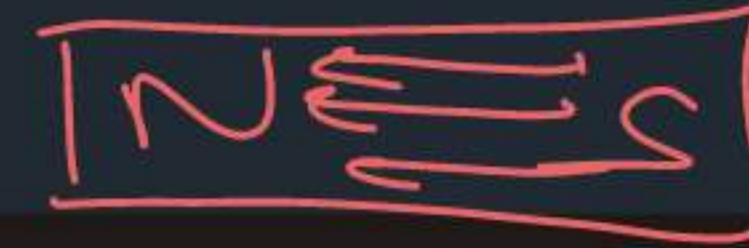
Q6. Why do iron filings align themselves along magnetic field lines when sprinkled around a magnet?

Because each iron filing becomes a tiny induced magnet and experiences a force along the direction of the magnetic field lines.

2025, 2024 (Sample Paper) 2021, 2018, 2015, 2014, 2012

Q. (a) State the relationship between the strength of the magnetic field and the degree of closeness of the field lines.

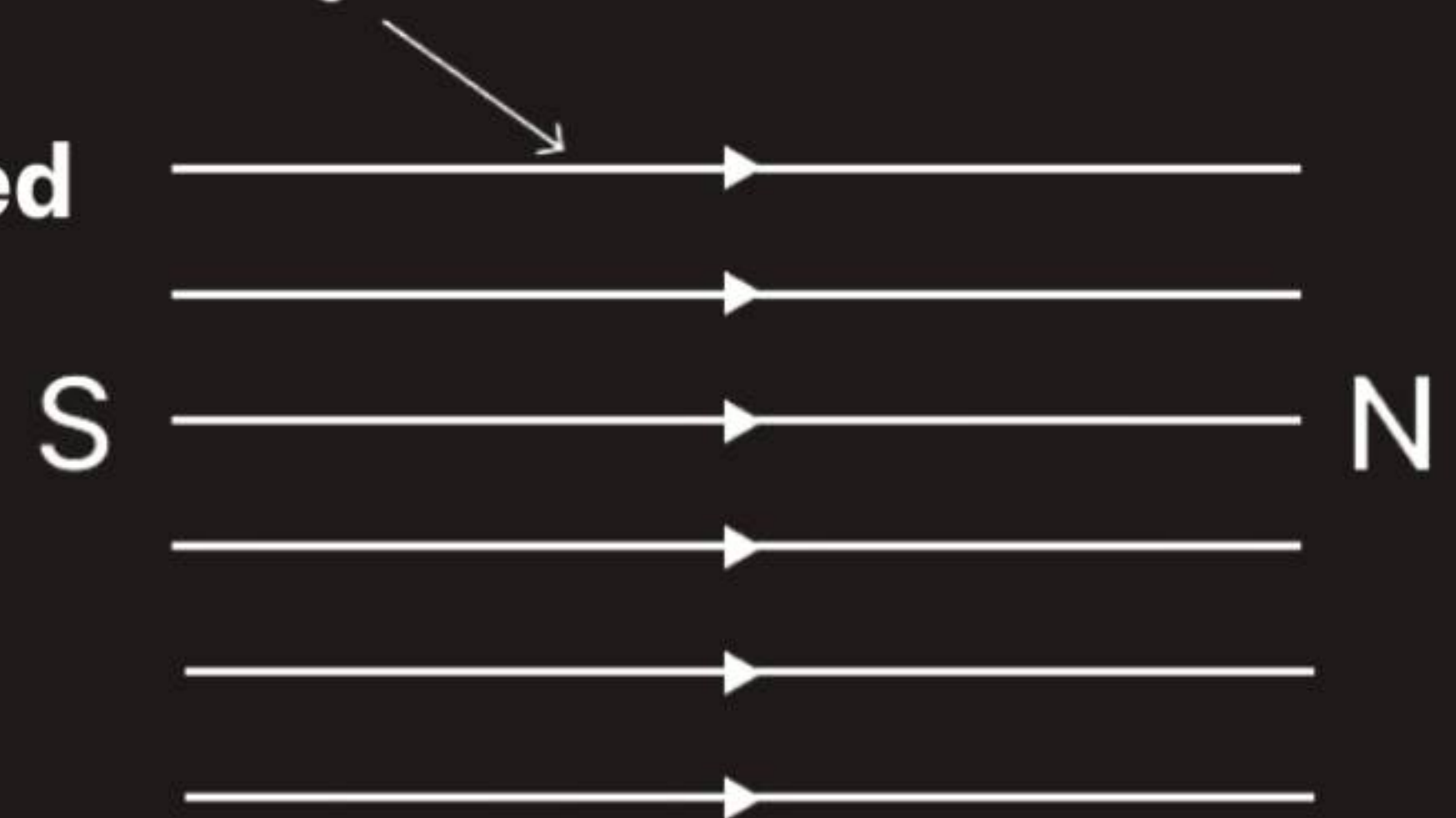
(b) The magnetic field in a given region is uniform. Draw a diagram to represent it.



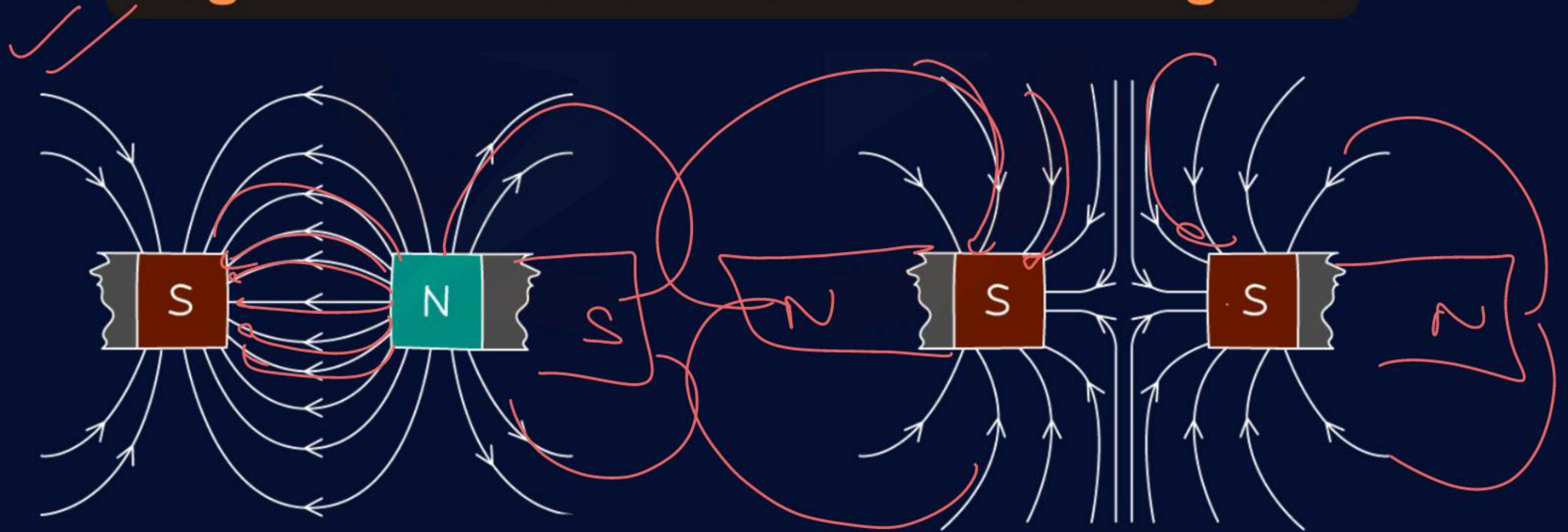
The strength of the magnetic field is directly related to the closeness of the magnetic field lines.

- **Closer field lines → Stronger magnetic field**
- **Lines farther apart → Weaker magnetic field**

Magnetic field lines



Magnetic Field Lines due to Two Bar Magnets



Attraction occurs between **unlike poles** and the field lines connect directly from N to S.

Repulsion occurs between **like poles**, and the field lines never intersect or connect in between.

2015, 2017, 2019 (Board), 2022, 2024 (Sample Papers)

Q. Name the poles P, Q, R and S of the magnets in the following figures 'a' and 'b':

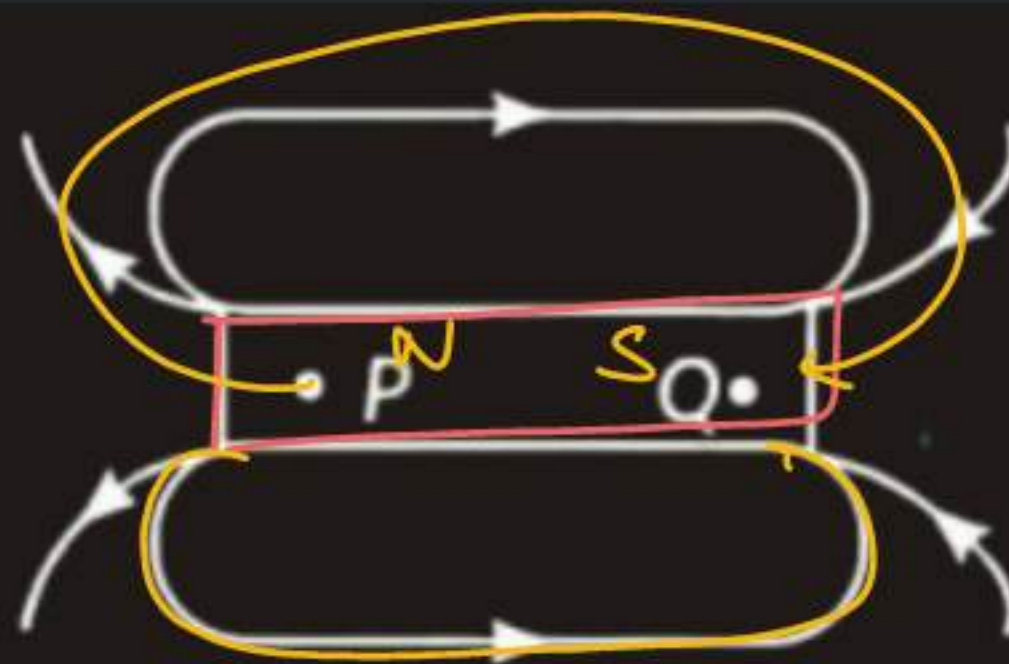


Figure 'a'

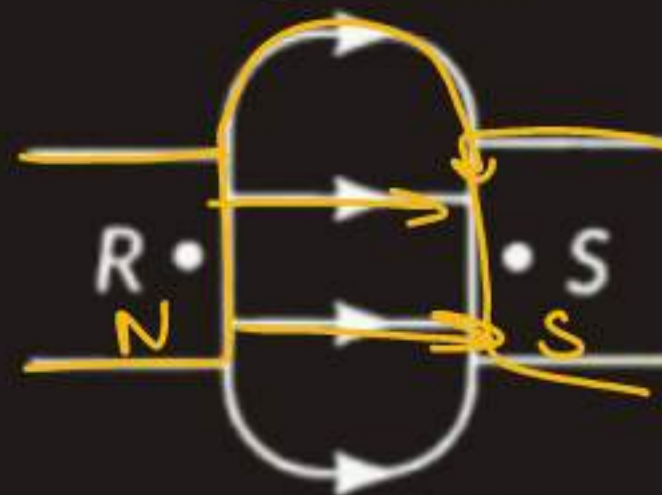


Figure 'b'

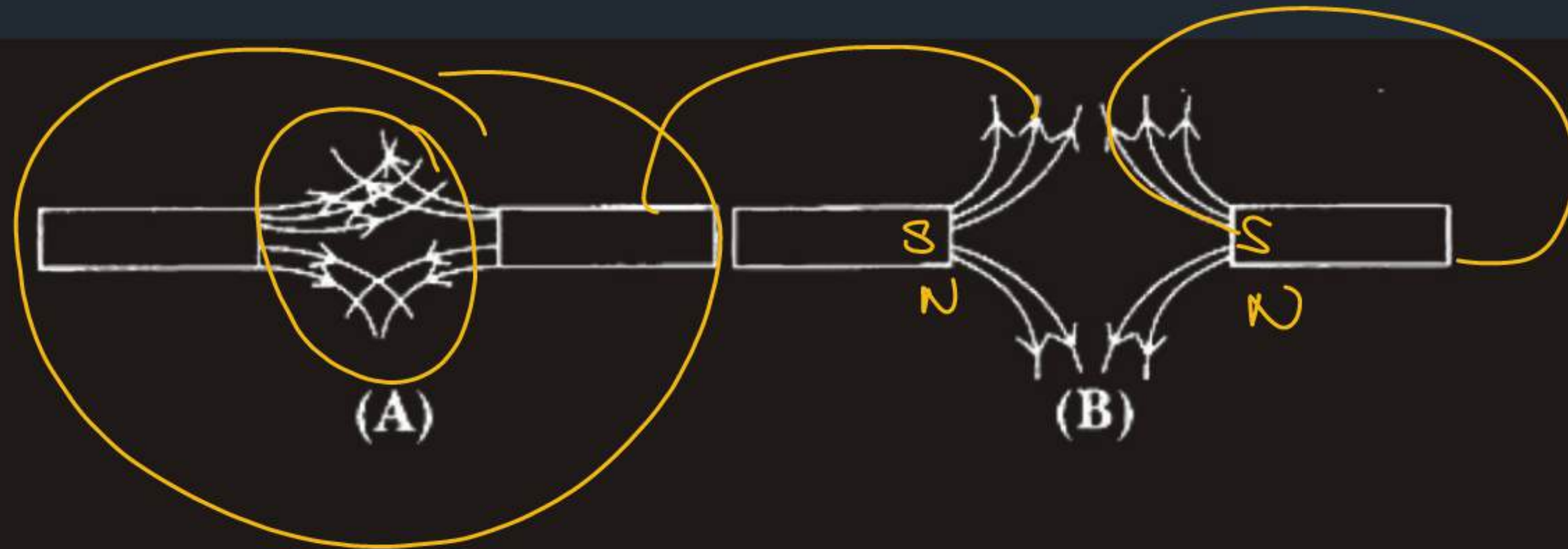
Ans: In Figure 'a', poles P and Q of the magnet represents north pole and south pole respectively. In figure 'b', poles R and S of the magnet also represents north pole and south pole respectively.

2016, 2018, 2020, 2023

Q. Magnetic field lines of two magnets are shown in Fig. A and Fig. B.

Select the figure that represents the correct pattern of field lines.

Give reasons for your answer. Also name the poles of the magnets facing each other

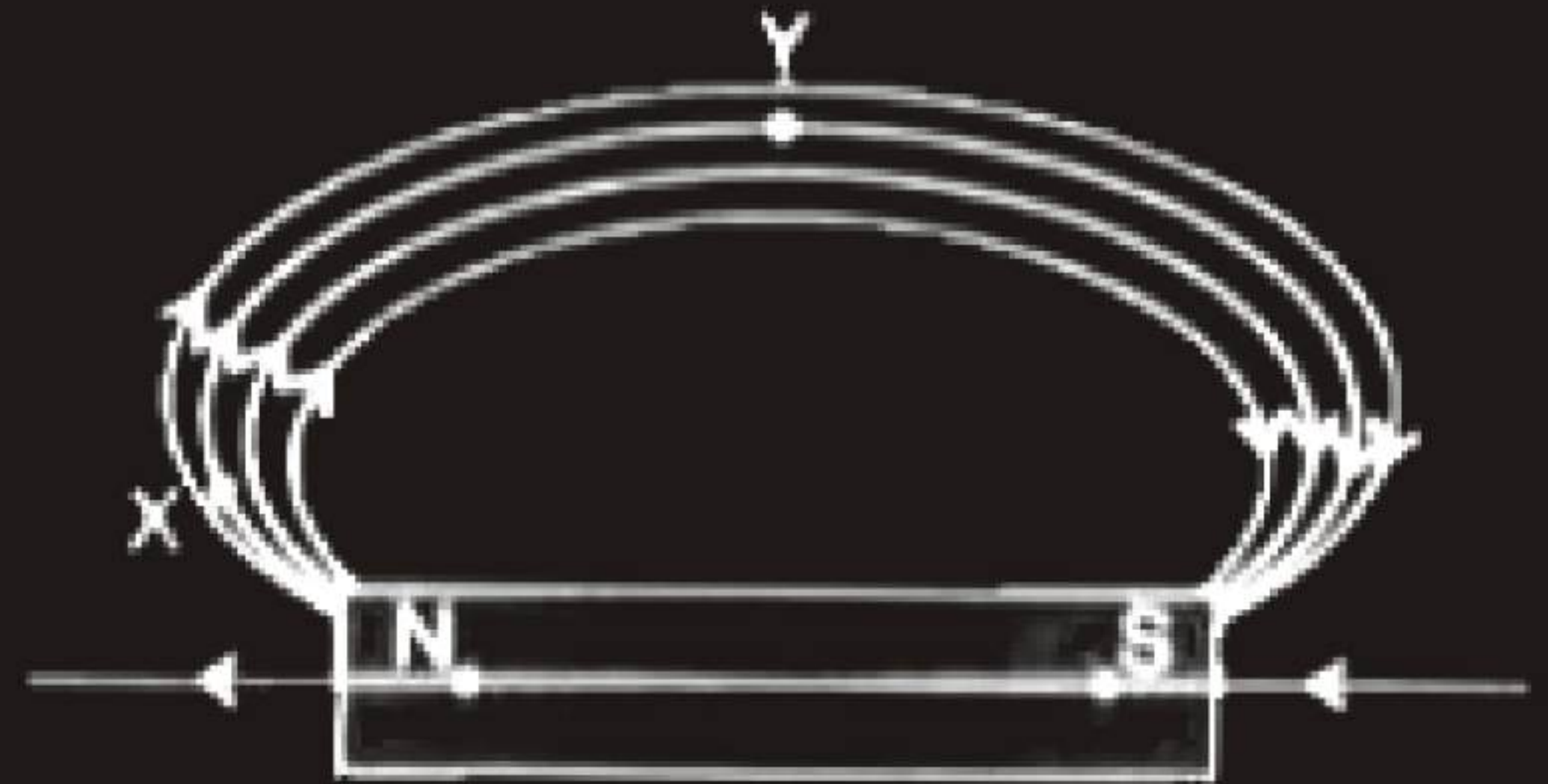


2012, 2014

Q. Magnetic field lines are shown in the given diagram. A student makes a statement that magnetic field at X is stronger than at Y. Justify this statement. Also, redraw the diagram and mark the direction of magnetic field lines.

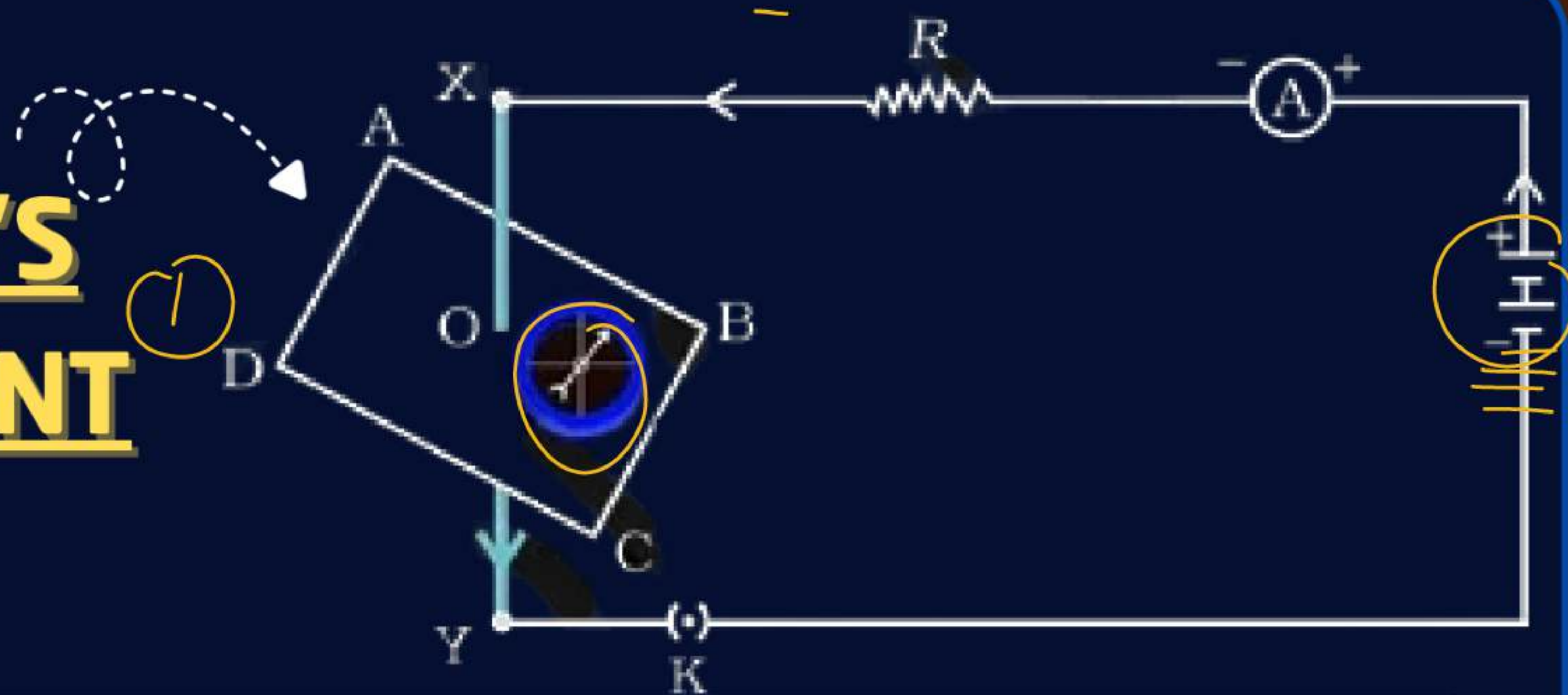


- Closeness of magnetic field lines is directly related to strength of magnetic field. Strength of magnetic field at point X (pole) is more than point Y.
- If the student redraws the diagram he/she should mark arrows correctly from North to South.



ACTIVITY

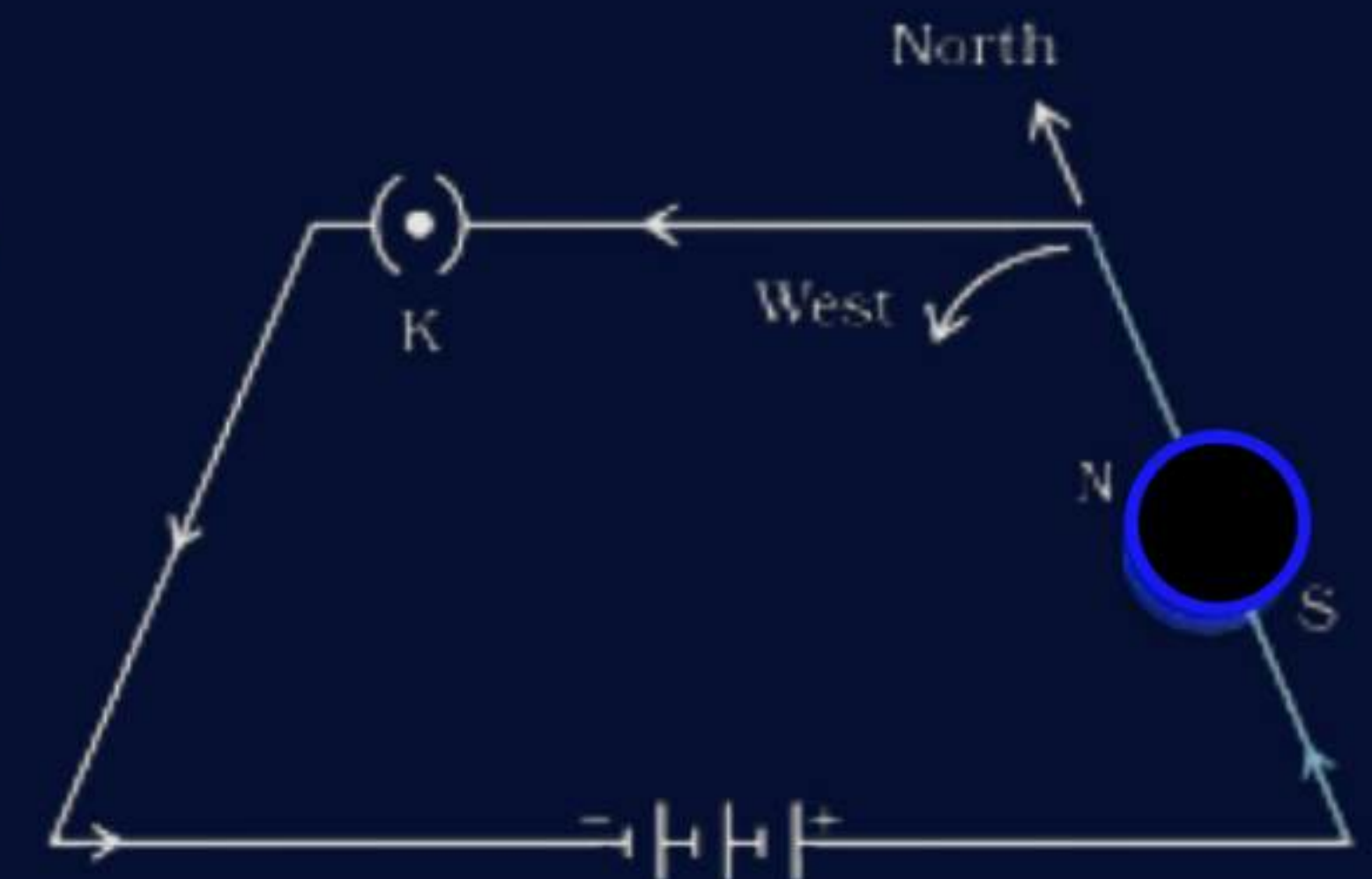
OERSTED'S EXPERIMENT



- Take a thick straight copper wire (XY) in a circuit.
- Place it perpendicular to the plane of paper.
- Keep a compass needle near the wire.
- When the key is inserted (current passes), the compass needle deflects.
- **The deflection shows that electric current produces a magnetic effect.**
- This proves a link between electricity and magnetism.

Magnetic Field due to a Current-Carrying Conductor

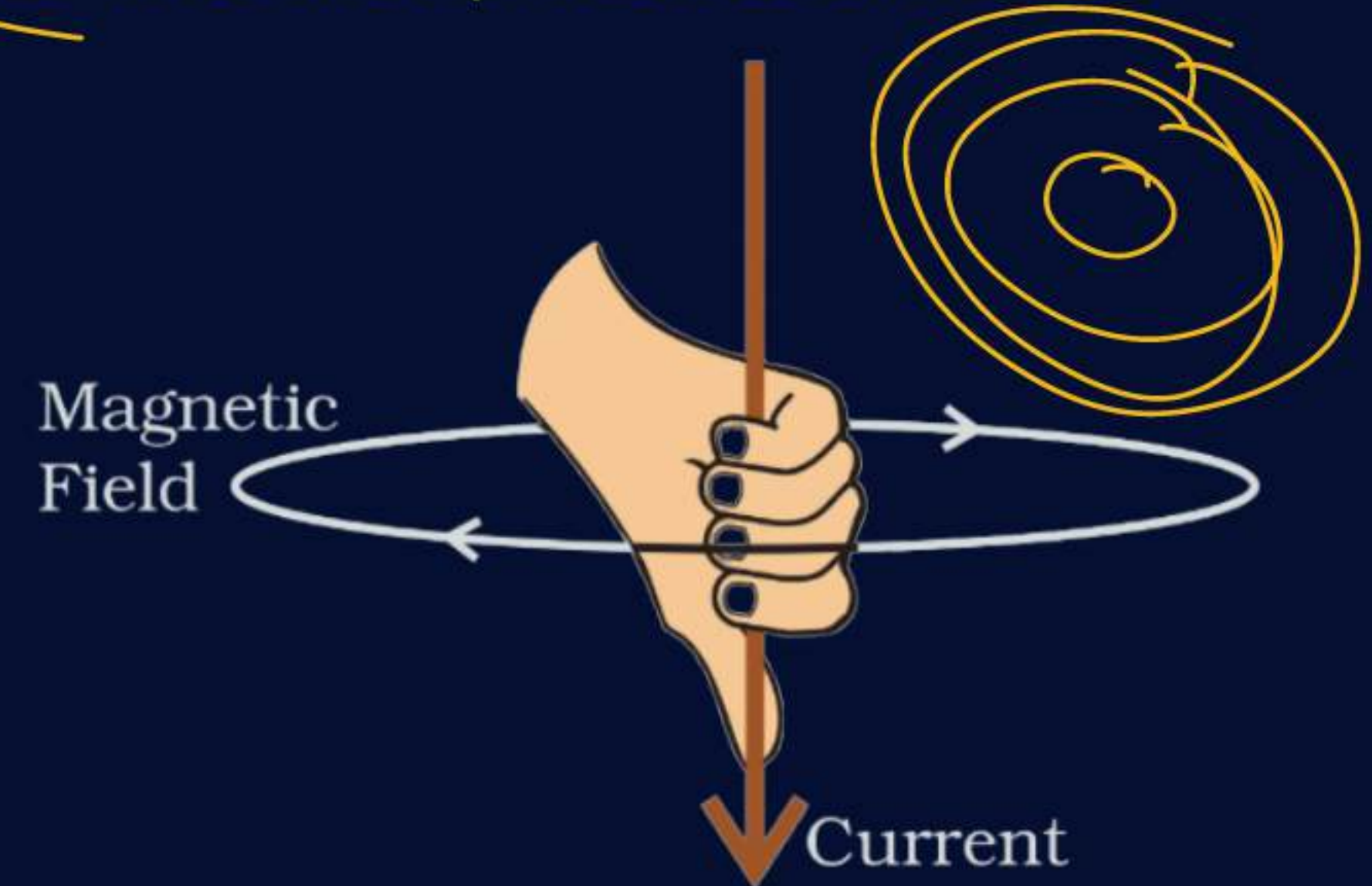
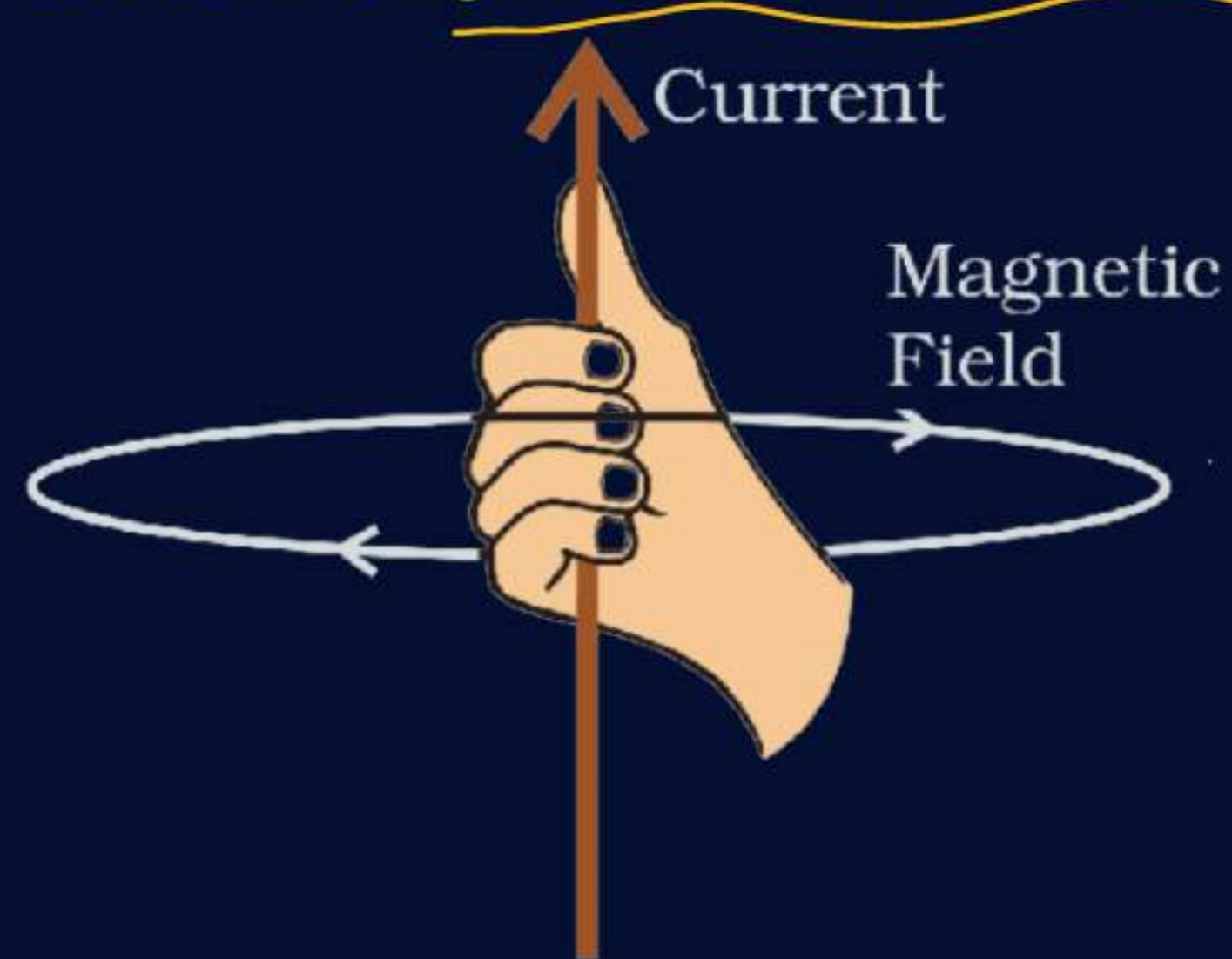
- **Magnitude of Current (I):** The magnetic field increases with an increase in current through the conductor.
- **Stronger current $\rightarrow$ Stronger magnetic field.**
- **Distance from the Conductor (r):** The magnetic field decreases as the distance from the conductor increases.
- **Farther point $\rightarrow$ Weaker magnetic field.**
- **Direction of Current:** The direction of the magnetic field around the conductor depends on the direction of current.
- **Reversing the current reverses the direction of the magnetic field.**



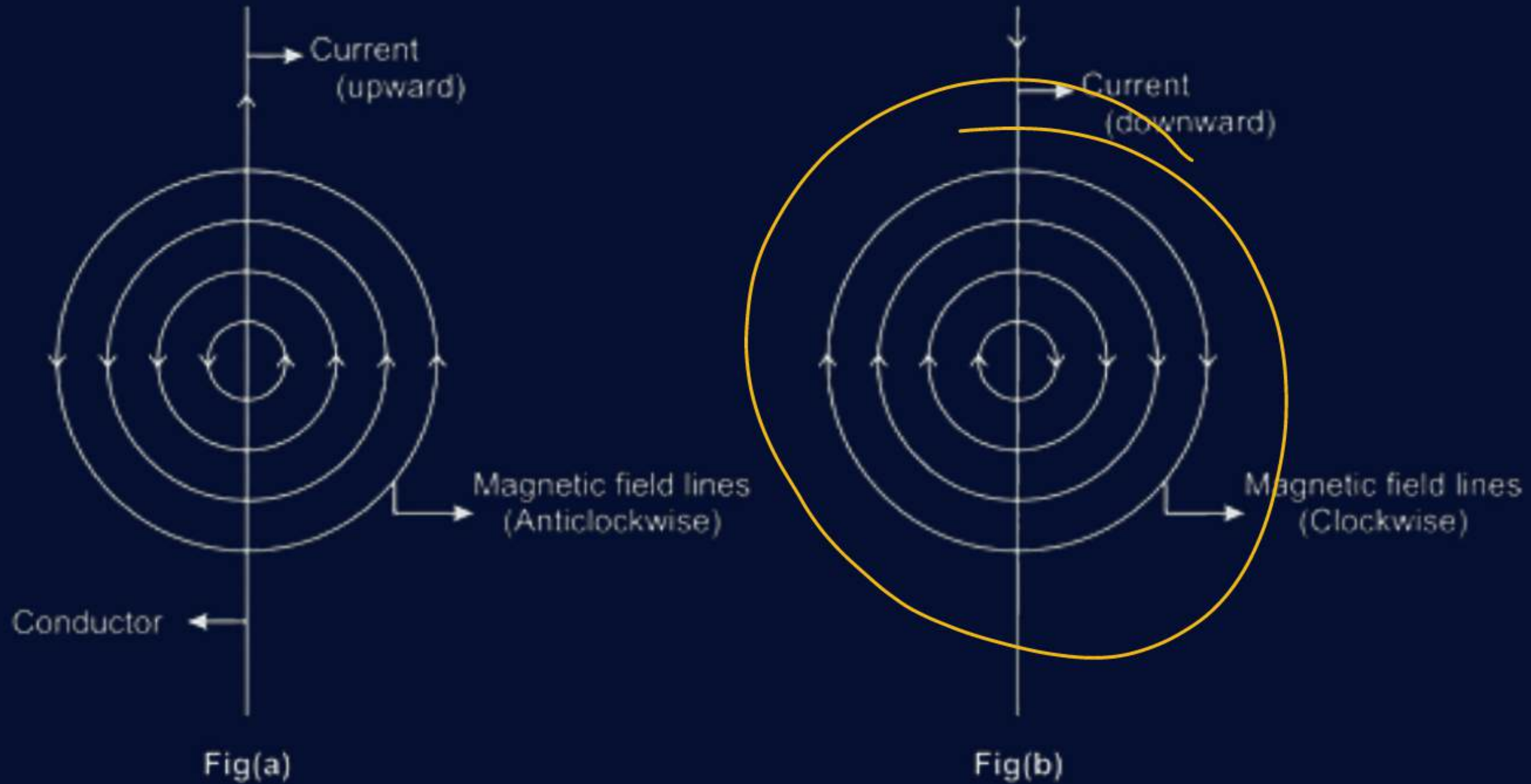
MAXWELL'S RIGHT HAND THUMB RULE

State the Right-Hand Thumb Rule. How is it used to find the direction of magnetic field around a straight current-carrying conductor?

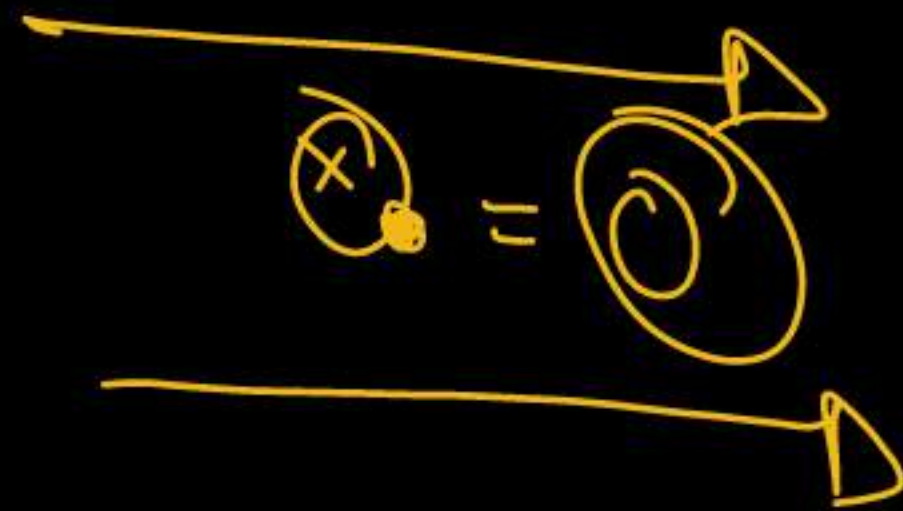
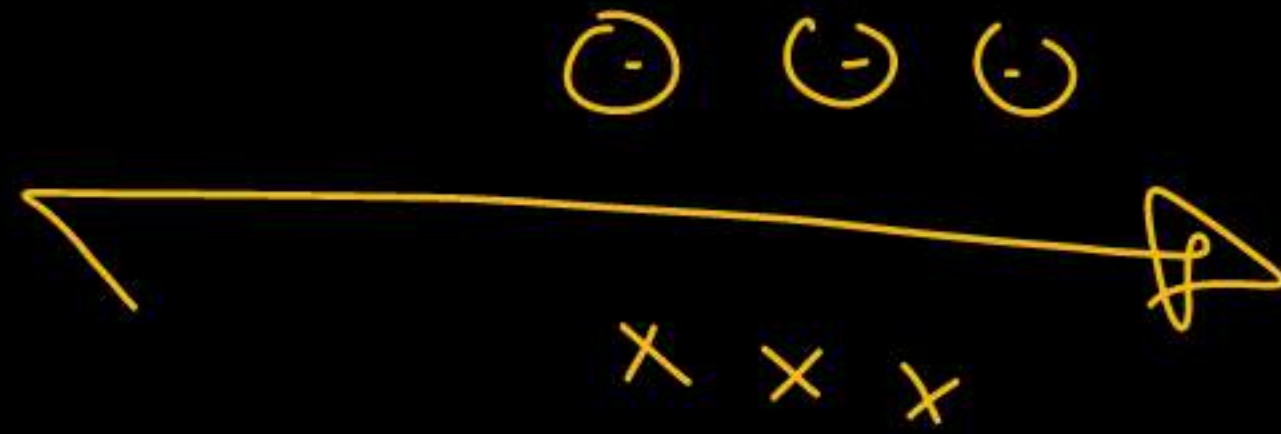
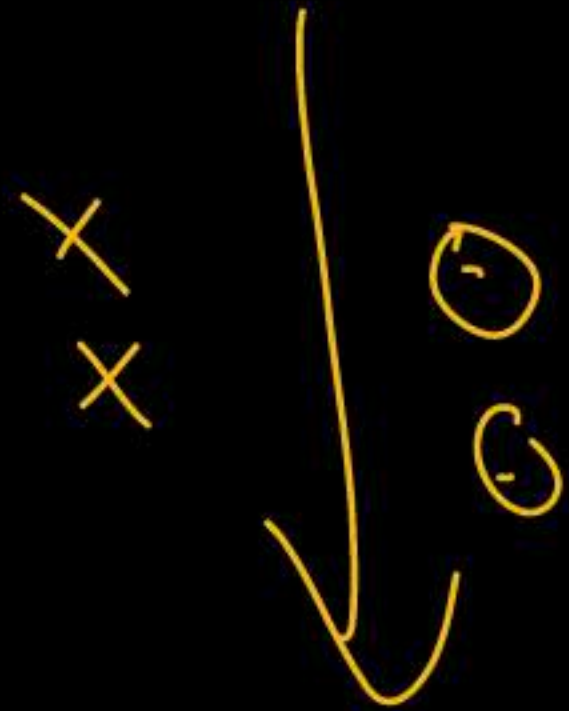
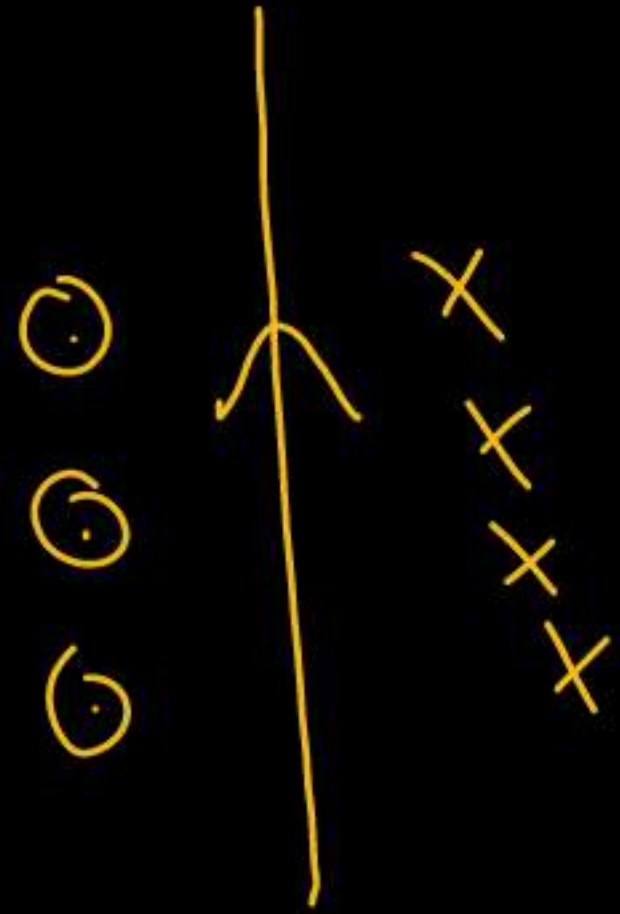
If a current-carrying straight conductor is imagined to be held in the right hand such that the thumb points in the direction of current, then the curl of the fingers around the conductor gives the direction of the magnetic field lines produced around it.



Shape of Magnetic Field due to a Straight Current-Carrying Conductor



X X X X X
O O O O



2014, 2015, 2016, 2019

Q. A compass needle is placed near a current-carrying wire.

State your observation for the following cases and give reason for the same in each case:

a. Magnitude of electric current in the wire is increased.

b. The compass needle is displaced away from the wire.

(a) When the current is increased, the deflection of the compass needle increases.

Reason: Increasing the current increases the strength of the magnetic field produced around the wire.

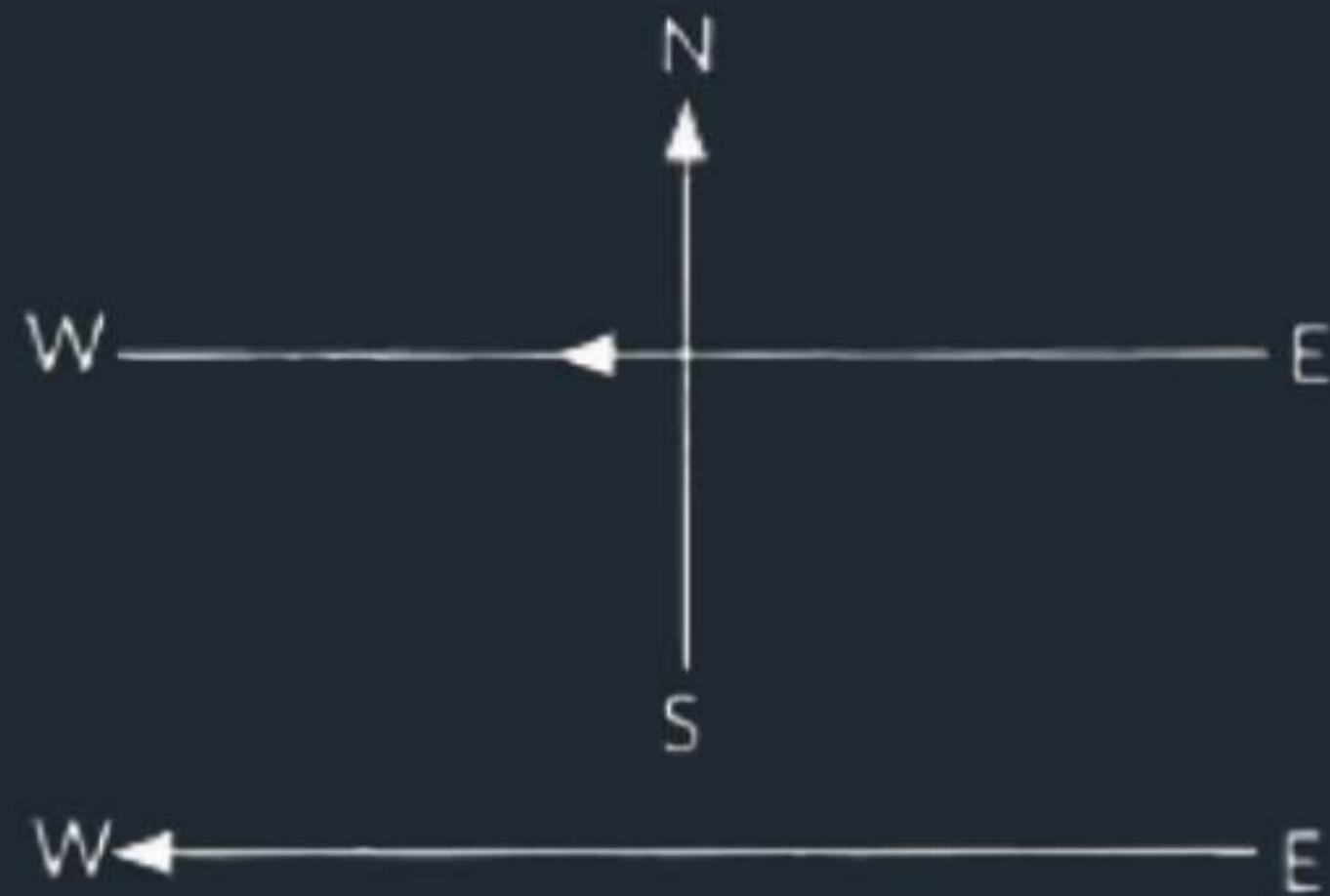
(b) When the compass needle is moved away from the wire, the deflection decreases.

Reason: The magnetic field strength decreases with distance from the current-carrying conductor.

2018 Board, 2021 Sample Paper

Q. A constant current flows in a horizontal wire in the plane of the paper from east to west as shown in the figure. The direction of the magnetic field will be north to south at a point N

- (a) Directly above the wire**
- (b) Directly below the wire**
- (c) Located in the plane of the paper on the north side of the wire**
- (d) Located in the plane of the paper on the south side of the wire**

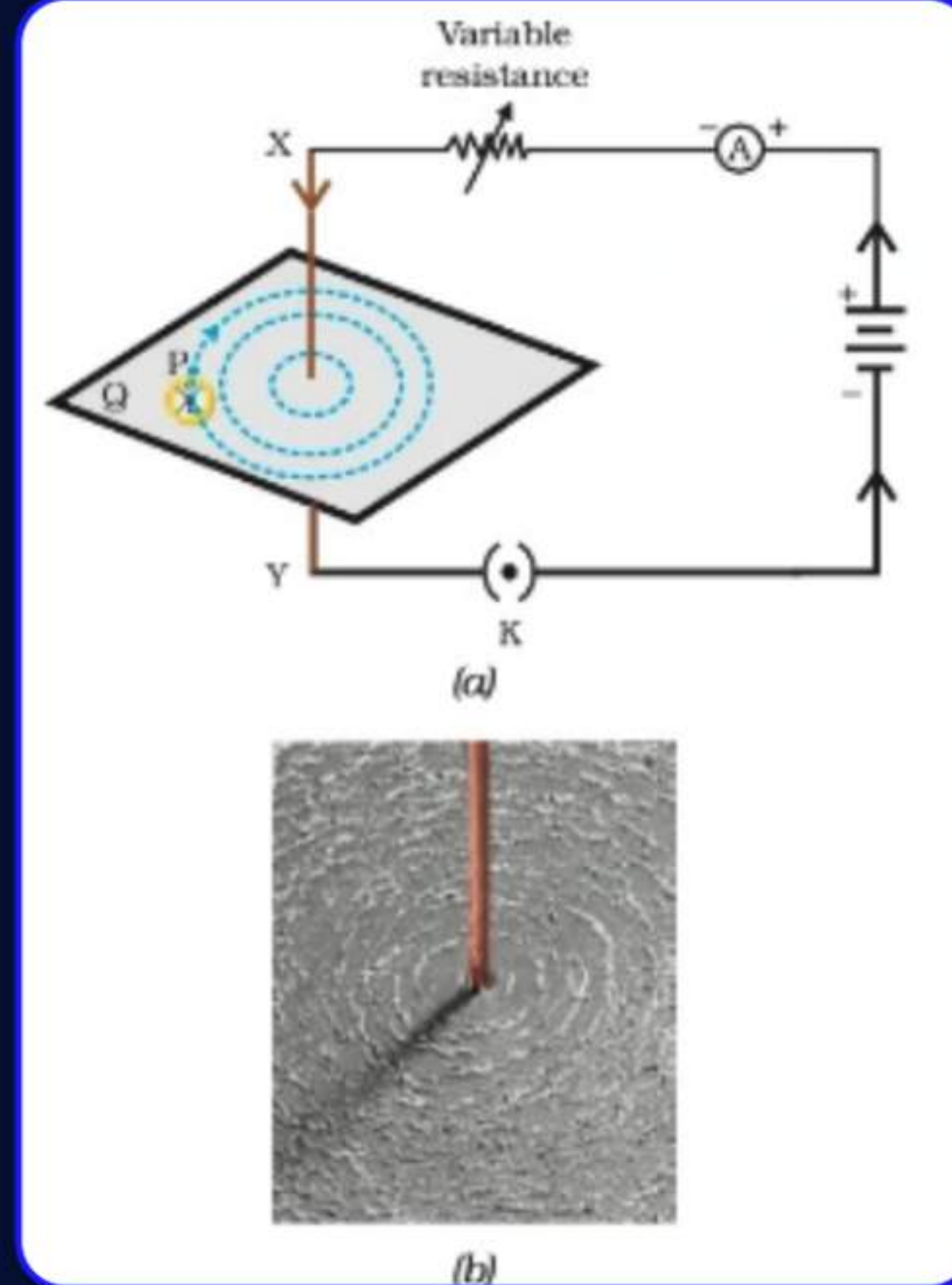


Magnetic Field due to a Current through a Straight Conductor

- Magnetic field lines around a straight current-carrying wire form **concentric circles centered on the wire**.
- The direction of the field is given by the deflection of a compass needle placed near the wire.

Factors Affecting the Magnetic Field:

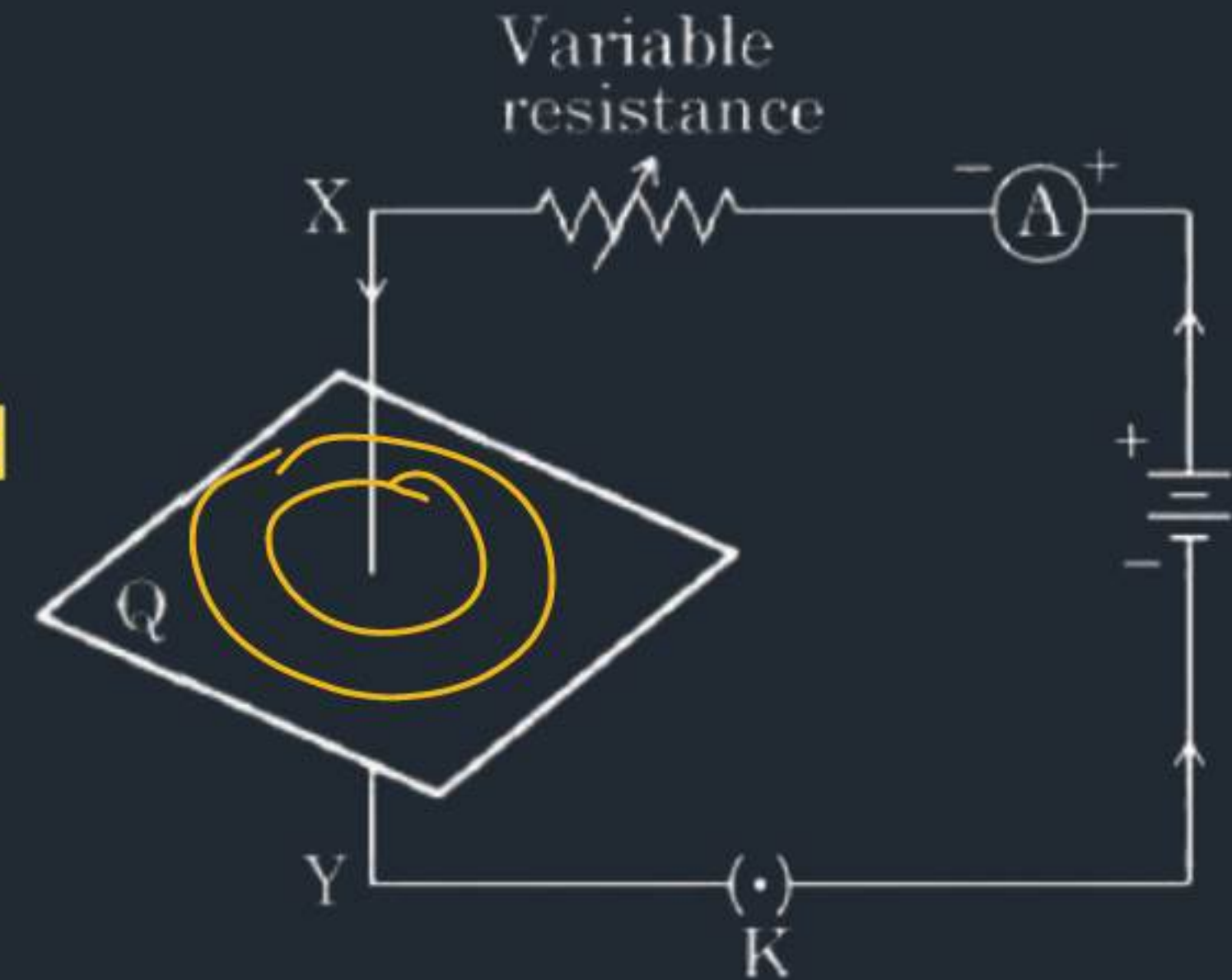
- **Reversing the direction of current** → Reverses the direction of magnetic field lines.
- **Increasing the current** → Increases the strength of the magnetic field (shown by greater compass needle deflection).
- **Moving away from the wire** → Decreases the magnetic field strength (smaller compass needle deflection).



2025

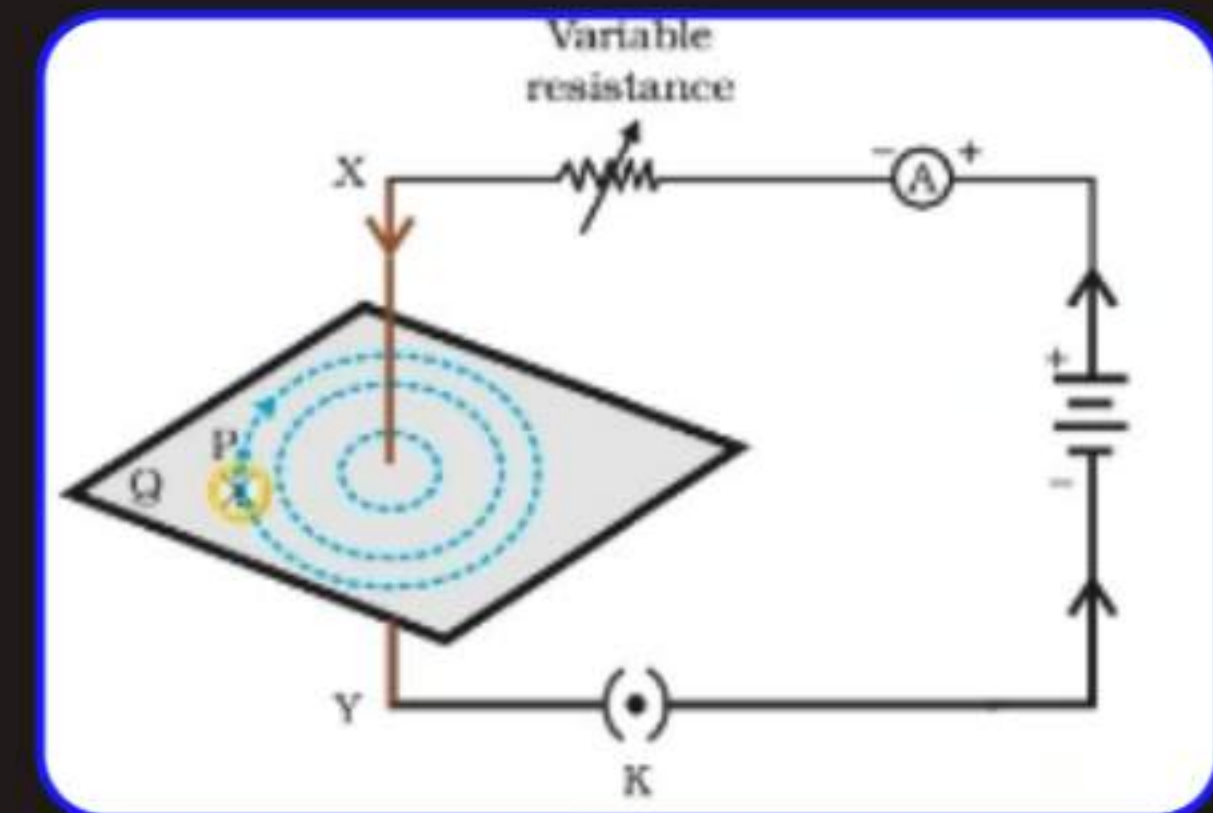
Q. The given figure shows the current passing through the straight conductor XY.

- (i) Copy the diagram and draw the magnetic field lines when current flows from conductor X to Y.**
- (ii) Name and state the rule used in determining the direction of the magnetic field lines in the situation given above.**



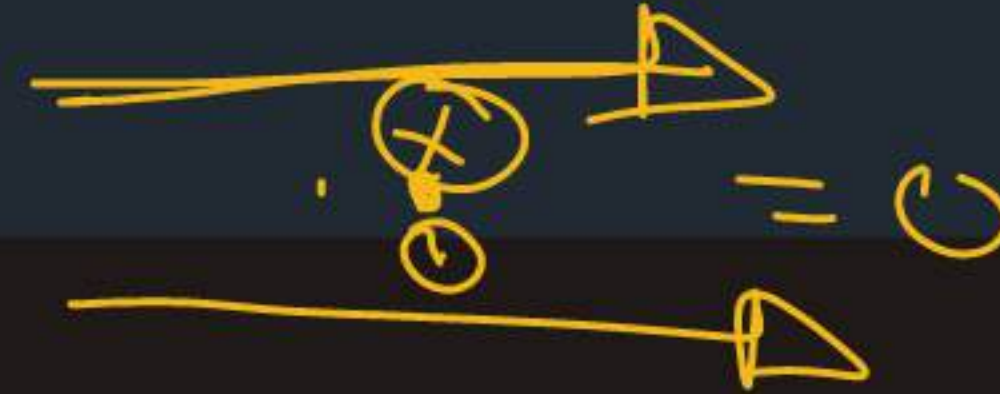
(ii) Name of the Rule: Right-Hand Thumb Rule (Maxwell's Rule)

If a current-carrying conductor is held in the right hand such that the thumb points in the direction of current, then the curl of the fingers around the conductor gives the direction of the magnetic field lines.



2015, 2017, 2019, 2020 Sample, 2022 Sample, and 2024 Sample Paper

Q. Two long, straight, parallel conductors carry equal currents in the same direction. Explain with reason what will be the magnetic field at a point midway between them

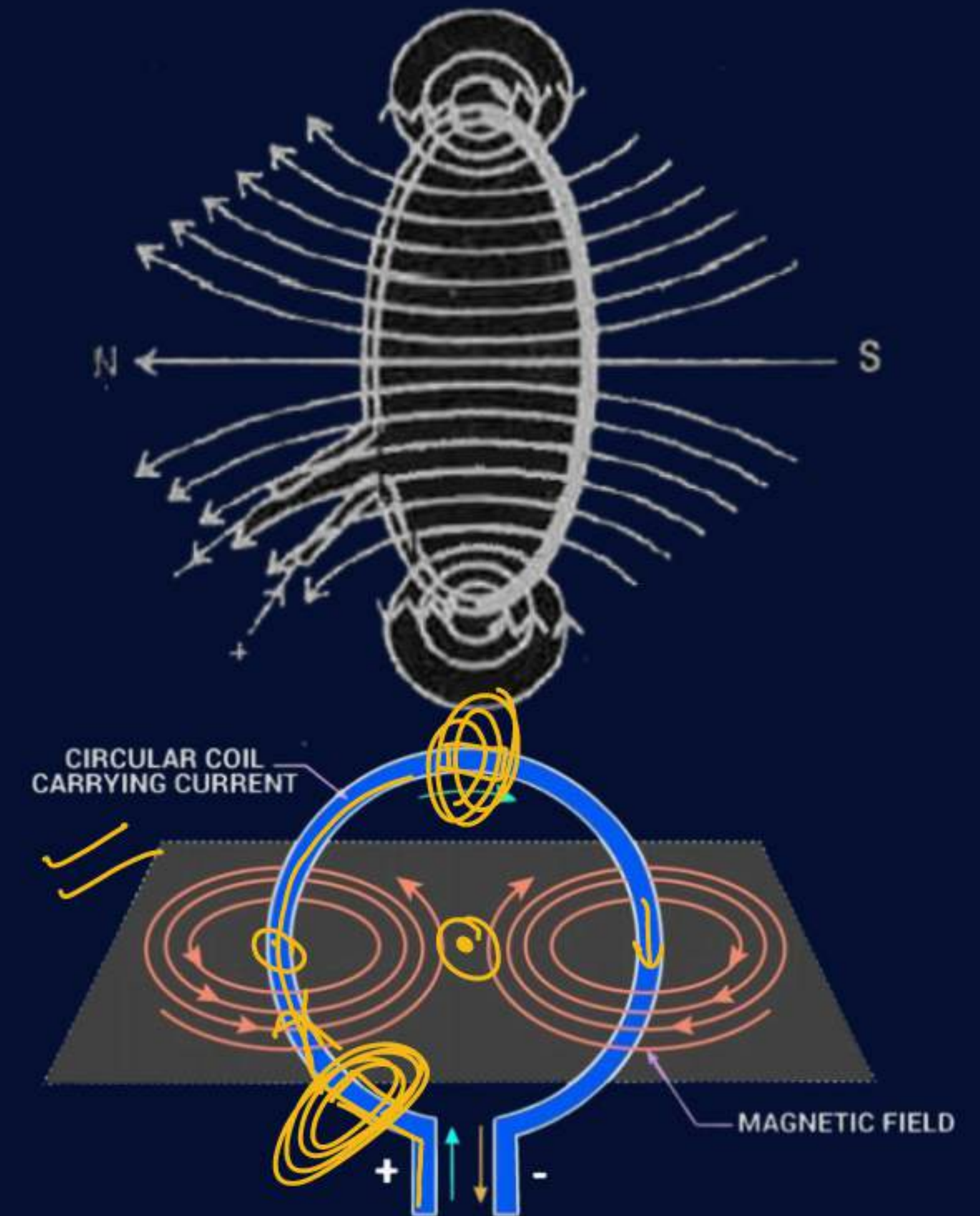


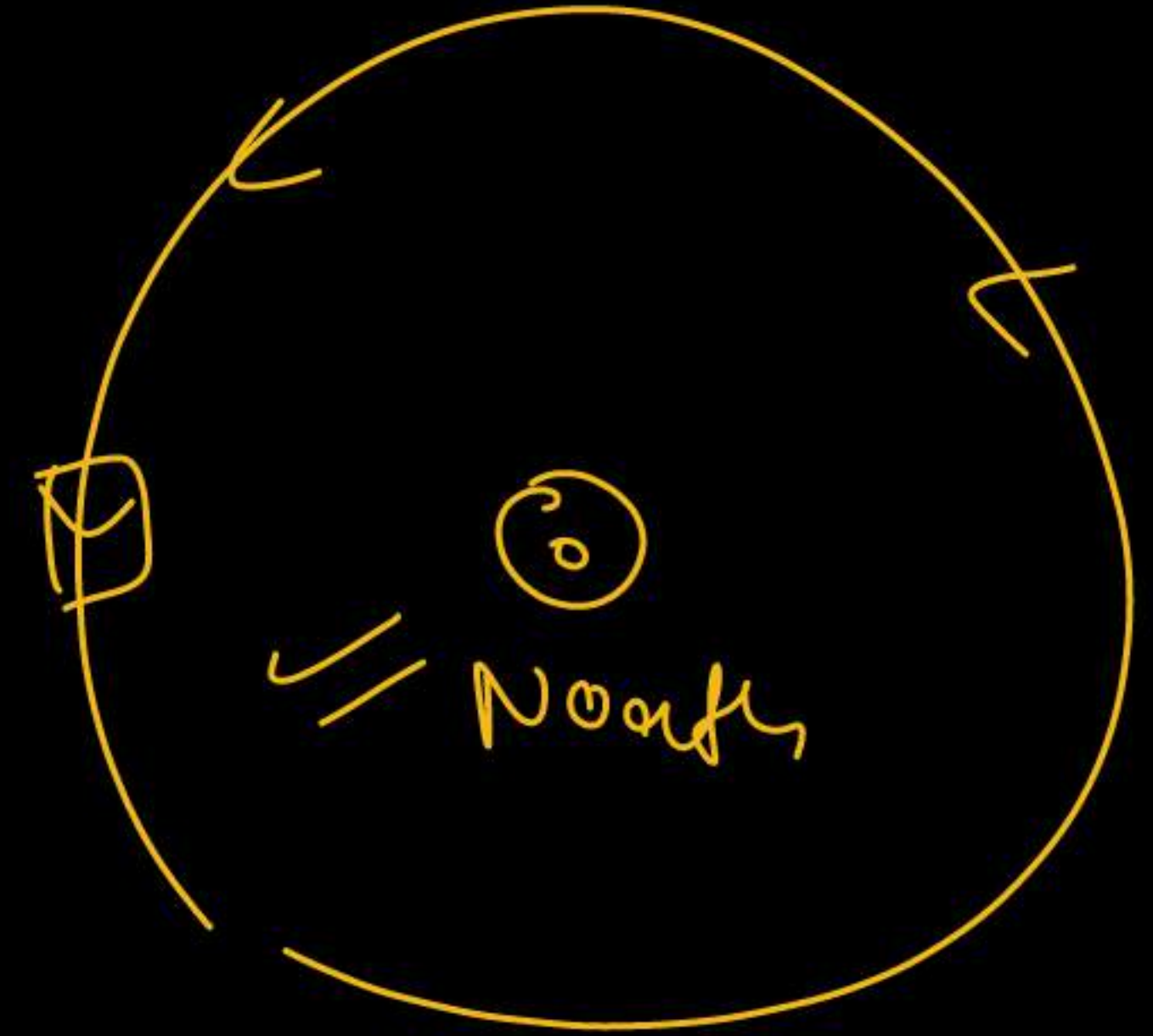
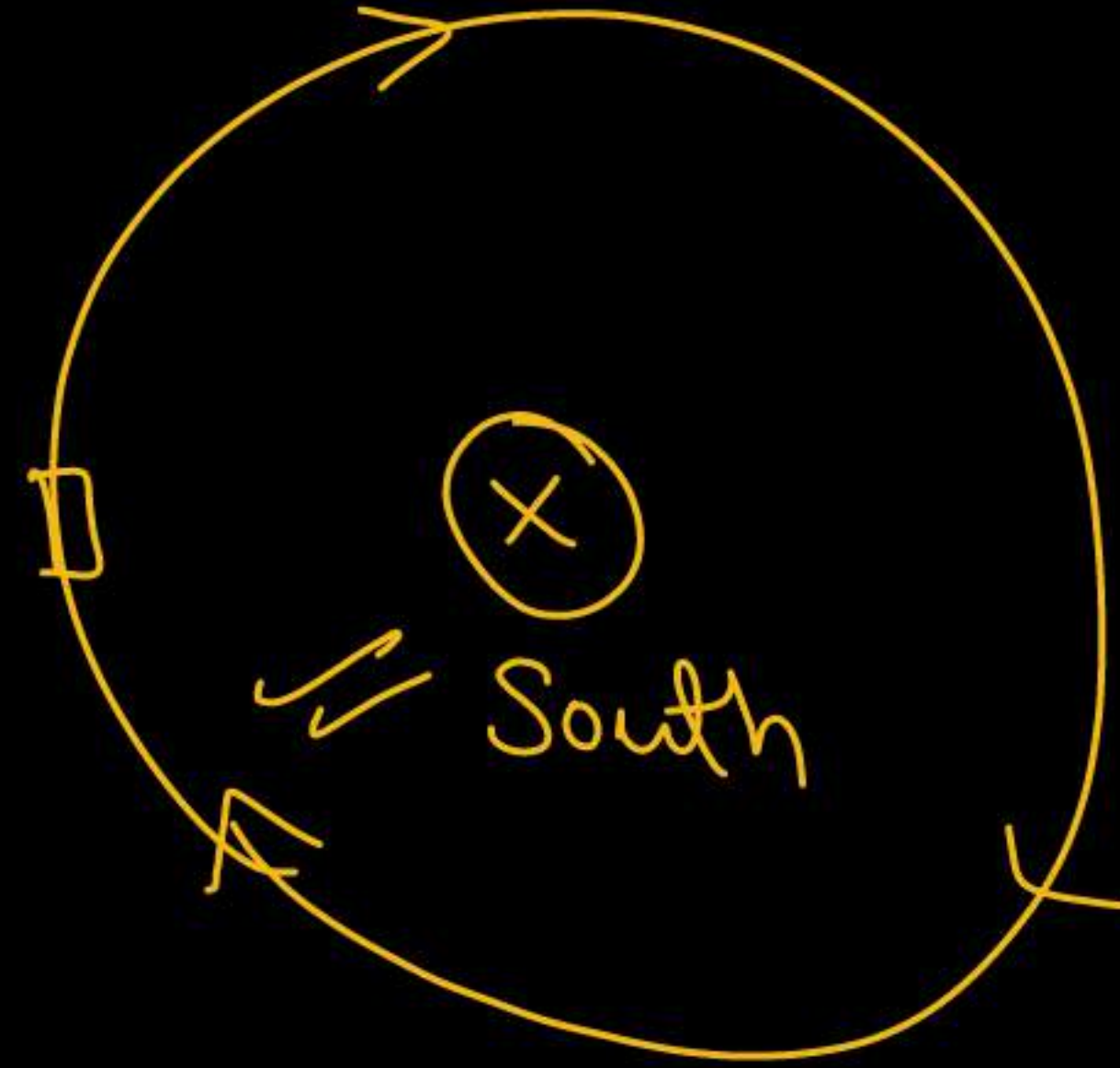
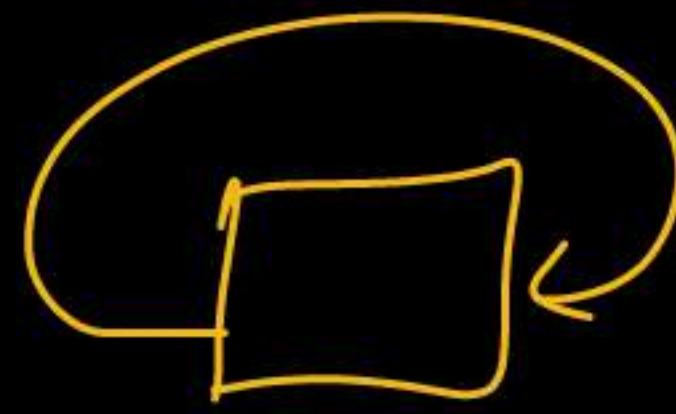
✓ When the currents flow in the same direction, the magnetic field lines due to both wires at the midpoint are in opposite directions.

If the currents are equal in magnitude, the two fields cancel each other, resulting in zero net magnetic field at the midpoint.

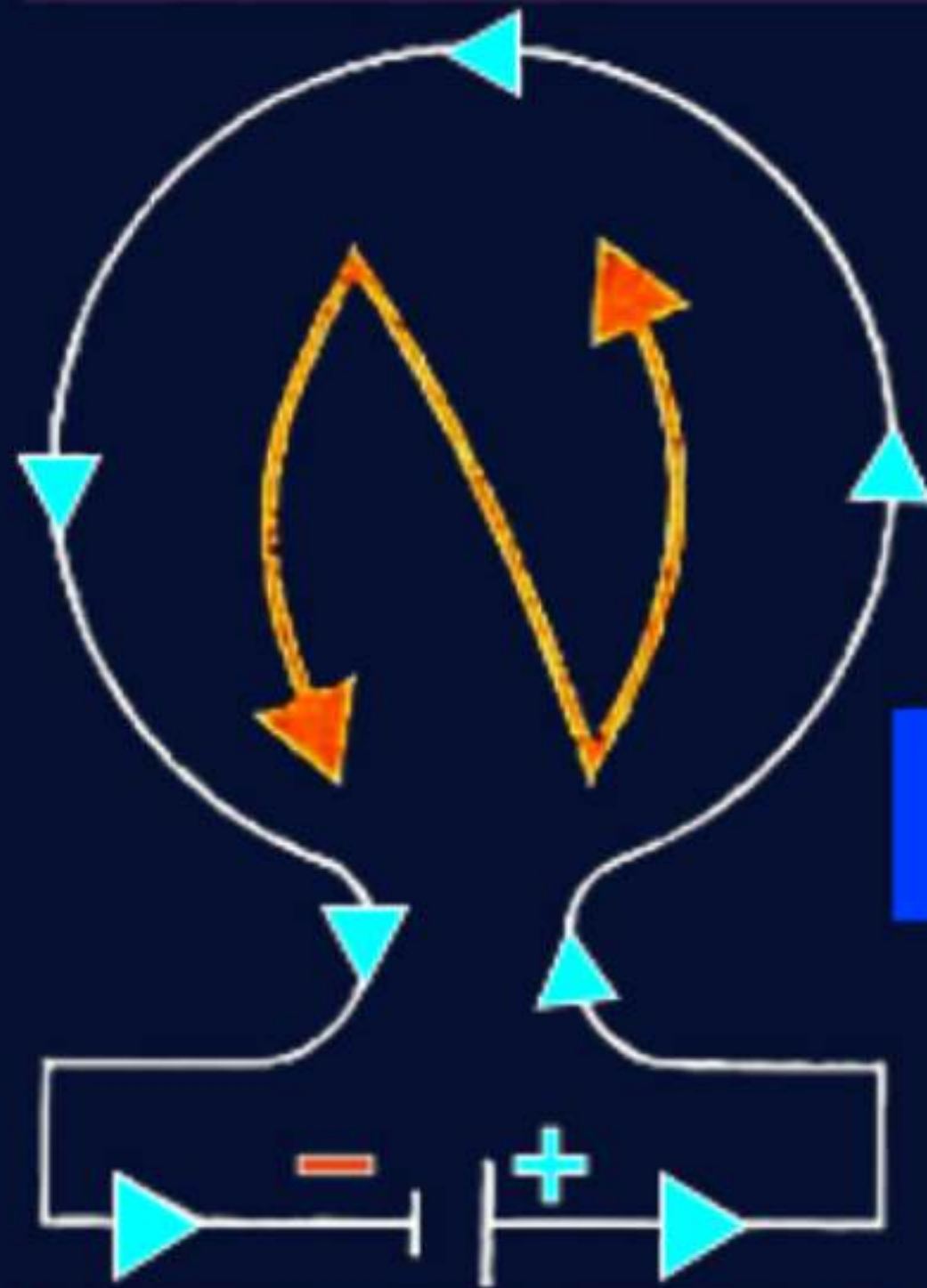
Magnetic Field due to a Current through a Circular Loop

- Each small portion of the loop behaves like a tiny straight conductor, producing concentric circles of magnetic field lines around it.
- These circles become larger and larger as we move away from the wire.
- At the centre of the loop, the arcs of these circles combine and appear as straight lines. Hence, the magnetic field at the centre is strong and nearly uniform.
- By applying the Right-Hand Thumb Rule, we find that each section of the loop produces a magnetic field in the same direction inside the loop.





Anti-clockwise Current



North Magnetic Pole

The direction of current in this face of circular wire is *Anti-Clockwise*, so this face of circular wire carrying Current will act as North magnetic pole (*N-pole*)

Clockwise Current



South Magnetic Pole

The direction of current in this face of circular wire is *Clockwise*, so this face of circular wire carrying Current will act as South magnetic pole (*S-pole*)

Clock - Face Rule

2025

Q. Draw the pattern of magnetic field lines of the magnetic field produced by a current-carrying circular loop. Mark on it the direction of (i) current and (ii) magnetic field lines. Name the two factors on which the magnitude of the magnetic field due to a current-carrying coil depends.



(ii) Factors Affecting Magnetic Field Strength of a Coil

1. **Magnitude of Current (I)**: The magnetic field increases as the current through the coil increases.

2. **Number of Turns (N)**: The magnetic field increases with the number of turns of the coil because the fields due to each turn add up.

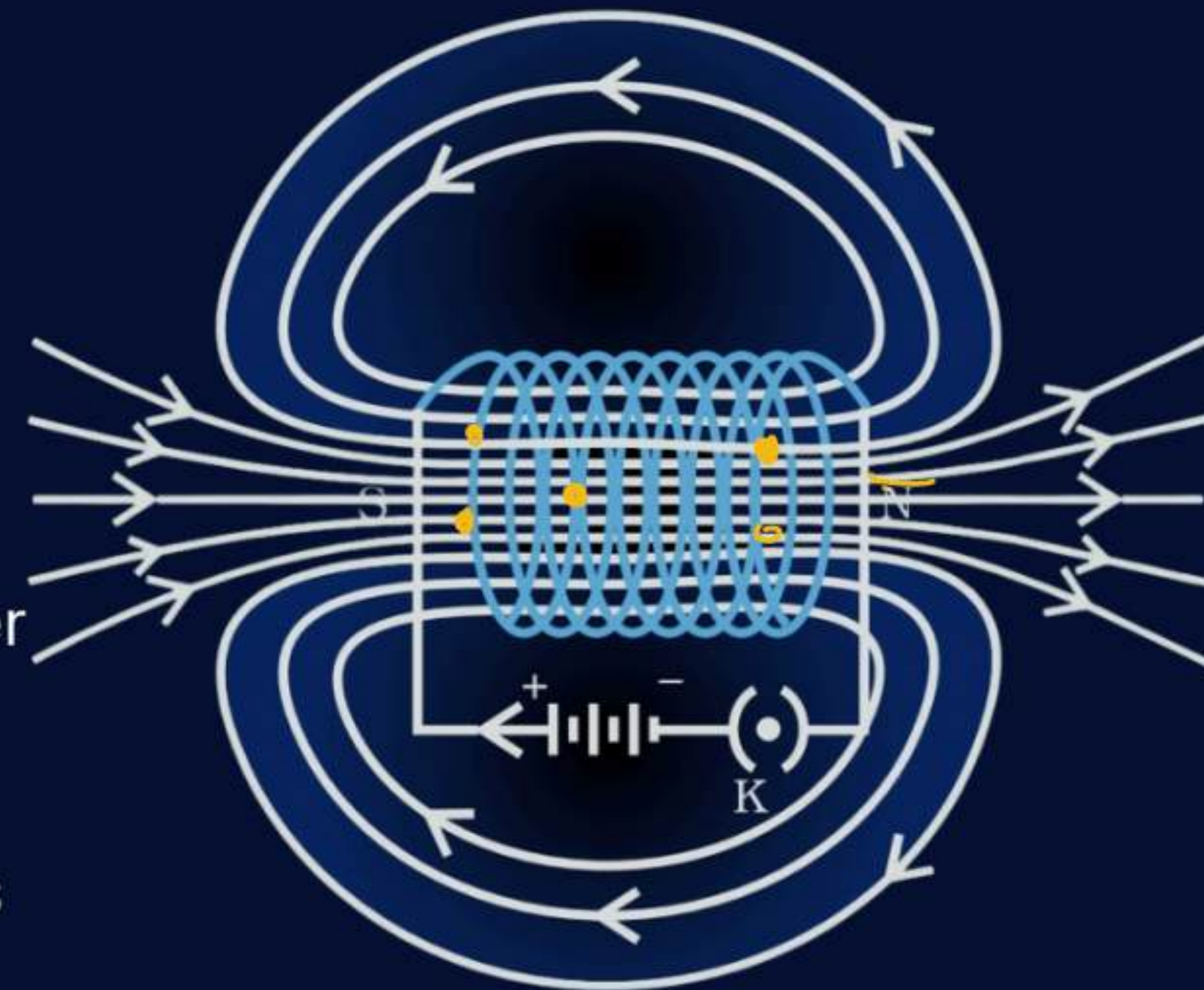
Magnetic Field due to a SOLENOID

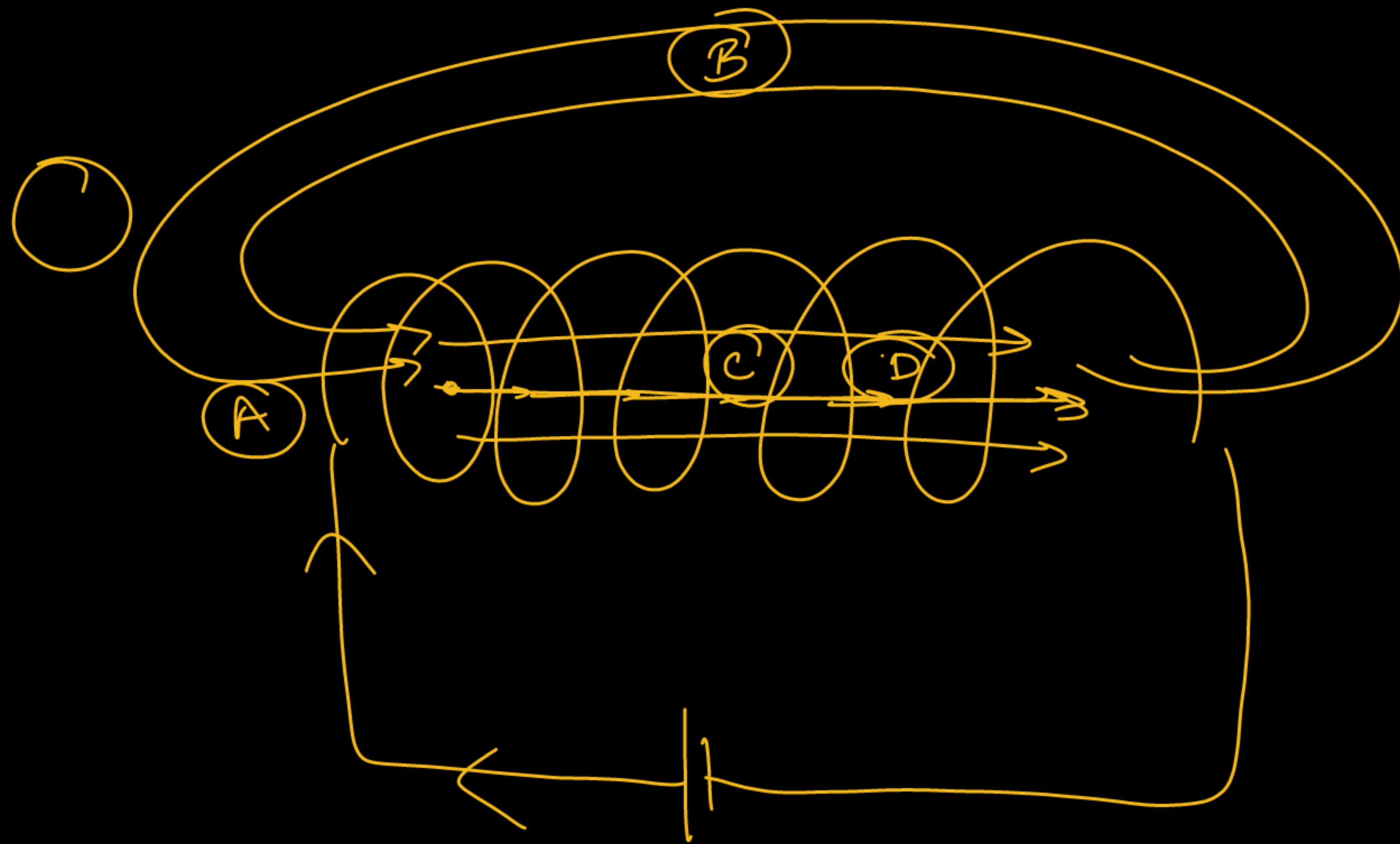
What is a solenoid? Draw the pattern of magnetic field lines around a current-carrying solenoid.

Solenoid: A solenoid is a coil of many circular turns of insulated copper wire wound closely in the shape of a cylinder.

Magnetic Field Pattern:

- The pattern of magnetic field lines around a current-carrying solenoid is similar to that of a bar magnet.
- One end of the solenoid acts as a **North Pole**, and the other end acts as a **South Pole**.
- Inside the solenoid, the magnetic field lines are **straight, parallel, and close to each other**, showing that the field is strong and uniform i.e. uniform magnetic field.
- Outside the solenoid, the field lines spread out, similar to a bar magnet's field pattern.

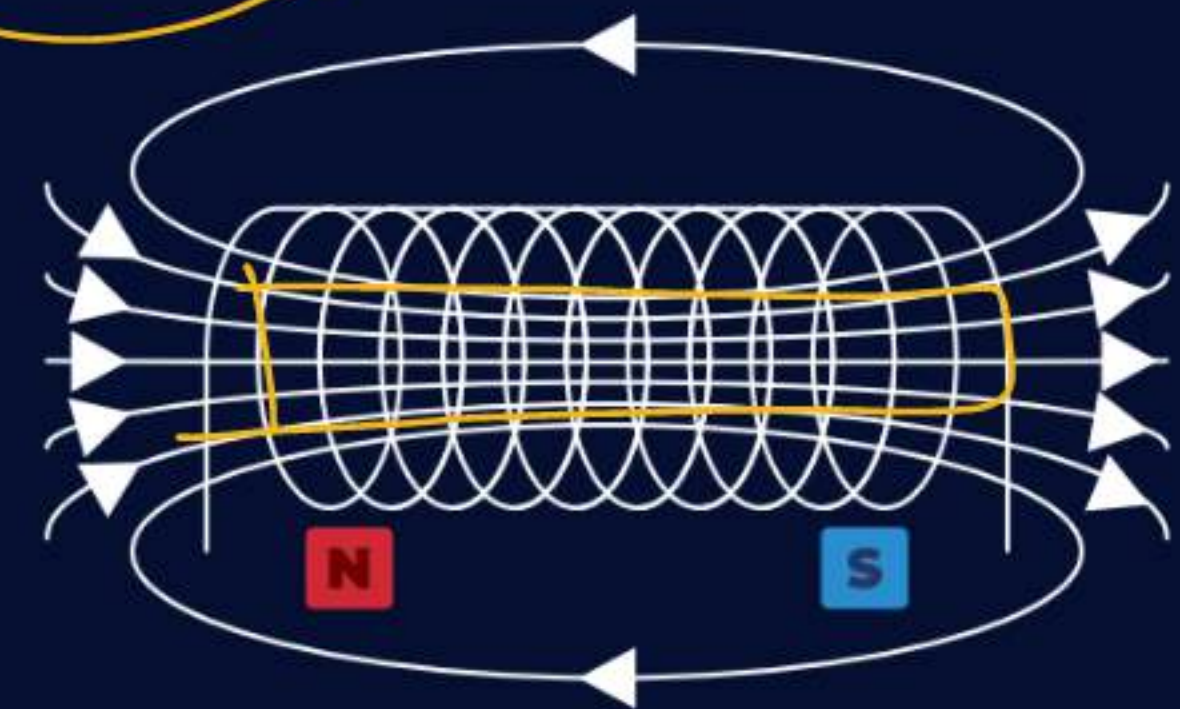




Factors on which Magnetic Field of Solenoid depends:

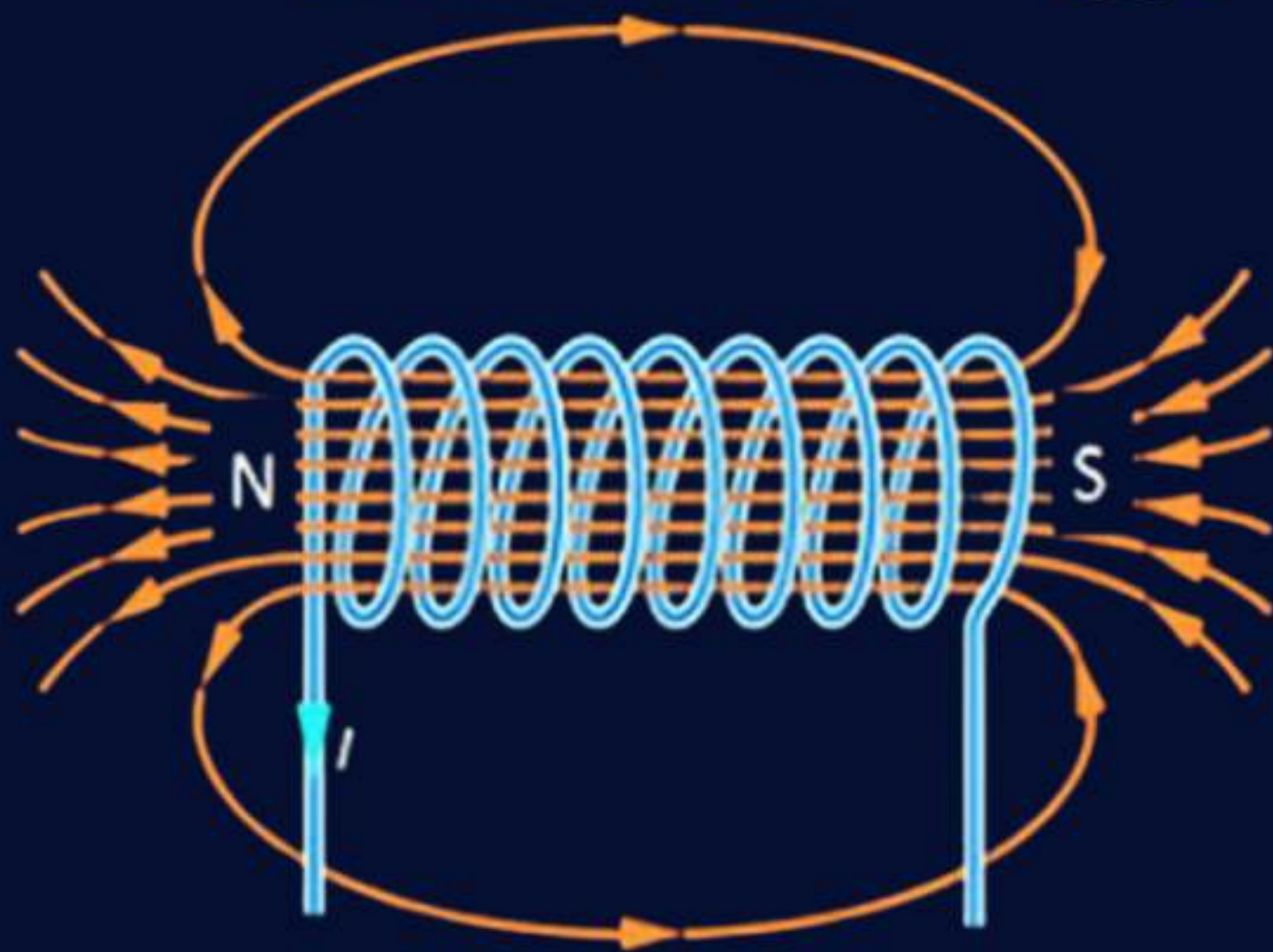
- Number of turns in the solenoid – *per unit length* more turns → stronger field.
- Current through the solenoid – higher current → stronger field.
- Nature of the core material inside the solenoid – soft iron core increases the strength of the field.

**When Current is passed through a Solenoid,
it behaves like a bar magnet!!**

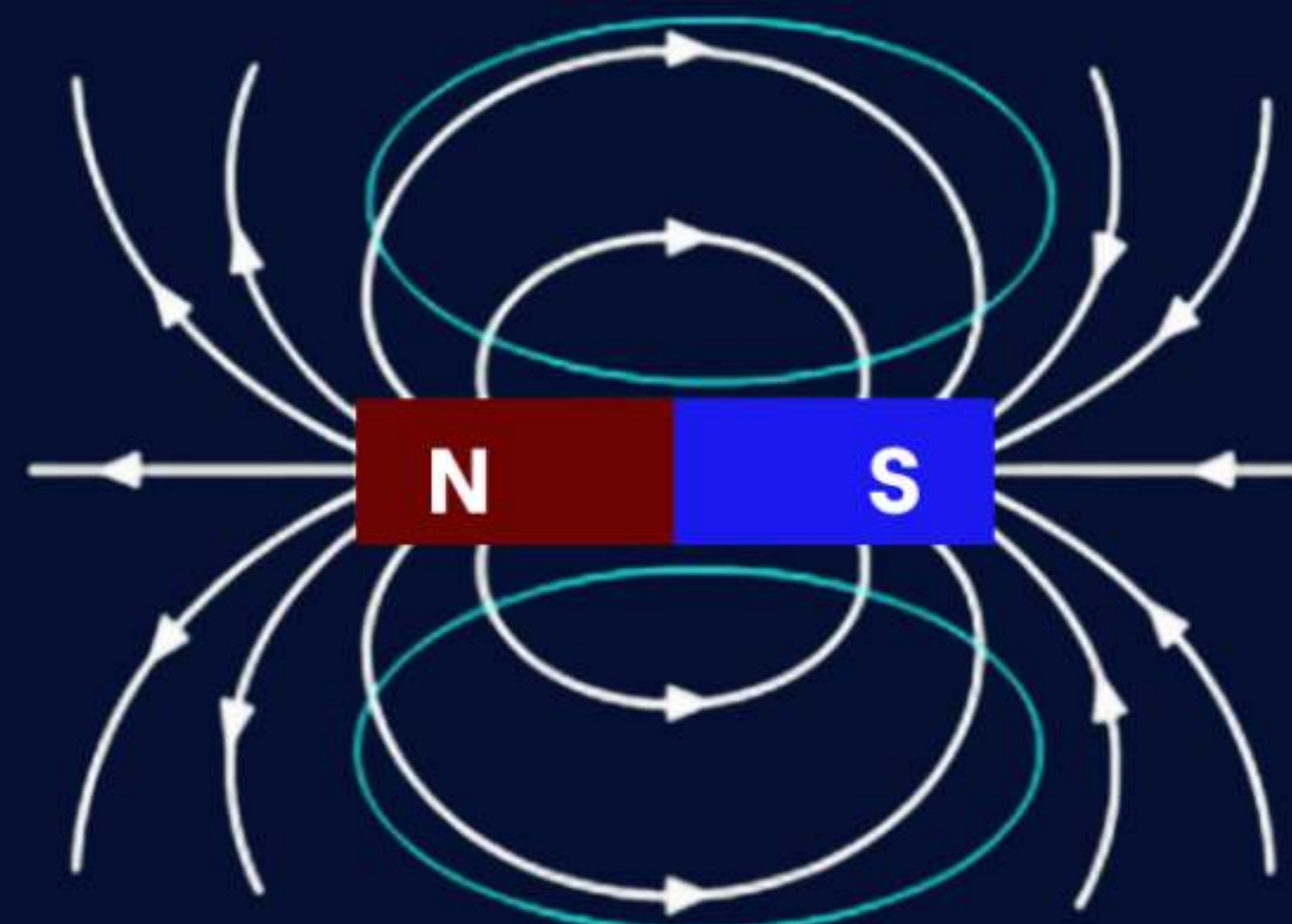


Why a Solenoid Acts as a Bar Magnet

- A current-carrying solenoid produces a **magnetic field pattern similar** to that of a bar magnet.
- One end of the solenoid behaves as a **north pole** and the other end as a **south pole**.
- The magnetic field lines inside the solenoid are nearly **parallel and uniform**, showing that the field is strong and similar to a bar magnet.
- The **polarity of the solenoid (north or south)** depends on the direction of current through it.



Solenoid



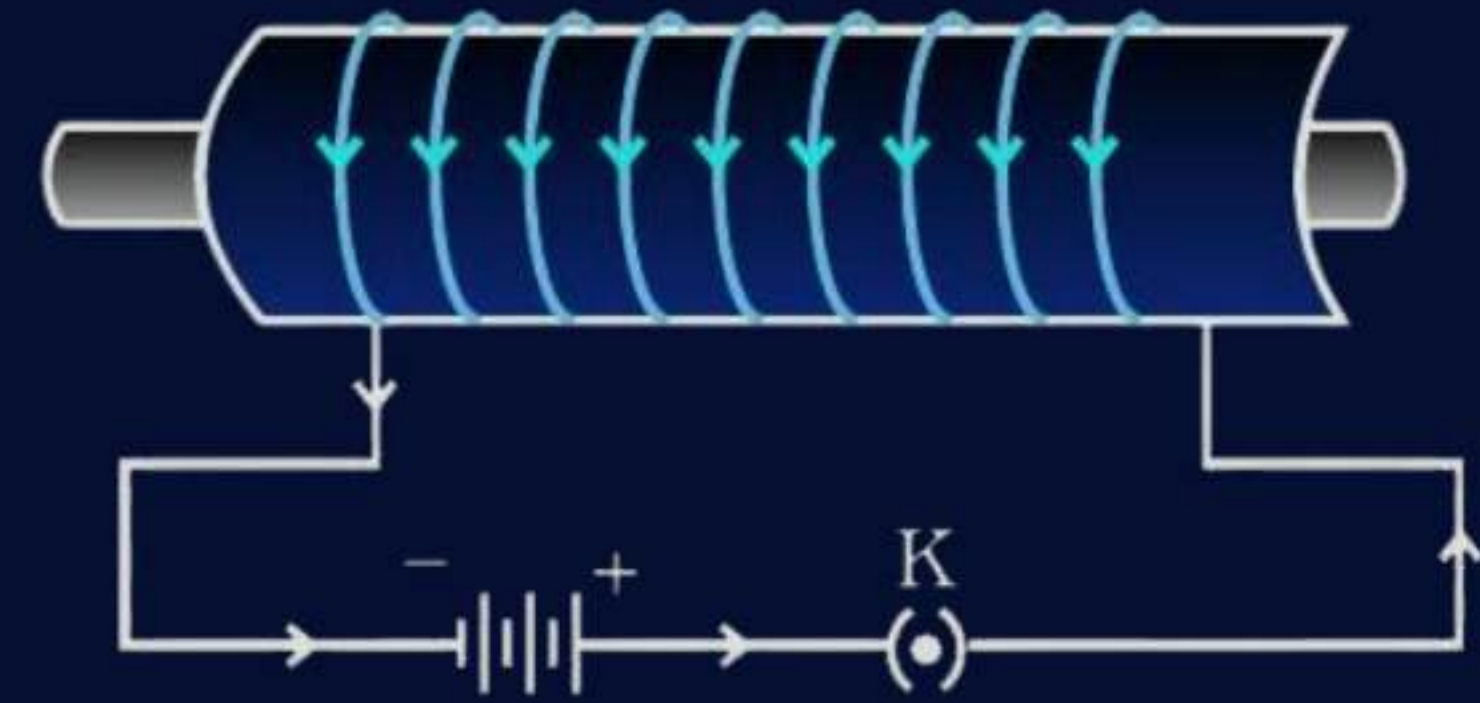
Magnet

North
South

Electromagnet:

An electromagnet is a magnet formed when a soft iron core is placed inside a current-carrying solenoid.

The strong magnetic field produced by the solenoid magnetises the iron core, making it behave like a temporary magnet.

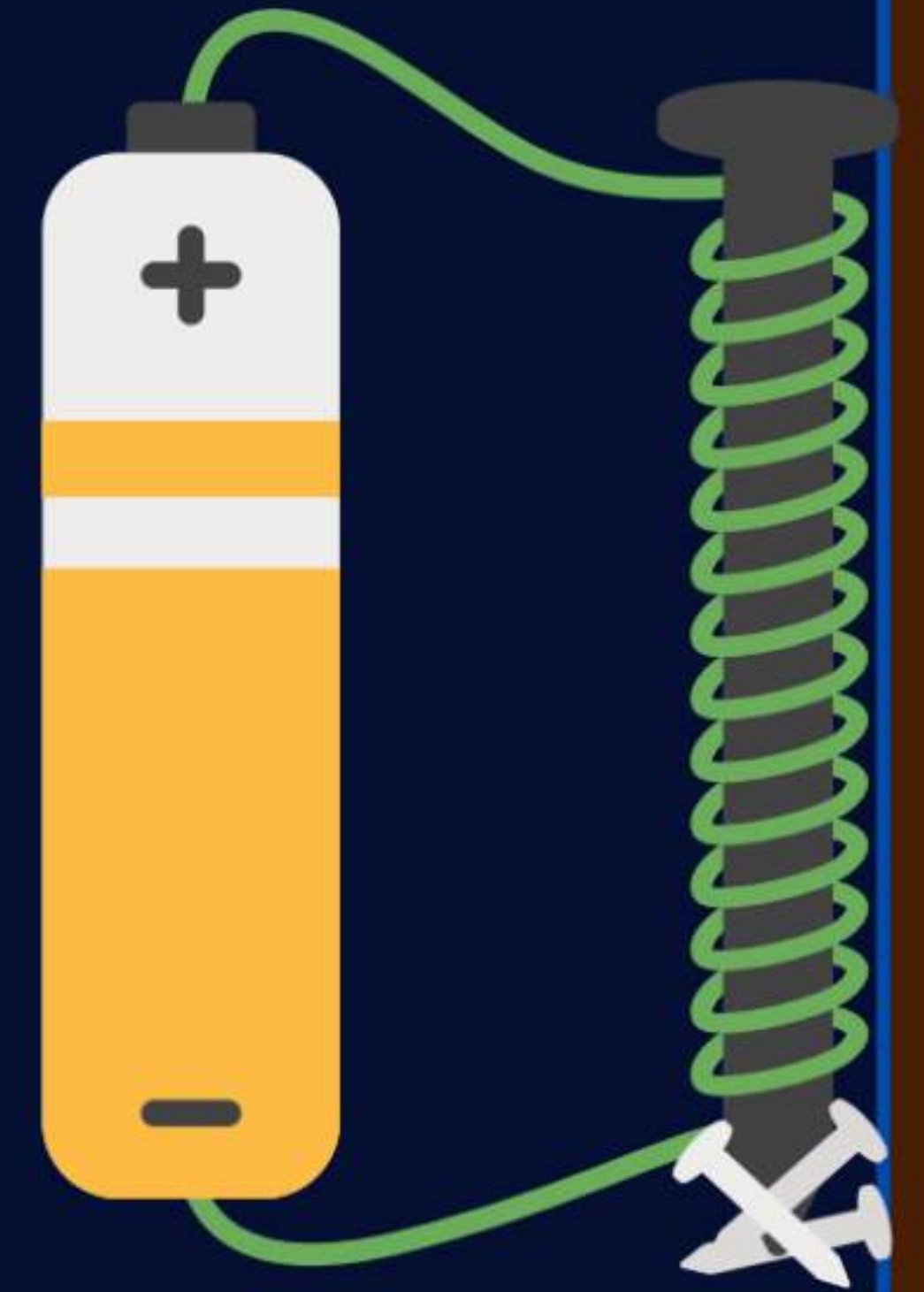


Uses of Solenoid and Electromagnet:

- Used in electric bells, cranes, motors, and magnetic locks.
- Helpful in creating controlled magnetic fields in laboratories.

Why an Electromagnet is Better? ✓

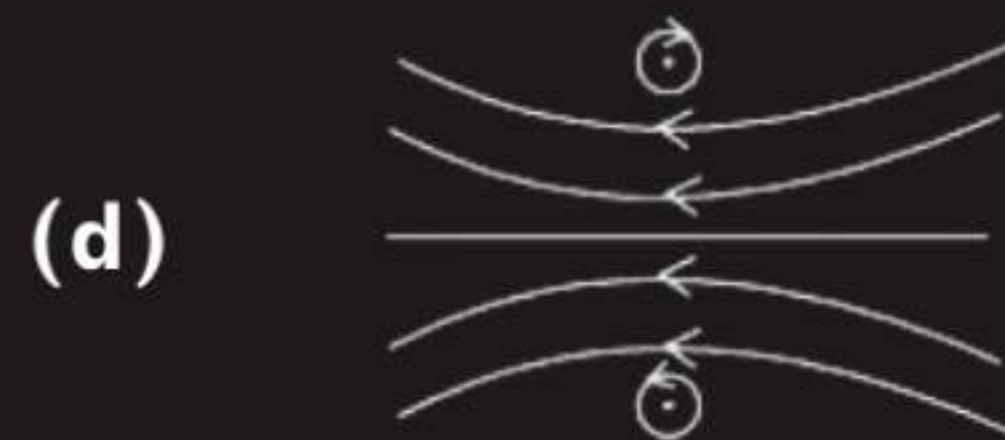
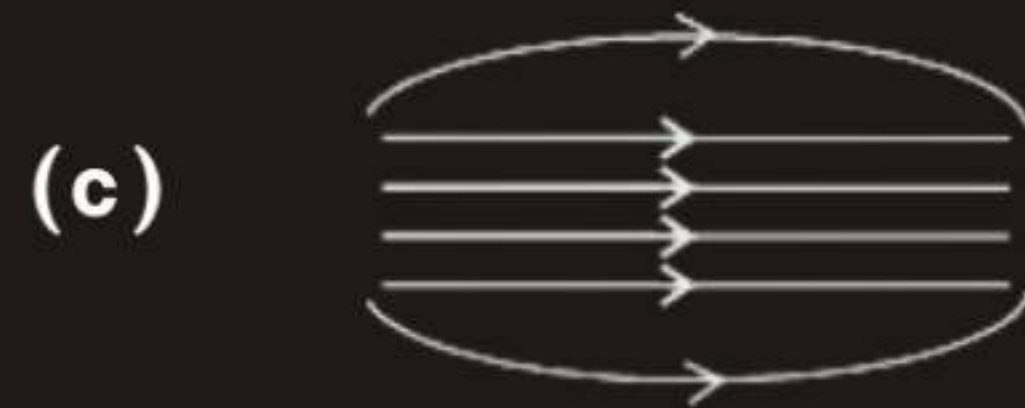
- **Stronger Magnetic Field:** An electromagnet produces a much stronger magnetic field than a permanent magnet when current flows through it.
- **Magnetism Can Be Controlled:** The strength of the magnetic field can be increased or decreased by changing the amount of current or the number of turns in the coil.
- **Temporary Magnetism:** It behaves as a magnet only when current flows, so it can be switched on or off as needed.
- **Polarity Can Be Reversed:** The north and south poles of an electromagnet can be reversed by reversing the direction of current.
- **Useful for Many Applications:** Used in electric bells, cranes, electric motors, relays, and speakers, where controlled magnetism is required.



Electromagnet	Permanent Magnet
<u>Produced only when electric current flows through the coil.</u>	Exists permanently, without electricity.
Can be increased or decreased by changing current.	Fixed strength, cannot be changed.
<u>Temporary</u> – vanishes when current is switched off.	Permanent – magnetic field is always present.
<u>Polarity can be reversed</u> by changing the direction of current.	✗ Polarity is fixed, cannot be changed.
Ex: Electric bell, electric crane, MRI machine.	Ex: Compass needle, refrigerator magnets, speakers.

2024

Q. The pattern of the magnetic field produced inside solenoid is:



2025

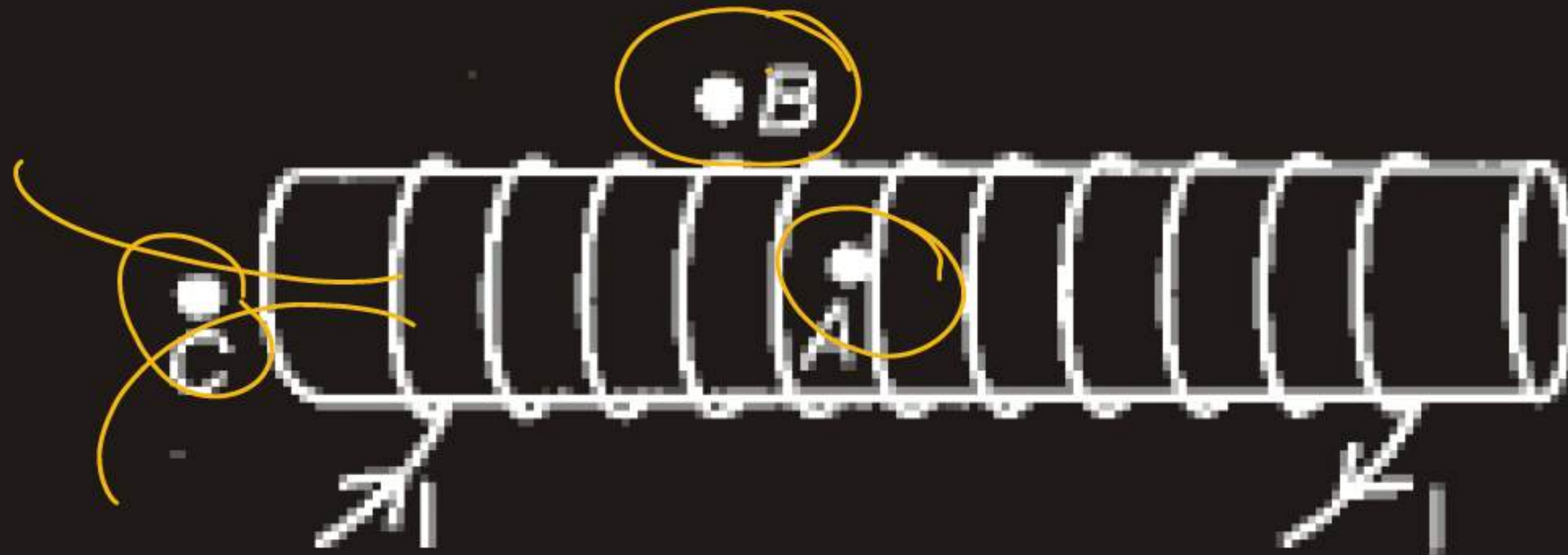
Q. The strength of the magnetic field produced inside a long straight current-carrying solenoid does not depend upon:

- (A) Number of turns in the solenoid**
- (B) Direction of current flowing through the solenoid**
- (C) Material of the core filled inside the solenoid**
- (D) Magnitude of current in the solenoid**



2015

Q. For the current carrying solenoid as shown below, draw magnetic field lines and giving reason explain that out of the three points A, B and C at which point the field strength is maximum and at which point it is minimum.



2013, 2015, 2017, 2018

Q. Can a freely suspended current-carrying solenoid stay in any direction? Justify your answer. What will happen when the direction of current in the solenoid is reversed? Explain.

- A freely suspended current-carrying solenoid cannot stay in any direction; it aligns along the north-south direction like a bar magnet.
- This happens because a current-carrying solenoid behaves as a bar magnet having north and south poles.

When the direction of current is reversed, the poles of the solenoid interchange – the north pole becomes south and south becomes north.

2017, 2020, 2022 Sample, 2025

Q. How does a current-carrying solenoid behave like a bar magnet? State the factors on which its magnetic field depends

A current-carrying solenoid behaves like a bar magnet because it produces a similar magnetic field pattern –

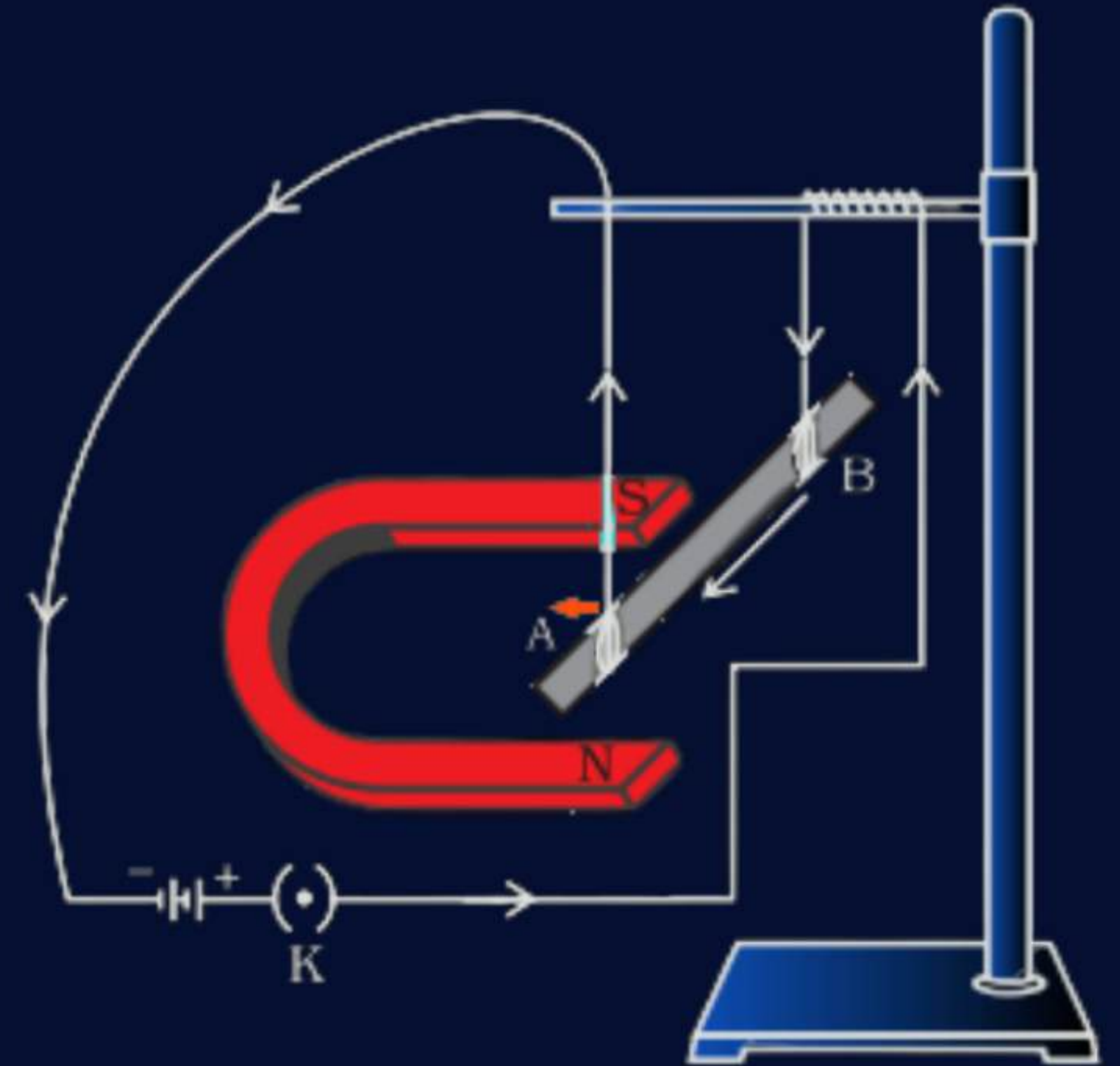
- One end acts as the north pole and the other as the south pole.
- The magnetic field lines inside the solenoid are nearly parallel and uniform, like those inside a bar magnet.

Factors on which the Magnetic Field of a Solenoid Depends:

- Number of turns per unit length – More turns produce a stronger magnetic field.
- Magnitude of current – Higher current gives a stronger field.
- Nature of core material – Inserting a soft iron core increases the strength of the magnetic field.

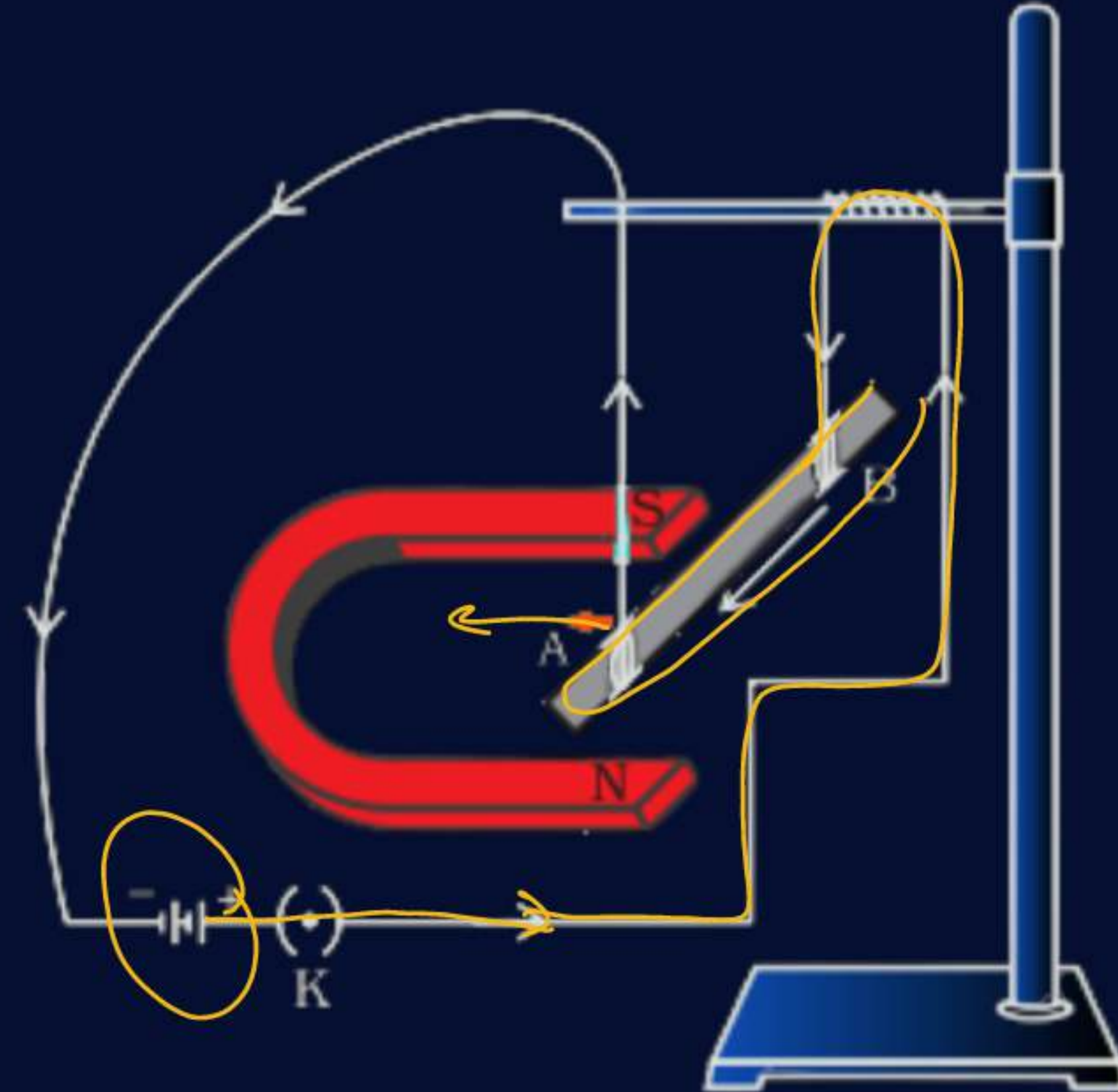
Force on a Current-Carrying Conductor in a Magnetic Field

- An electric current flowing through a conductor produces a magnetic field.
- This magnetic field exerts a force on a magnet placed nearby.
- According to the French scientist André-Marie Ampère (1775–1836), → **the magnet also exerts an equal and opposite force on the current-carrying conductor.**



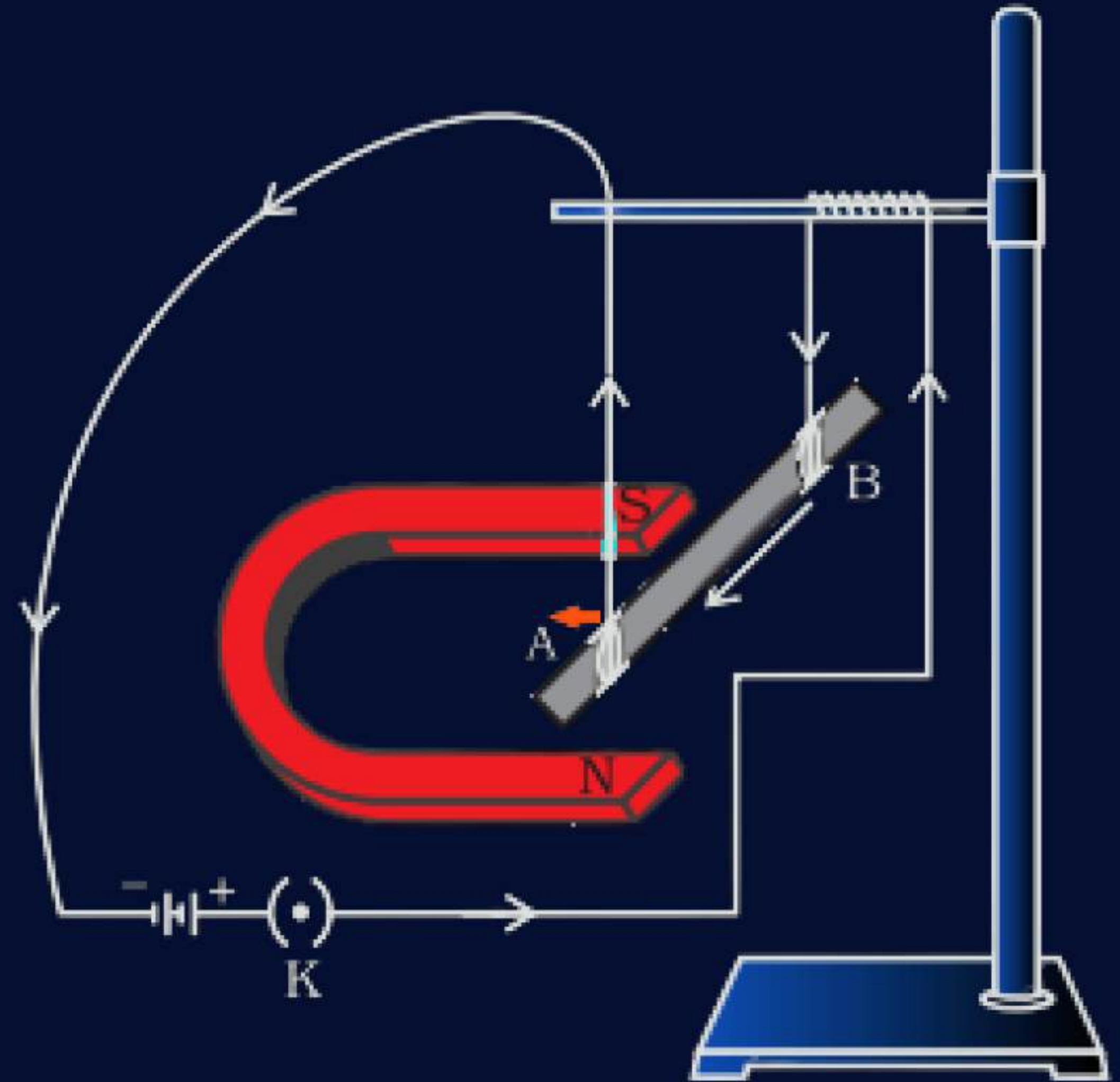
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Activity 12.7

- Take a small aluminium rod AB (of about 5 cm). Using two connecting wires suspend it horizontally from a stand, as shown in Fig. 12.12.
 - Place a strong horse-shoe magnet in such a way that the rod lies between the two poles with the magnetic field directed upwards. For this put the north pole of the magnet vertically below and south pole vertically above the aluminium rod (Fig. 12.12).
 - Connect the aluminium rod in series with a battery, a key and a rheostat.
 - Now pass a current through the aluminium rod from end B to end A.
 - What do you observe? It is observed that the rod is displaced towards the left. You will notice that the rod gets displaced.
 - Reverse the direction of current flowing through the rod and observe the direction of its displacement. It is now towards the right.
- Why does the rod get displaced?



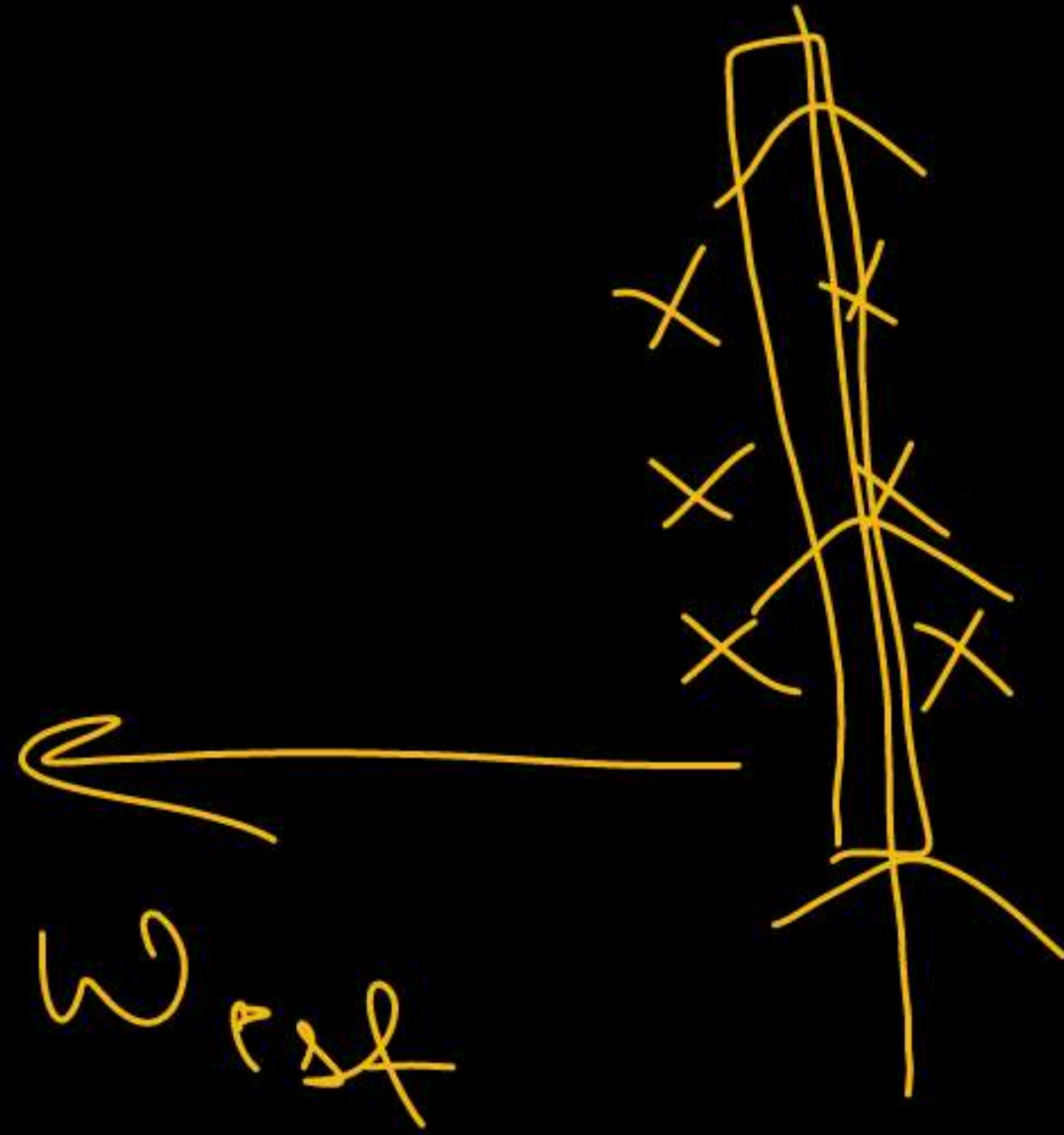
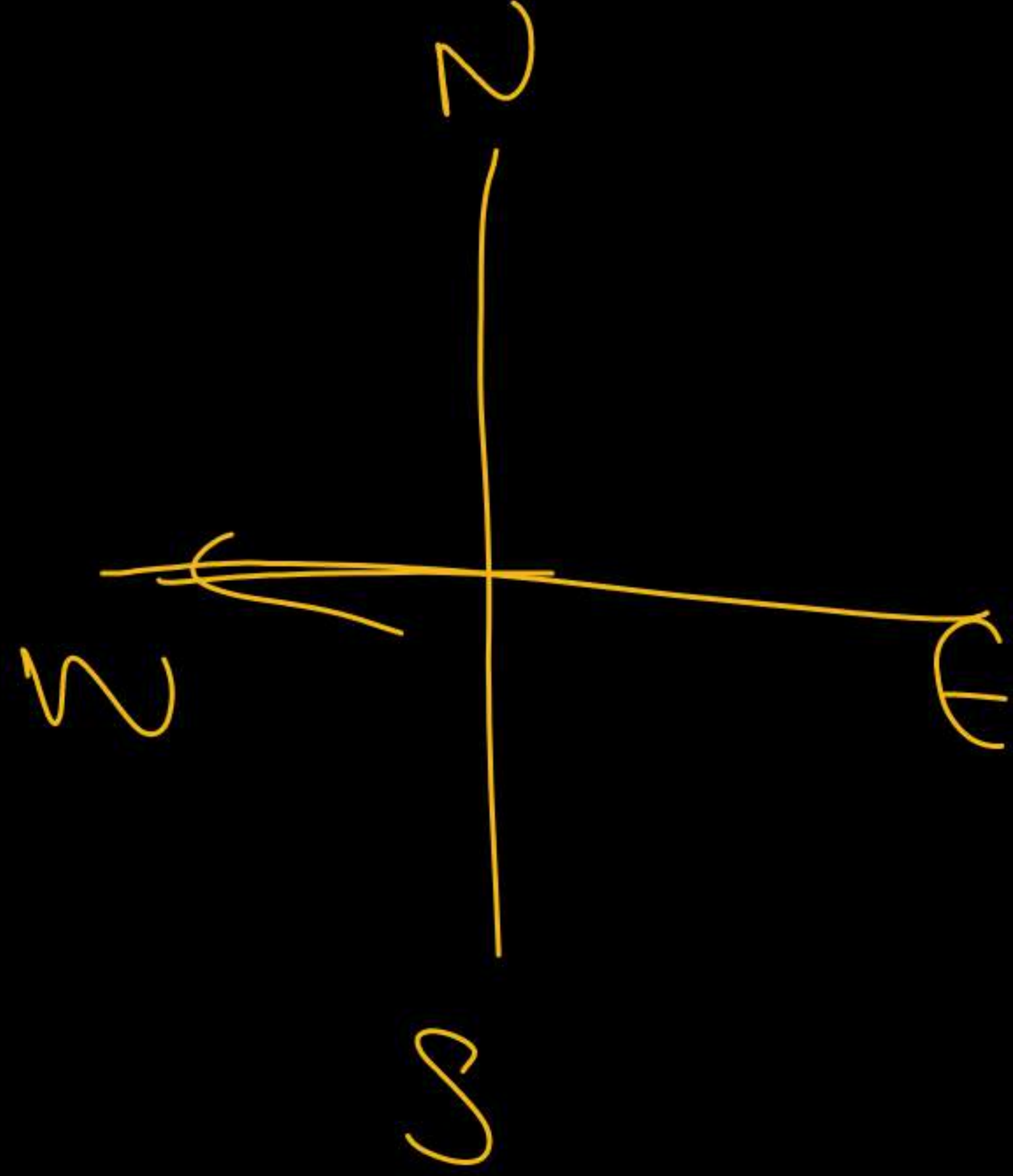
FLEMING'S LEFT HAND RULE

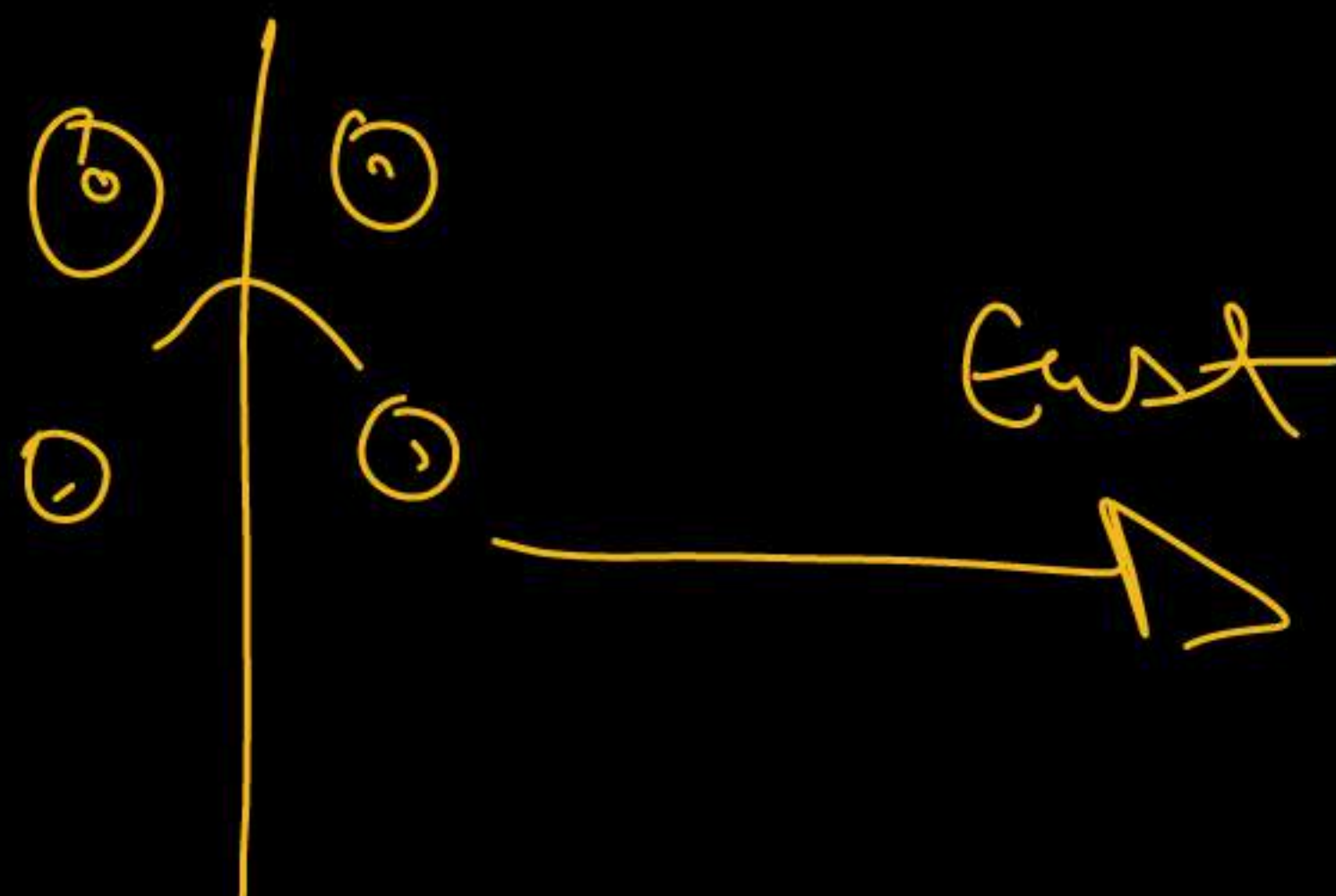
State Fleming's Left-Hand Rule. Name one device that works on this principle.

If the forefinger, middle finger, and thumb of the left hand are stretched mutually perpendicular to each other, and the **forefinger** points in the **direction of magnetic field**, the **middle finger** points in the **direction of current**, then the **thumb** will point in the **direction of motion (force) of the conductor**.

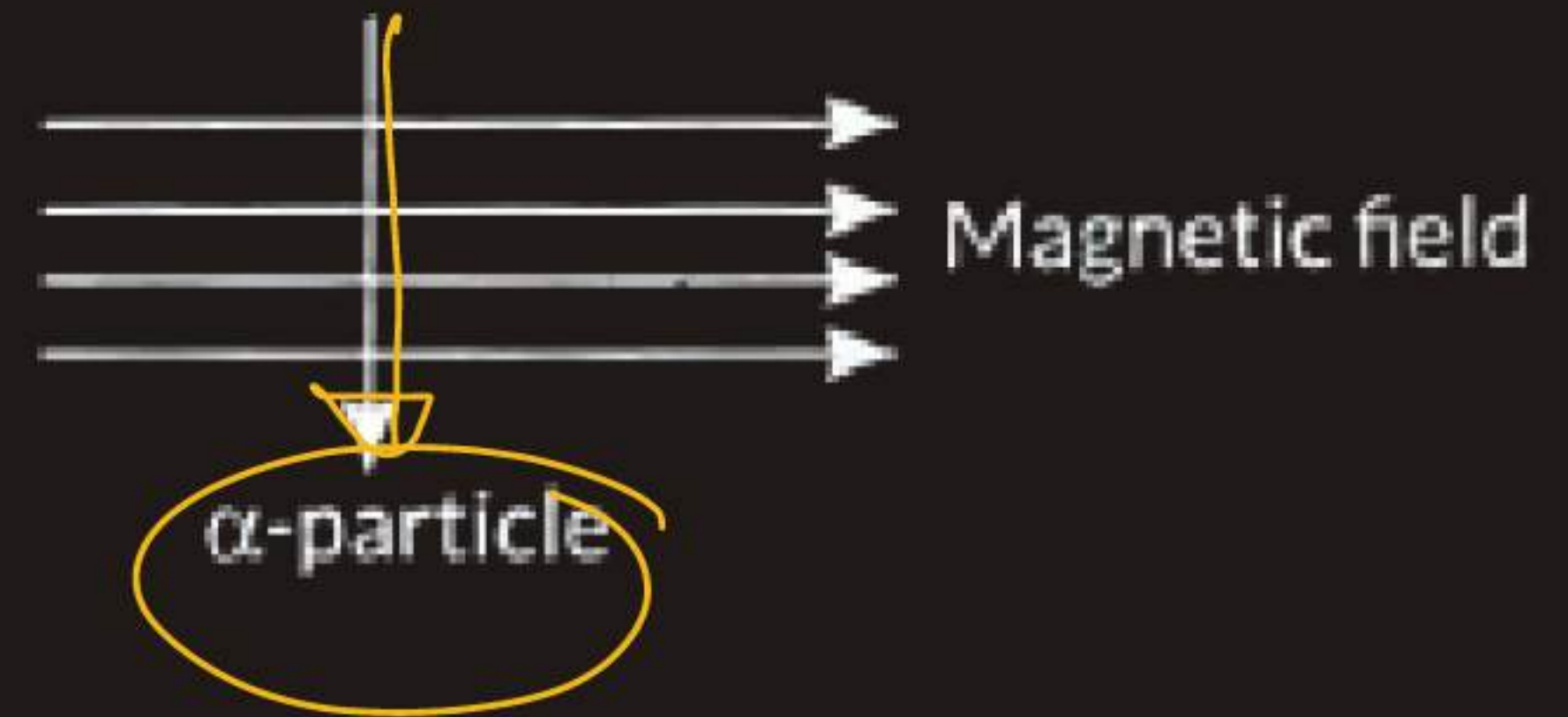
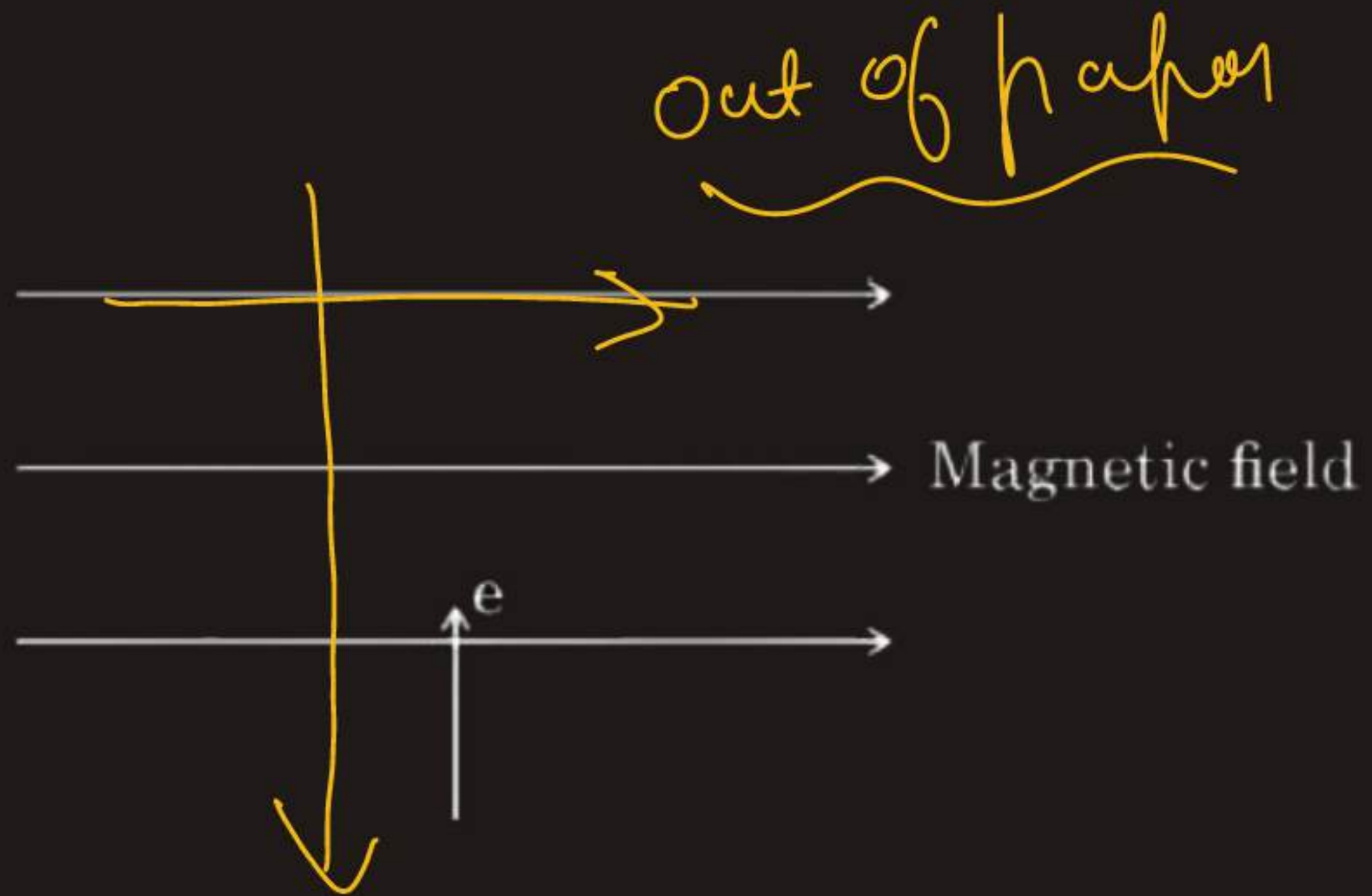


Fleming's left-hand rule





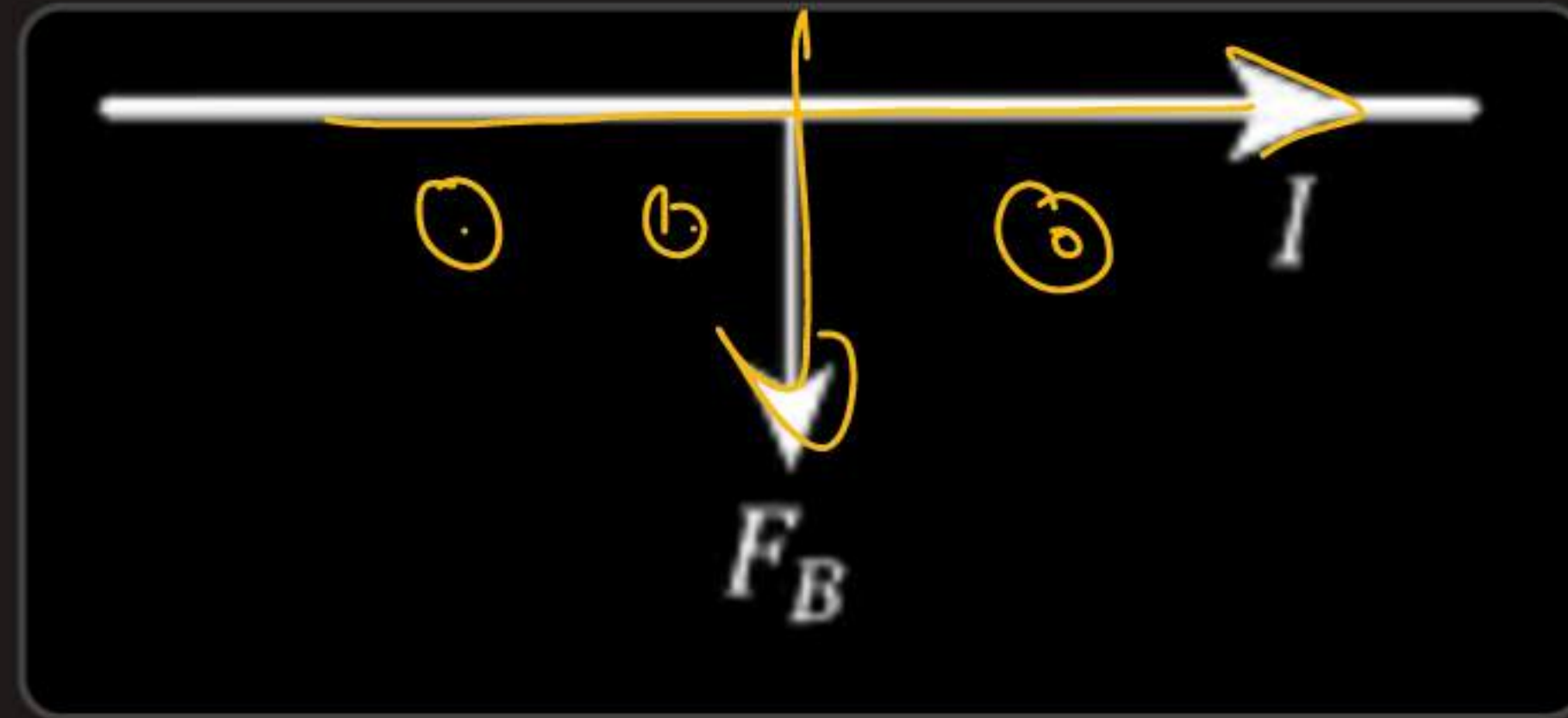
Q. Determine the direction of force in below cases:





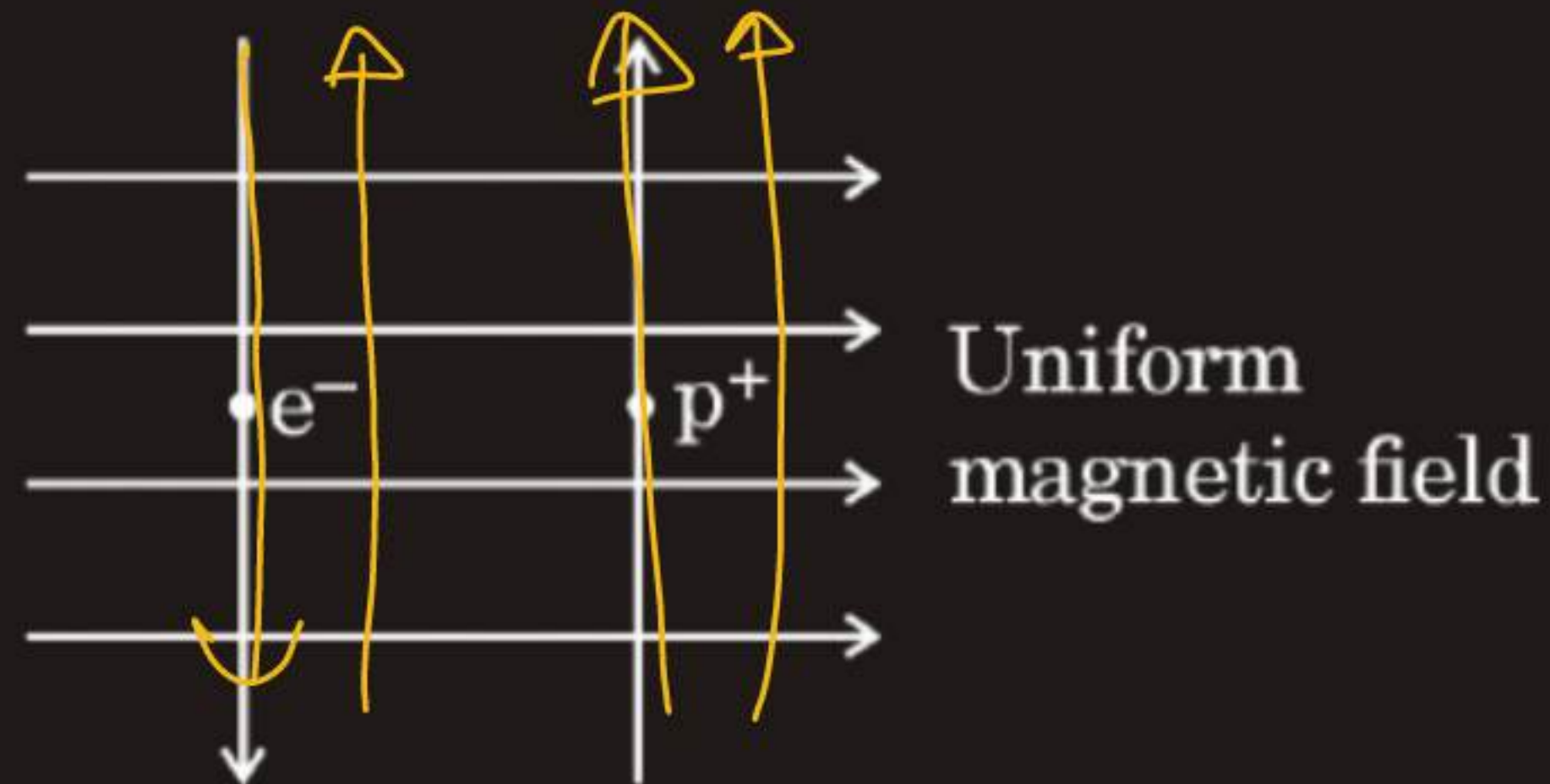
2015, 2017

Q. Determine the direction of the uniform magnetic field according to the given current and magnetic force..



2024: Q. A uniform magnetic field exists in the plane of the paper as shown in the diagram. In this field, an electron (e^-) and a positron (p^+) enter as shown. The electron and positron experience forces:

- ~~(a)~~ both pointing into the plane of the paper.
- (b) both pointing out of the plane of the paper.
- (c) pointing into the plane of the paper and out of the plane of the paper respectively.
- (d) pointing out of the plane of the paper and into the plane of the paper respectively.



2015, 2016, 2017, 2019

Q. A uniform magnetic field is directed vertically upwards.

In which direction in this field should an α -particle (which is a positively charged particle) be projected so that it is deflected southward?

Name and state the rule you have used to find this direction.

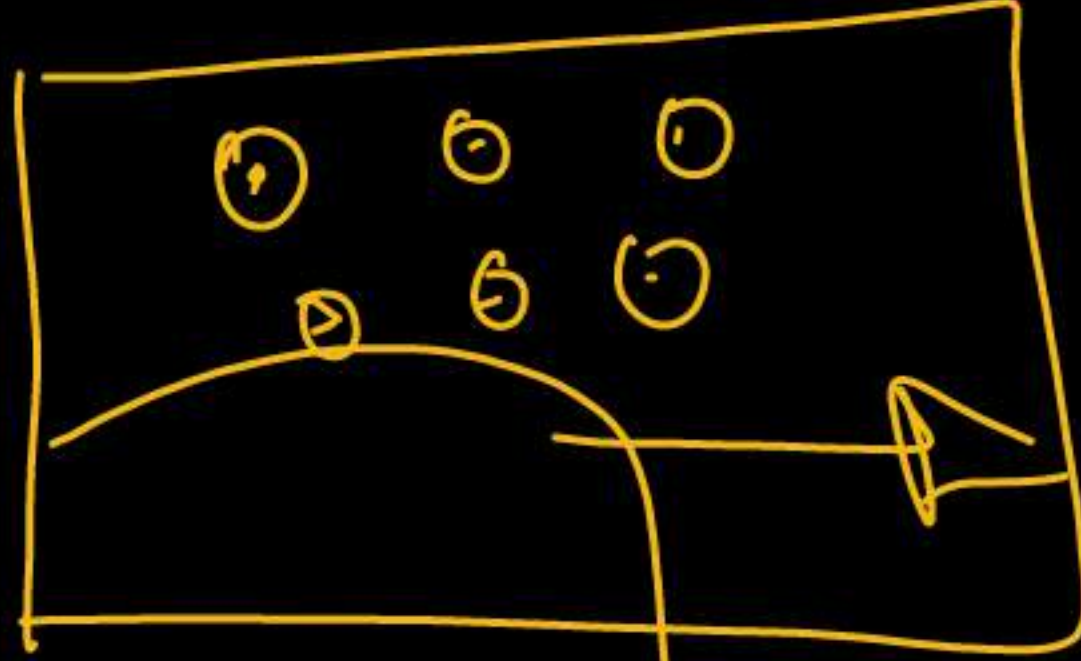
When a current-carrying conductor or a moving positive charge is placed in a magnetic field, it experiences a force perpendicular to both the direction of current (motion of positive charge) and the magnetic field.

- Name of the Rule: Fleming's Left-Hand Rule

If the forefinger, middle finger, and thumb of the left hand are stretched mutually perpendicular to each other:

- The forefinger shows the direction of magnetic field,
- The middle finger shows the direction of current (motion of positive charge), and
- The thumb shows the direction of force (motion/deflection of the particle).

N



Cash

S

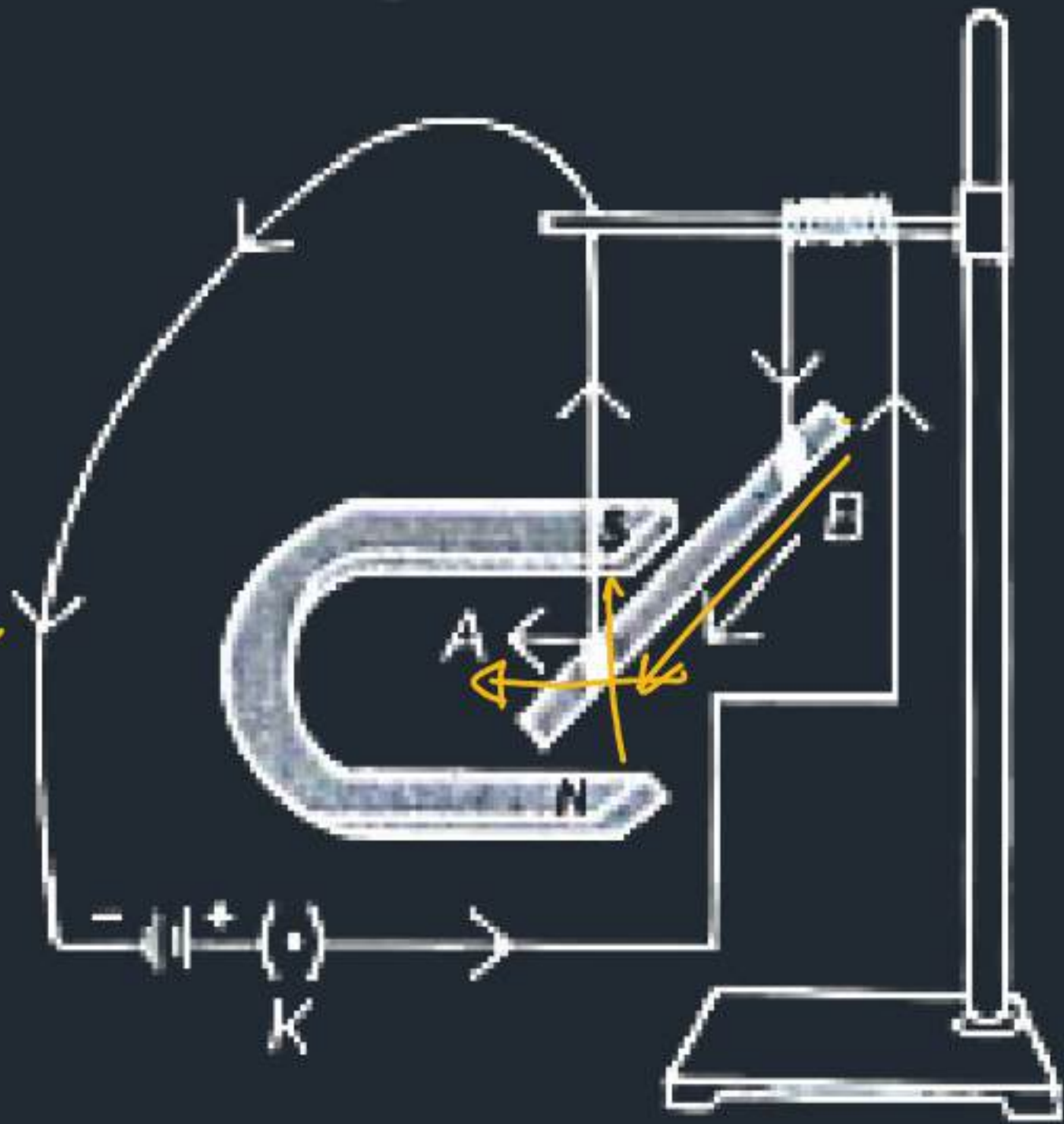
2022

Q. A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminium rod AB, a strong horseshoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions

(a) Why does the rod get displaced on passing current through it?

(b) State the rule that determines the direction of the force on the conductor AB.

(c) If the U shaped magnet is held vertically and the aluminium rod is suspended horizontally with its end B towards due north, then on passing current through the rod B to A as shown, in which direction will the rod be displaced?



Ans: (a) On passing current, the rod gets displaced because of a magnetic force exerted on the rod when it is placed in the magnetic field.

(b) Fleming's left hand rule is used to determine the direction of magnetic force exerted on the conductor AB.

(c) The rod will be displaced towards left according to Fleming's left-hand rule.

Factors on Which Force on a Current-Carrying Conductor Depends:

✓ **Current (I):** Greater the current, greater the force.

✓ **Magnetic Field (B):** Stronger field → larger force.

✓ **Length (L):** Longer conductor → more force.

• **Angle (θ):**

✓ **Maximum at 90° (conductor $\perp$ field)**

✓ **Zero at 0° or 180° (conductor $\parallel$ field)**

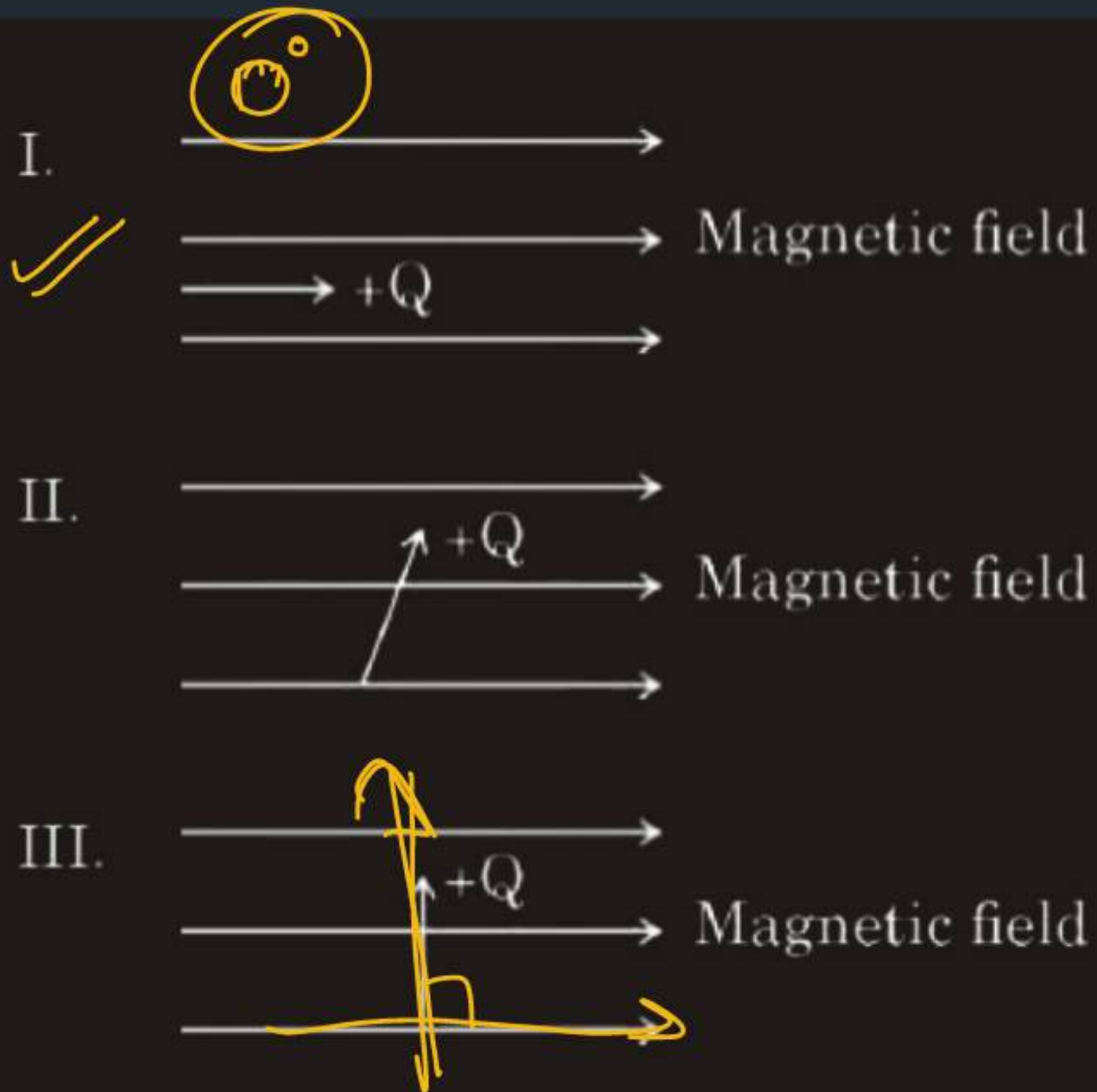
$$F = \underline{I B L \sin \theta}$$



$$F = BIL \sin \theta$$

2025

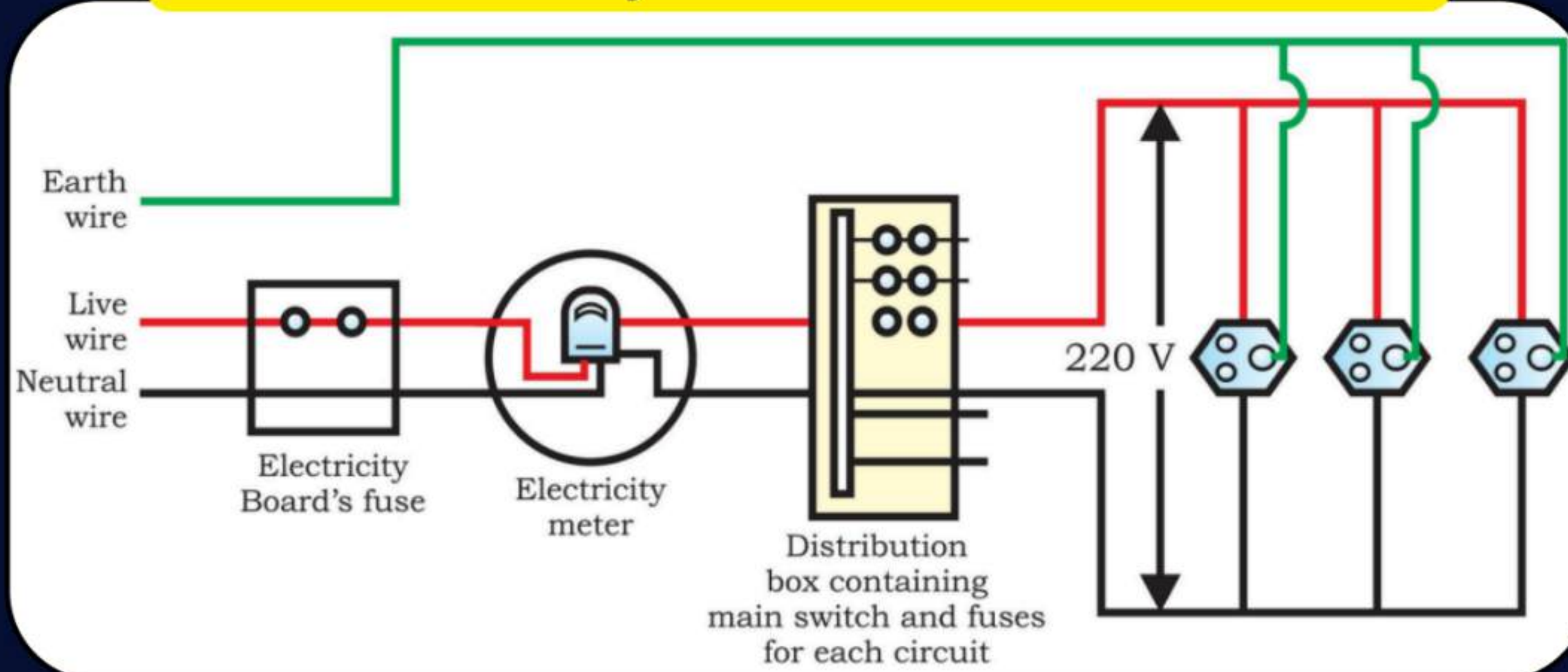
Q. Consider three diagrams in which the entry of a positive charge ($+Q$) in a magnetic field is shown. Identify the case in which the force experienced by the charge is (i) maximum, and (ii) minimum.



DOMESTIC ELECTRIC CIRCUIT

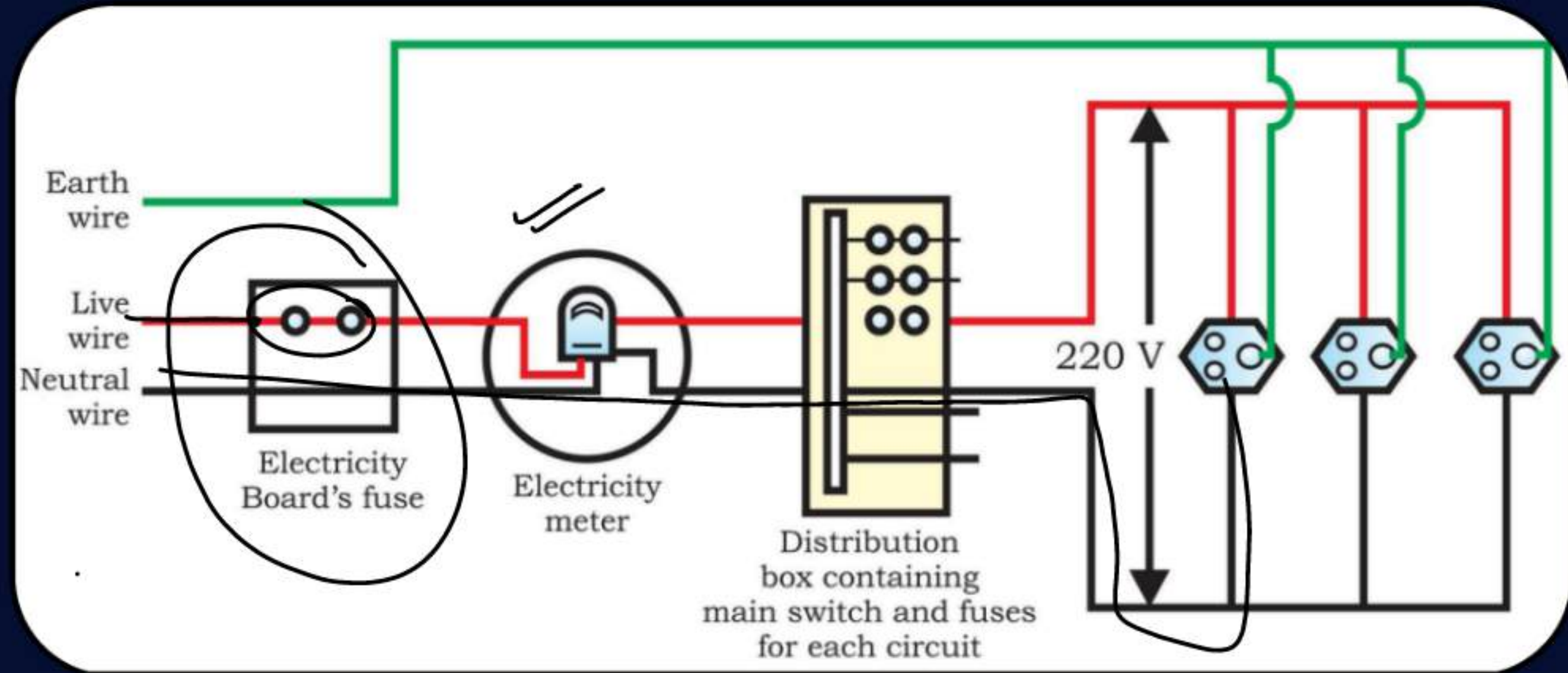
The system through which we receive electric power in our homes is called the domestic electric circuit. It is designed to safely distribute electricity for running household appliances.

Draw a labelled diagram of a domestic electric circuit.



Advantages of Parallel Connection:

- ✓ 1. Each appliance gets the same potential difference equal to the supply voltage (220 V).
- ✓ 2. Each appliance can be switched on or off independently without affecting other appliances.
- ✓ 3. Devices of different power ratings can be used together as they draw the required current according to their resistance.



3 TYPES OF WIRE

// **Live Wire (Phase Wire):** Usually covered with red insulation.

- Carries current from the power station to the household.

C **Neutral Wire:** Usually covered with black insulation.

- Completes the circuit by carrying current back from the appliance.

Earth Wire: has green insulation and is connected to a metal plate deep in the earth near the house.

- ✎ It acts as a safety measure for appliances with metallic bodies (e.g., iron, toaster, fan, refrigerator).
- ✎ The metal body is connected to the earth wire, providing a low-resistance path for leakage current.
- It ensures that the metallic body remains at earth potential, preventing electric shocks.



Q. Why is it dangerous to touch a live wire but safe to touch a neutral wire?

220V

0V

A live wire has a high potential of 220 V, and touching it allows current to flow through the body to the earth, which can cause an electric shock.

The neutral wire is at zero potential, so touching it does not complete the circuit, and no current flows through the body.



In Domestic Electric Circuit:

- In India, the potential difference maintained between the live wire and the neutral wire is **220 volts**.
- At the meter board, these wires first pass through an electricity meter and a main fuse.
- Then they are connected to the main switch, which distributes current to different circuits in the house.

Usually, there are two separate circuits:

- **15 A circuit** → for high power appliances like geysers, air coolers, etc.
- **5 A circuit** → for low power appliances like bulbs and fans.

Electric Fuse:

- An electric fuse prevents overloading and short-circuiting in circuits.
- The fuse wire has a **low melting point and high resistance**.
- It is made of an alloy of **lead (Pb) and tin (Sn) or copper (Cu) and tin (Sn)**.
- The fuse is connected in **series** with the circuit.
- When excessive current flows, the fuse heats up (Joule's law: $H = I^2Rt$), melts, and breaks the circuit, protecting the appliances.

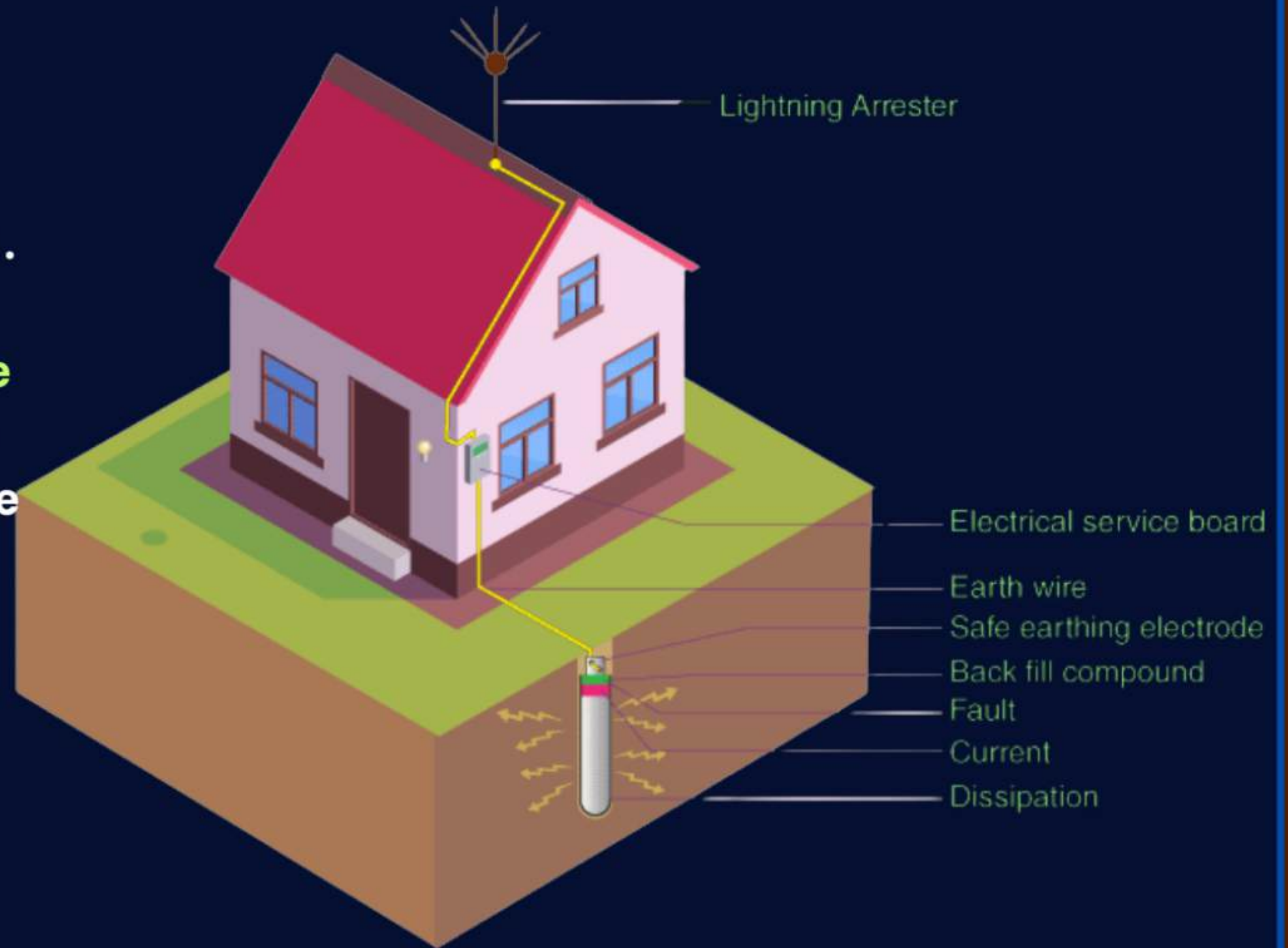
Fuse ratings:

- **5 A** → Low-power devices (bulbs, fans, TV)
- **15 A** → High-power devices (iron, geyser, cooler)



Earthing of Electrical Appliances:

- Earthing is a safety measure for appliances with metallic bodies (iron, toaster, fan, refrigerator, etc.).
- The **metallic body is connected to the earth wire** (green insulation), which is linked to a metal plate buried in the ground.
- In case of current leakage, the earth wire provides a low-resistance path for current to flow to the ground.
- This prevents electric shocks and protects appliances from damage.



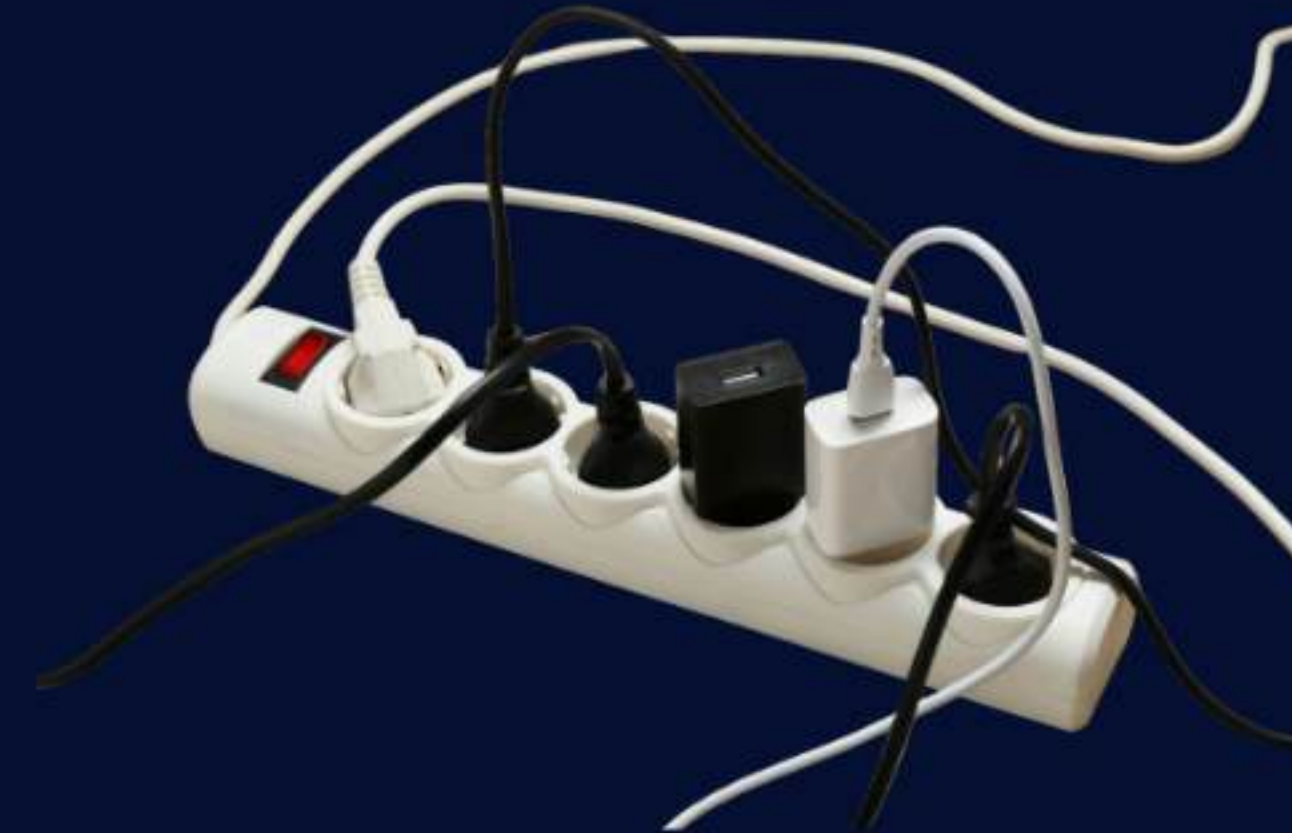
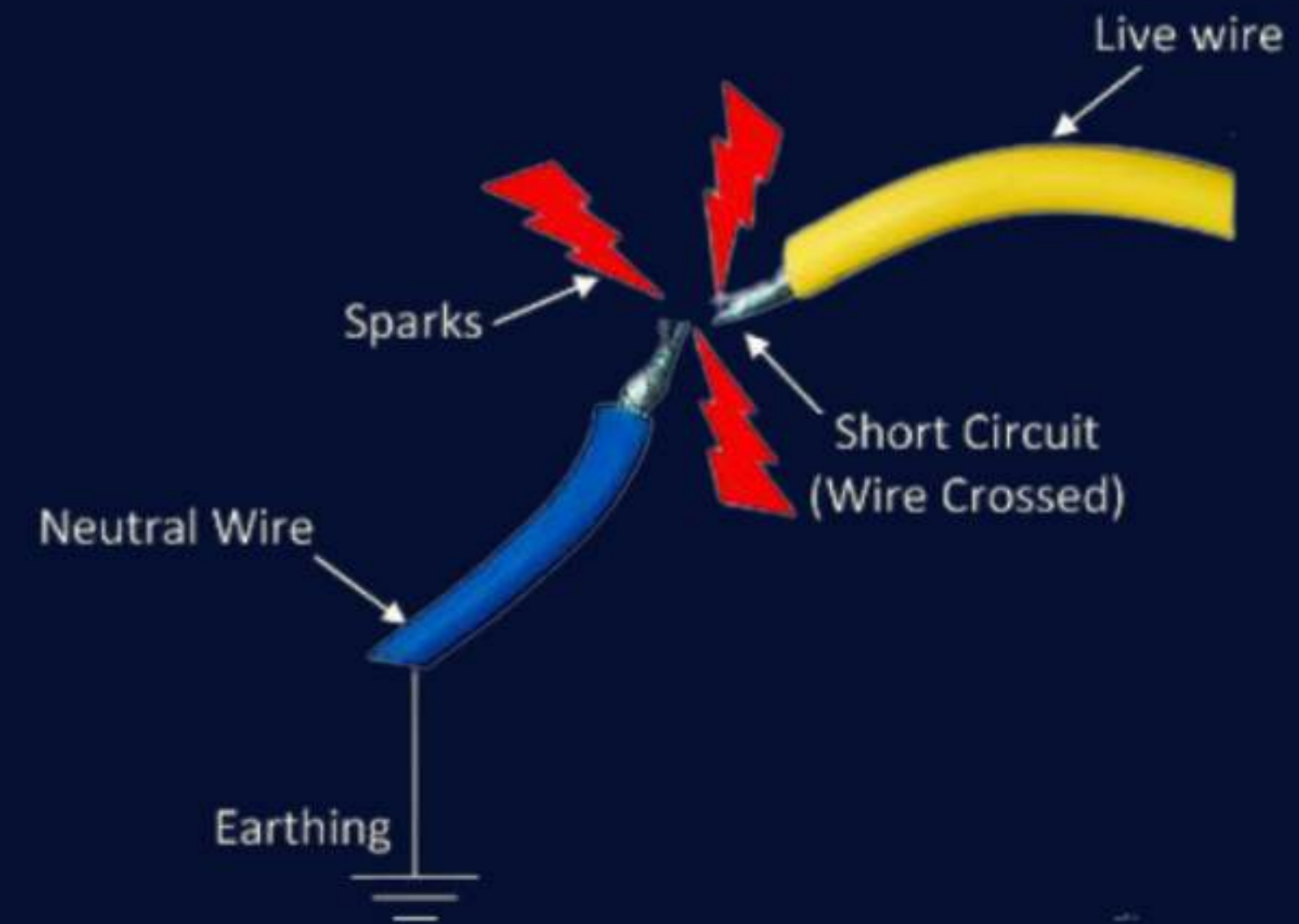
Short-Circuiting and Overloading:

- **Overloading** occurs when the circuit carries more current than it can safely handle.
- Overload: Resistance $\downarrow$ (low) $\rightarrow$ Current $\uparrow$ (high)

Overloading can also occur when:

- There is a sudden voltage hike, or
- Too many appliances are connected to a single socket.

- **Short-circuiting** happens when the live wire and neutral wire come into direct contact (due to damaged insulation or a fault in the appliance).
- In a short circuit, the current increases abruptly and can cause overheating or fire.
- Short Circuit: Resistance $\rightarrow 0 \rightarrow$ Current $\rightarrow \infty$ (theoretically)





$$R = 0$$

$$I = \infty$$

2016, 2017, 2019, 2022, 2024

Q. What is the function of an electric fuse in a domestic circuit? How does it work?

A fuse is a safety device made of a thin wire of low melting point (usually tin or lead alloy).

Function: It protects the circuit and appliances from damage due to excessive current.

Working: When current exceeds the safe limit, the fuse wire melts and breaks the circuit, stopping current flow and preventing fire or damage.

2015, 2018, 2020 Sample, 2023

**Q. What is meant by short-circuiting and overloading in an electric circuit?
How does a fuse prevent damage due to them? //**

- Short-circuiting: Occurs when live and neutral wires come into direct contact, causing a sudden large current flow.
- Overloading: Happens when too many appliances are connected in the same circuit, drawing excess current.
- Fuse protection: The fuse wire melts and breaks the circuit when excess current flows, thus preventing overheating and fire.

2014, 2016, 2019, 2022, 2025

Q. Why is the earth wire necessary? Where is it connected?

- The earth wire provides a low-resistance path for leakage current to flow safely into the ground.
- It is connected to the metal body of the appliance and the earth terminal at the ground, preventing electric shock.

2014, 2018, 2020, 2024

Q. Distinguish between the live wire and the neutral wire.

Live Wire	Neutral Wire
Carries current from the supply to the appliance.	Completes the circuit by carrying current back.
Maintained at 220 V potential. ✓	Maintained at 0 V potential. ✓
Usually red in colour.	Usually black in colour.

2015, 2019, 2021, 2024

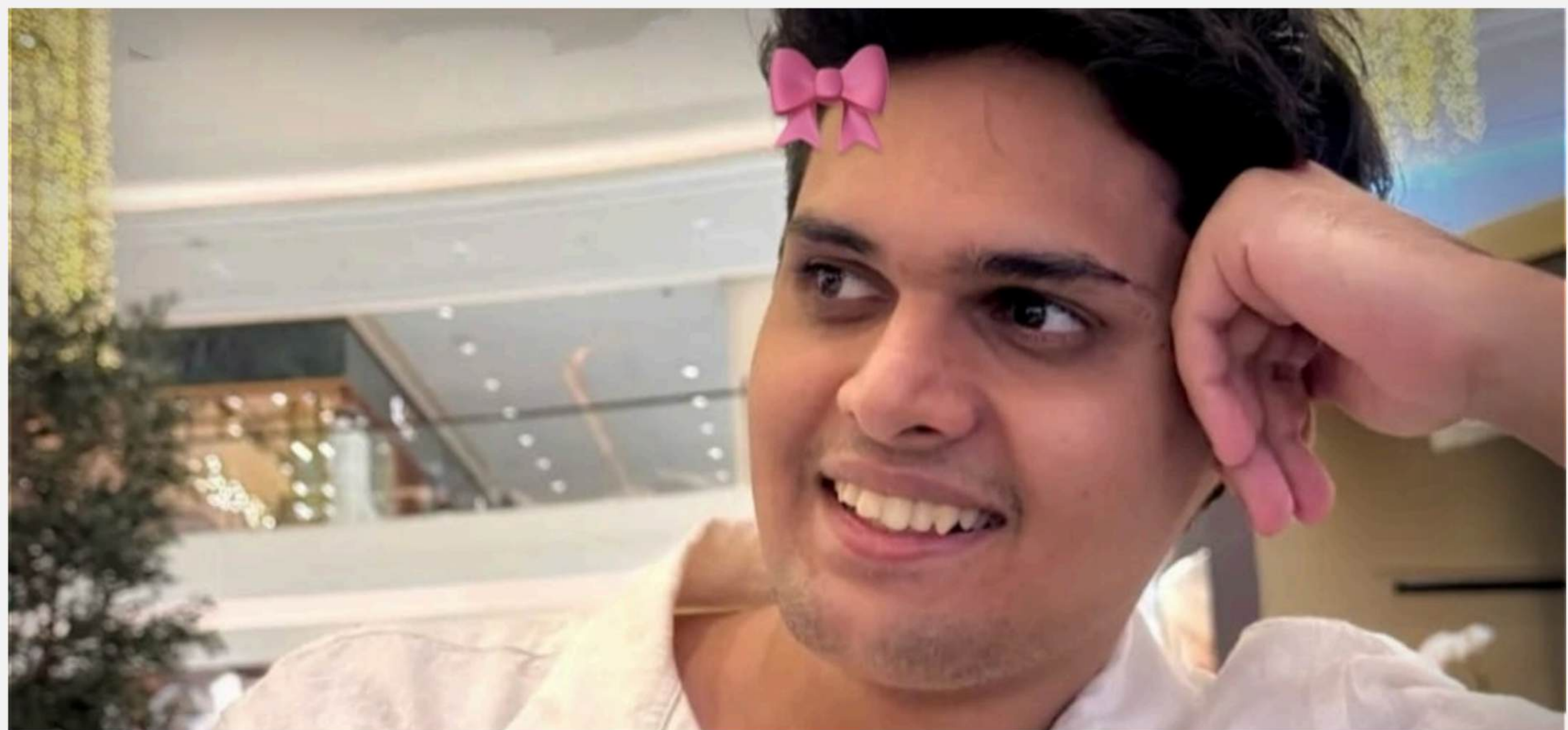
Q. Why are switches and fuses always connected in the live wire?

- Because the live wire carries current to the appliance.
- Connecting switches and fuses in the live wire ensures that when the circuit is open or fuse blows, the appliance is disconnected from the high-voltage supply, ensuring safety.

AC v/s DC



Alternating Current (AC)	✓ Direct Current (DC)
AC can travel safely over long distances, even <u>between cities</u> .	DC cannot travel long distances; it loses power.
Frequency is 50 or 60 Hz, depending on the country.	DC has zero frequency.
Current direction reverses periodically.	✓ Current flows steadily in one direction.
✓ Cheaper than DC generation	✓ Expensive than AC generation



SAMAJH AYA NA MURKH BALAK OR BALIKAO??